Summary Report, v4: Nine-Component Composite, Three Sampled Designs, Honest Selection

2026

1 Setup

1.1 Shared setup (identical across versions)

- **Suite.** 31 datasets: 11 entity matching (EM), 6 schema matching (SM), 11 table question answering and understanding (TQA), 3 financial documents (FIN). Official test or dev splits only; 150 items per dataset by seeded stratified sampling, 100 flagged for answering (OfficeQA Pro V2: 90; Alaska schema: 93); 3,083 answered items per model. Entity-matching pools are 50/50 match / non-match (Papadakis–Christen keeps its real 1.8 % ratio); schema matching is one question per source column with a candidate list; BIRD is graded on the executed result of the gold SQL.
- **Answers.** Eleven models (Claude Opus 5, Claude Sonnet 5, GPT-5.5, GPT-5.4-mini, Gemini 3.1 Pro, Gemini 3.8 Flash, Grok 4, DeepSeek-chat, DeepSeek-R1, gpt-oss-120b, Llama 3.3 70B); one prompt per item ending in `FINAL ANSWER: <answer>`; temperature 0 where accepted (Claude 5 and GPT-5.5 ran at provider defaults, recorded per row); output budget 700 tokens with mechanical re-asks at 2,500, 8,000, 16,000 and 32,000 for replies cut off before a final line (the budget is counted in each provider's own tokens and, for reasoning models, includes the chain of thought, so it is a termination test rather than an equal text allowance; 55 DeepSeek-R1 answers never terminate at 32,000 and are scored as given); every answer cached with tokens, cost and finish reason. The same cache feeds every version, so leaderboards are directly comparable.
- **Scoring primitives.** Extraction of the final line, Unicode / case / punctuation normalisation, number parsing with sign, percent and scale words, alias lists per gold (best alias wins), list golds, a hedge flag. Native metric per dataset: numeric tolerance (1 %) for numeric golds, exact match for text and boolean, ROUGE-L for free-form.
- **Coverage rule.** A model is in the study only if it covered ≥ 90 % of the suite; a dataset counts only items every model answered and only with ≥ 20 such items.
- **Leaderboard machinery.** Bradley–Terry strengths from item-level pairwise wins (MLE, ridge 0.01, SE from the observed information) as the headline pooling; Borda, Kemeny (with cost), Copeland, mean-z, reciprocal-rank fusion and family-balanced z as checks; 1,000 item bootstraps of the Borda and Bradley–Terry ranks; Friedman with Iman–Davenport, Nemenyi CD, Holm-corrected pairwise Wilcoxon across datasets; paired permutation and Wilcoxon per item within datasets (Holm); Kendall's W among datasets; majority-class baselines for boolean sets; post-stratified official-split estimates with a weighted bootstrap and positive-class F1 at the official match ratio; floor / ceiling / self-stability diagnostics per dataset.
- **Sampled designs.** Uniform-lattice, Dirichlet-snapped and Latin-hypercube samples on the weight simplex at step 0.05 (20 seeds each) with centred L_2 discrepancy and nearest-neighbour

distances, compared with the exhaustive optimum where one exists.
- **Reporting.** Each version's `run_all.sh` regenerates its `REPORT.md` from the cached answers; every number in this document is read from those output files.

1.2 What v4 searches

- **Per-answer rule.** A weighted sum of nine lexical components per answer: exact match, numeric tolerance (1 %), numeric decay $e^{-2.5 \cdot \text{rel. err}}$, token precision, token recall, token F1, edit similarity, ROUGE-L, Jaccard. The v1 rubric is the point (F1 0.35, decay 0.35, precision 0.15, recall 0.15) embedded in nine dimensions.
- **Candidates.** The step-0.05 lattice on the nine-weight simplex has 3,108,105 points, too many to enumerate, so three sampled designs of 100 points each (Dirichlet-snapped, Latin hypercube on the simplex, uniform lattice) plus five baselines (exact match only, F1 only, numeric tolerance only, uniform, v1 rubric): 305 candidates. Design comparison over 20 seeds with coverage diagnostics; regret is relative to the best over all seeds and designs since no exhaustive optimum exists.
- **Objective.** The same as v2: J = mean Kendall τ between each dataset's model ranking under the candidate and the Borda pool of the 31 rankings (also under Kemeny, Copeland and Bradley–Terry pooling; τ to native metrics beside). J measures consistency between datasets, not correctness.
- **Correctness.** 31,380 synthetic anchors of known utility (same generator as v2 and v5); Spearman ρ, AUC correct vs wrong, mean score of wrong answers, hedge gap; (J, ρ) Pareto front and knee.
- **Honesty and transfer.** Nested item bootstrap (B = 1,000: select in-bag, score out-of-bag; optimism gap; plateau), leave-one-dataset-out and leave-one-family-out, per-family optima, one-step sensitivity (neighbours one lattice step away within the candidate set).
- **Leaderboards.** Exact match, v1 rubric (headline for comparability with v2), the five best by J, the best by ρ and the Pareto knee, each pooled by Bradley–Terry with the full significance battery.

2 Results

2.1 A single typed metric matches the nine-component composite

The best candidate is exact match 1.00 ($J = 0.437$); the single-metric baselines exact match (0.437, rank 1) and numeric tolerance (0.434, rank 2) sit at or near the top, and the best mixed candidate reaches 0.432 (Figures 1, 2, 3). The v1 rubric sits at rank 230 of 305, but the whole candidate range is 0.039 wide. The three designs find the same region (Table 11); uniform lattice has the highest best-J and the uniform lattice the best coverage.

Table 1: v4 audit summary.

Candidates	305 (3 designs × 100 + 5 baselines) on a lattice of 3,108,105 points; J range 0.398–0.437
Best	exact match 1.00 (em_only), $J = 0.437$, τ native 0.92; exact match alone $J = 0.437$ (rank 1); v1 rubric $J = 0.411$ (rank 230 of 305)
Best per design	Dirichlet 0.436, LHS 0.438, uniform lattice 0.441; regret over seeds $\leq$ 0.015
Nested selection ($B = 1{,}000$)	in-bag J of selected 0.382, out-of-bag 0.334 (optimism gap 0.048); v1 rubric out-of-bag 0.339
Selected vs v1 rubric, out of bag	-0.005, 95 % CI $[-0.033, +0.017]$; 230 distinct candidates selected
Plateau	305 of 305 not significantly below the best; practical plateau (within 0.02): 304; v1 rubric inside: yes
Transfer (LODO)	mean held-out τ: re-fitted weights +0.42, full best +0.42, v1 rubric +0.38
Anchors	v1 rubric ρ 0.878; best-J 0.828; best-ρ 0.884; knee 0.884; exact match nan; Spearman(J, ρ) across candidates +0.00
Sensitivity	one step moves J by at most 0.013 around the v1 rubric and 0.020 around the best
Leaderboard (v1 rubric, BT)	first Gemini 3.8 Flash, then Claude Opus 5, GPT-5.5; last Llama 3.3 70B; Friedman p = 1.3e-14; $W = 0.27$
Leaderboard (best J = exact match 1.00, BT)	first Claude Opus 5; τ to the exact-match ranking 1.00

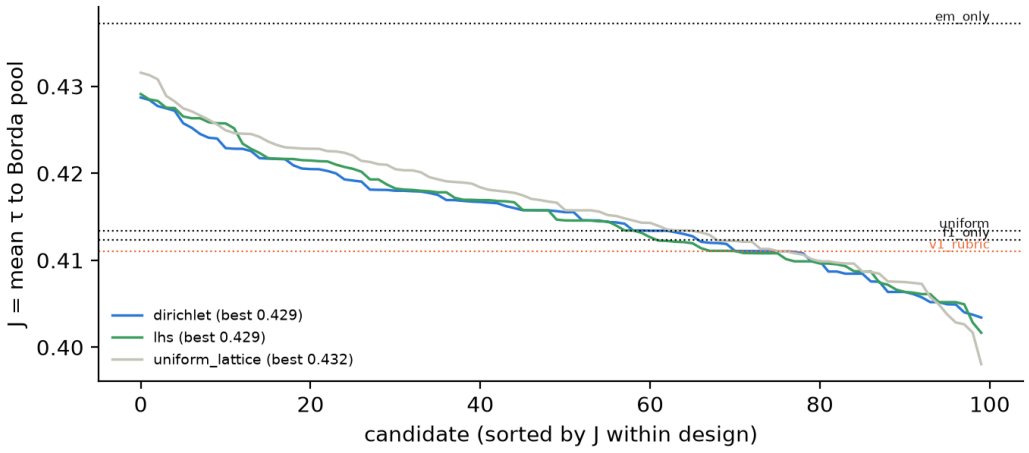


Figure 1: All 305 candidates sorted by J, with the baselines marked.

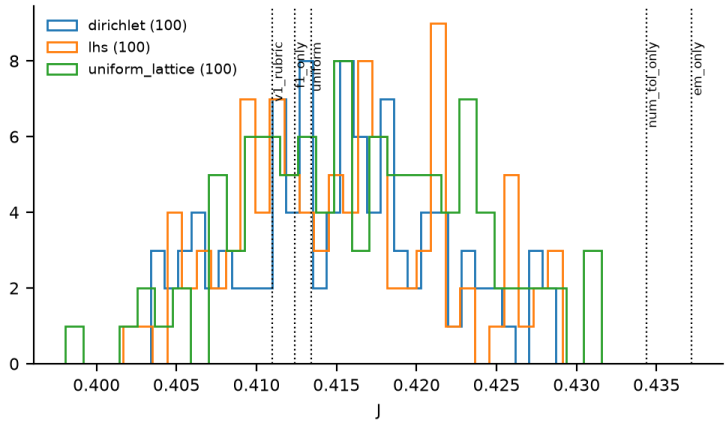


Figure 2: Distribution of J over the 100 candidates of each design, with the five baselines marked.

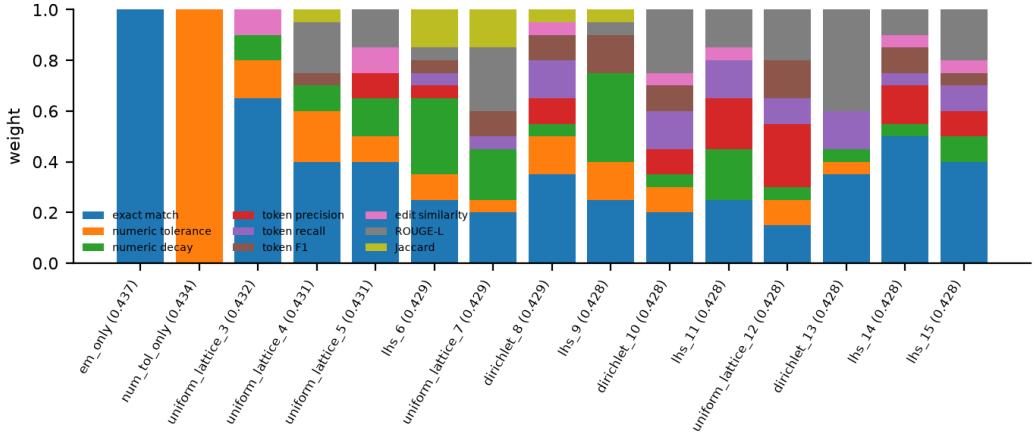


Figure 3: Component weights of the fifteen best candidates.

Table 2: Sampled designs: summary over 20 seeds.

design	best_J_mean	best_J_sd	best_J_max	mean_J	regret_mean	regret_max	min_pairwise_L1	mean_nearest_L1	discrepancy	zero_share
dirichlet_lattice	0.4307	0.0025	0.4357	0.4159	0.0105	0.0145	0.1950	0.4027	2.6738	0.8820
lhs_simplex	0.4325	0.0033	0.4383	0.4160	0.0087	0.0143	0.2150	0.4096	2.6718	0.8780
uniform_lattice	0.4330	0.0034	0.4412	0.4158	0.0082	0.0131	0.2350	0.4650	2.5904	0.9705

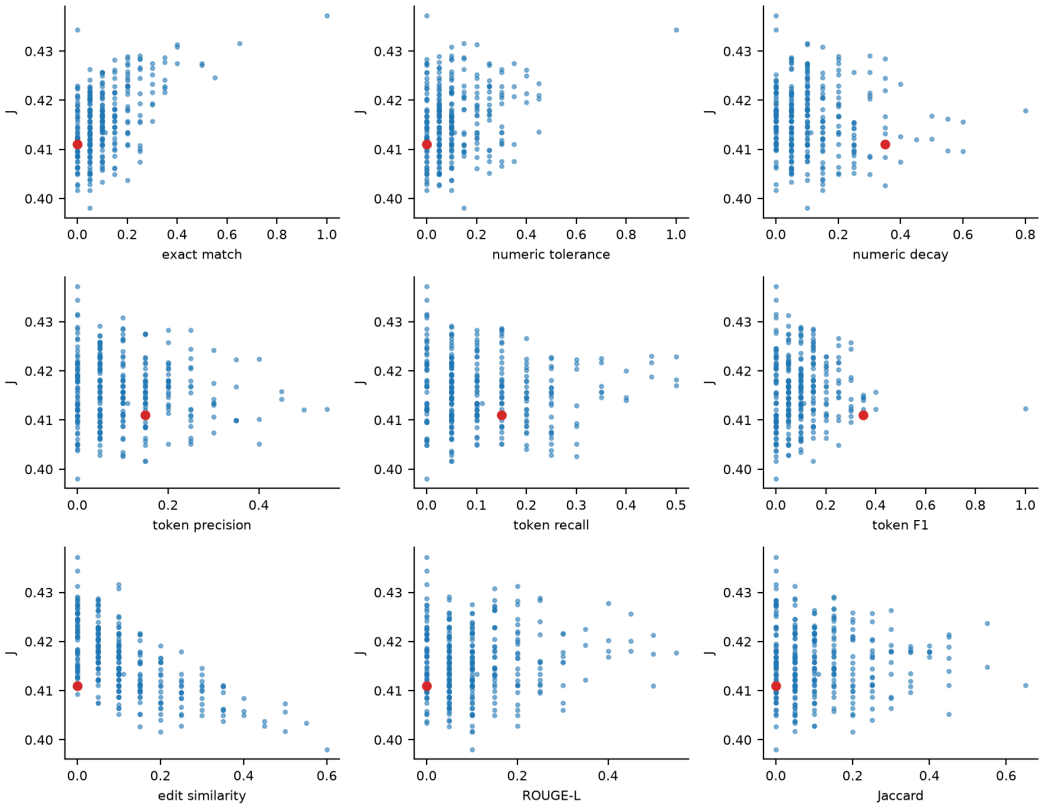


Figure 4: J against each component weight over all candidates; red = v1 rubric.

2.2 Honest selection

As in v2, the in-bag winner loses 0.048 out of bag and ends -0.005 from the v1 rubric with an interval that includes zero; 230 different candidates are selected across resamples and the plateau covers 305 of 305 (Figures 5, 6, 7).

2.3 Transfer

Re-fitted weights predict a held-out dataset no better than the v1 rubric (Figure 8); the per-family optima differ by family (Figure 9, Table 41).

Table 3: Per-family optimum.

family	best weights	J own best	J global best
EM	exact match 1.00	0.358	0.358
SM	numeric tolerance 0.15, numeric decay 0.60, edit similarity 0.25	0.544	0.541
TQA	exact match 1.00	0.523	0.523
FIN	numeric tolerance 0.15, numeric decay 0.05, token precision 0.25, token recall 0.05, ROUGE-L 0.15, Jaccard 0.35	0.881	0.738

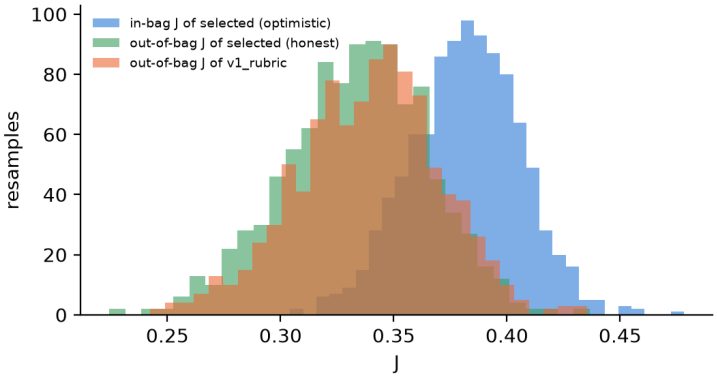


Figure 5: In-bag vs out-of-bag J of the selected candidate against the v1 rubric.

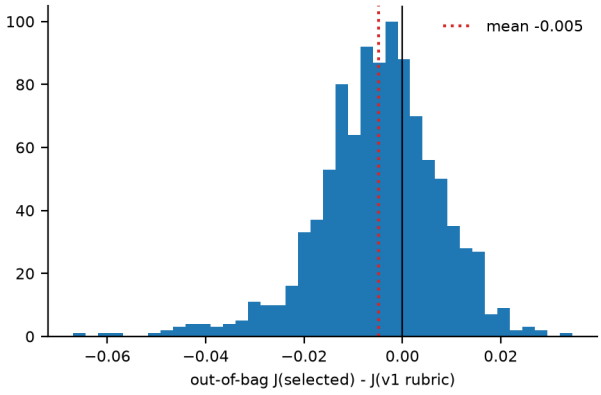


Figure 6: Paired out-of-bag difference between the selected candidate and the v1 rubric over 1,000 resamples.

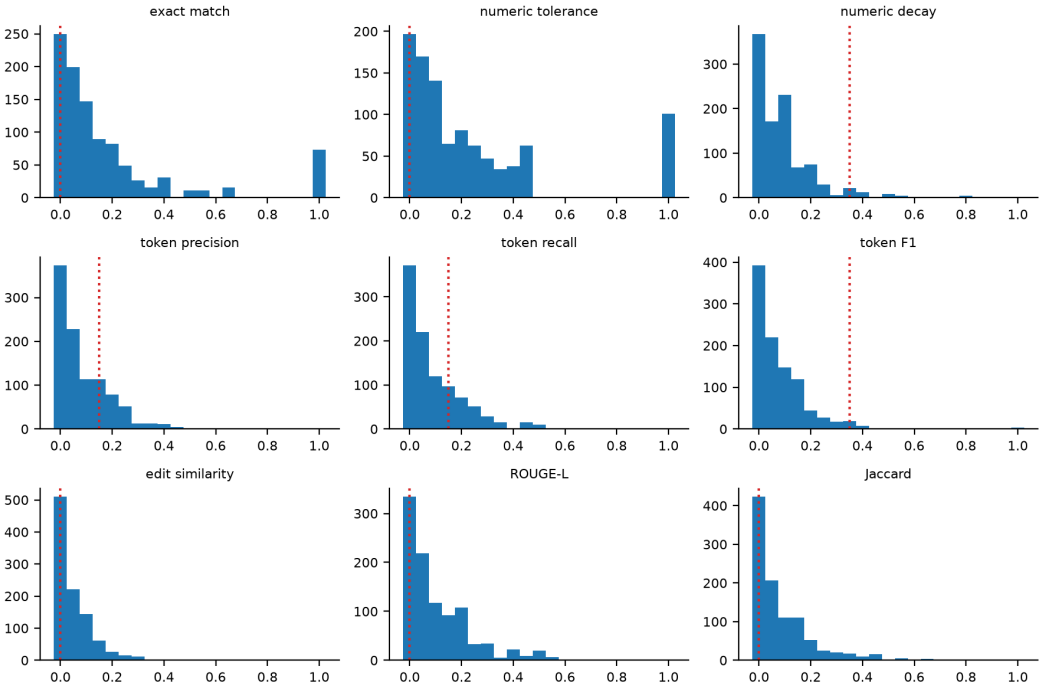


Figure 7: Weights selected as in-bag best across 1,000 bootstrap resamples; dotted = v1 rubric.

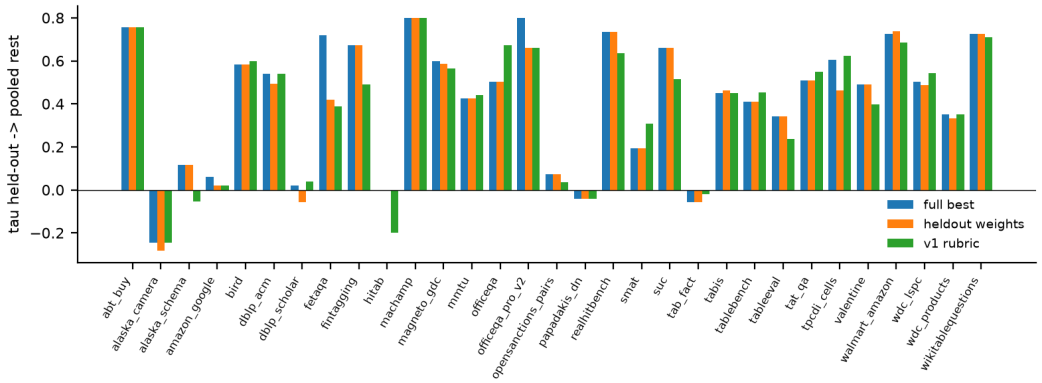


Figure 8: Leave-one-dataset-out: agreement of the held-out dataset with the pooled rest under re-fitted weights, the full best and the v1 rubric.

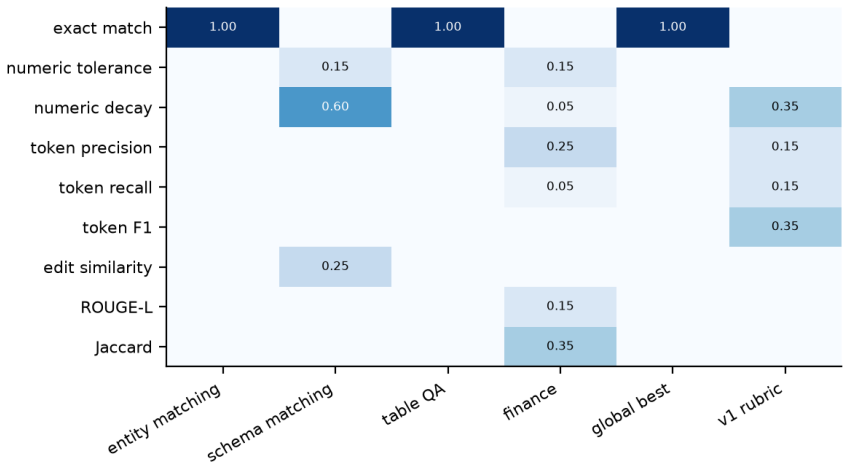


Figure 9: Per-family optimum, global optimum and the v1 rubric over the nine components.

2.4 Correctness

The best-by-J candidate (exact match 1.00) is the worst labelled rule on the anchors ($\rho = 0.828$ vs 0.878 for the v1 rubric): a binary metric gives no partial credit to near misses, which the anchors reward (Figure 11). The Pareto knee (em 0.40, num_tol 0.10, num_decay 0.15, tok_prec 0.10, edit_sim 0.10, rouge_l 0.15) keeps $J = 0.431$ and $\rho = 0.884$. Across candidates J and ρ correlate +0.00: agreement and correctness pull apart here as in v2.

Table 4: Anchor correctness for the labelled candidates.

label	w_em	w_num_tol	w_num_decay	w_tok_prec	w_tok_rec	w_tok_f1	w_edit_sim	w_rouge_l	w_jaccard	J	rho	rho_ds	auc	wrong_mean	hedge_gap
em_only	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.437	0.828	0.830	0.989	0.011	-0.807
num_tol_only	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.434	0.796	0.808	0.989	0.022	-0.595
uniform	0.111	0.111	0.111	0.111	0.111	0.111	0.111	0.111	0.111	0.413	0.878	0.861	0.994	0.046	-0.555
f1_only	0.000	0.000	0.000	0.000	0.000	1.000	0.000	0.000	0.000	0.412	0.856	0.846	0.989	0.030	-0.496
v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.411	0.878	0.868	0.994	0.034	-0.506

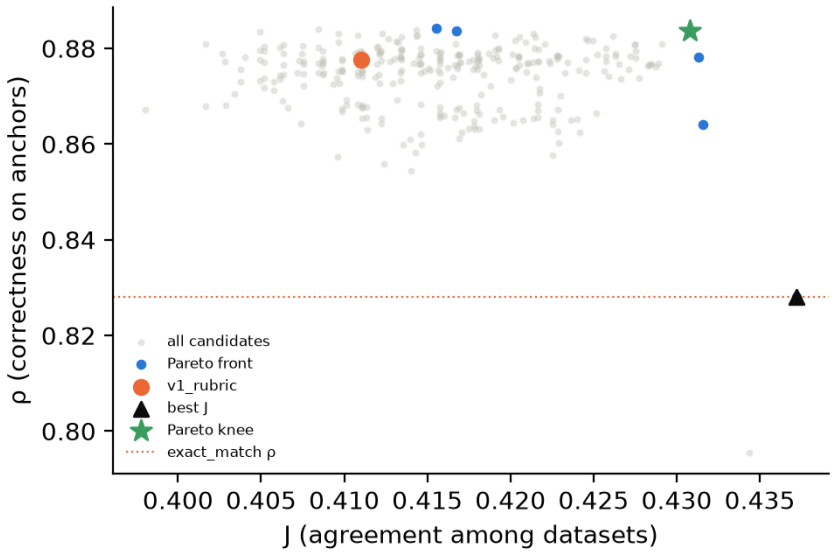


Figure 10: Agreement J against anchor correctness ρ for every candidate.

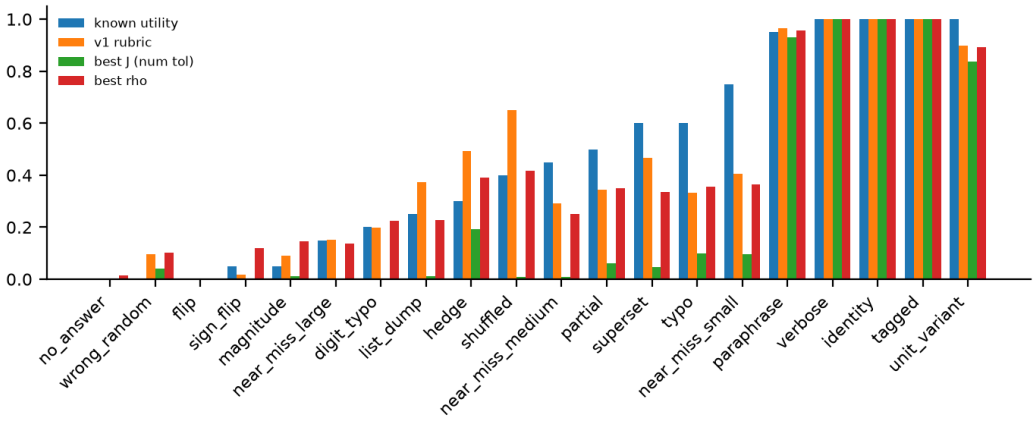


Figure 11: Anchors: known utility of each operator against the mean score it receives under the v1 rubric, the best-J and the best-ρ candidates.

2.5 Leaderboard

Under the v1 rubric the ranking is the one v2 reports (same answers, same rule): Gemini 3.8 Flash first by Bradley–Terry, GPT-5.5 first by Borda, Llama 3.3 70B last. Under the best-J rule (exact match 1.00) Claude Opus 5 is first, as under exact match. Rule choice moves the top three and nothing else.

Table 5: Leaderboard under the headline rule (v1 rubric).

rank	model	BT strength	SE	Borda rank	Borda 95%	P(first) BT	mean score
1	Gemini 3.8 Flash	0.112	0.015	3	[1, 4]	0.920	0.812
2	Claude Opus 5	0.097	0.000	2	[1, 3]	0.080	0.812
3	GPT-5.5	0.042	0.015	1	[1, 3]	0.000	0.808
4	DeepSeek-R1	0.037	0.015	4	[3, 7]	0.000	0.801
5	Claude Sonnet 5	0.023	0.015	5	[4, 7]	0.000	0.796
6	Gemini 3.1 Pro	0.021	0.015	6	[4, 7]	0.000	0.799
7	Grok 4	0.016	0.015	7	[4, 7]	0.000	0.797
8	GPT-5.4-mini	-0.022	0.015	9	[7, 9]	0.000	0.790
9	DeepSeek-chat	-0.031	0.015	8	[7, 9]	0.000	0.784
10	gpt-oss-120b	-0.111	0.015	10	[10, 11]	0.000	0.767
11	Llama 3.3 70B	-0.183	0.015	11	[10, 11]	0.000	0.738

Table 6: Bradley–Terry rank (Borda rank in brackets) under each reported rule, Friedman p and Kendall's W among datasets.

rule	Gemini 3.8 Flash	Claude Opus 5	GPT-5.5	DeepSeek-R1	Claude Sonnet 5	Gemini 3.1 Pro	Grok 4	GPT-5.4-mini	DeepSeek-chat	gpt-oss-120b	Llama 3.3 70B	Friedman p	W
exact match	3 (3)	1 (1)	2 (2)	4 (4)	7 (5)	5 (7)	6 (6)	8 (9)	9 (8)	10 (10)	11 (11)	2.7e-15	0.28
v1 rubric	1 (3)	2 (2)	3 (1)	4 (4)	5 (5)	6 (6)	7 (7)	8 (9)	9 (8)	10 (10)	11 (11)	1.3e-14	0.27
top1	3 (3)	1 (1)	2 (2)	4 (4)	7 (5)	5 (7)	6 (6)	8 (9)	9 (8)	10 (10)	11 (11)	2.7e-15	0.28
top2	3 (3)	1 (1)	2 (2)	4 (4)	7 (5)	5 (7)	6 (6)	8 (9)	9 (8)	10 (10)	11 (11)	4.2e-15	0.27
top3	2 (3)	1 (1)	3 (2)	4 (4)	5 (5)	6 (6)	7 (7)	8 (8)	9 (9)	10 (10)	11 (11)	2.0e-15	0.29
top4	1 (3)	2 (1)	3 (2)	4 (4)	6 (5)	5 (6)	7 (7)	8 (9)	9 (8)	10 (10)	11 (11)	9.9e-16	0.29
top5	1 (3)	2 (2)	3 (1)	4 (4)	5 (5)	6 (6)	7 (7)	8 (8)	9 (9)	10 (10)	11 (11)	1.6e-15	0.29
best-rho	1 (3)	2 (2)	3 (1)	4 (4)	5 (5)	6 (6)	7 (7)	8 (8)	9 (9)	10 (10)	11 (11)	1.0e-14	0.28
knee	1 (3)	2 (2)	3 (1)	4 (4)	5 (5)	6 (6)	7 (7)	8 (8)	9 (9)	10 (10)	11 (11)	1.6e-15	0.29

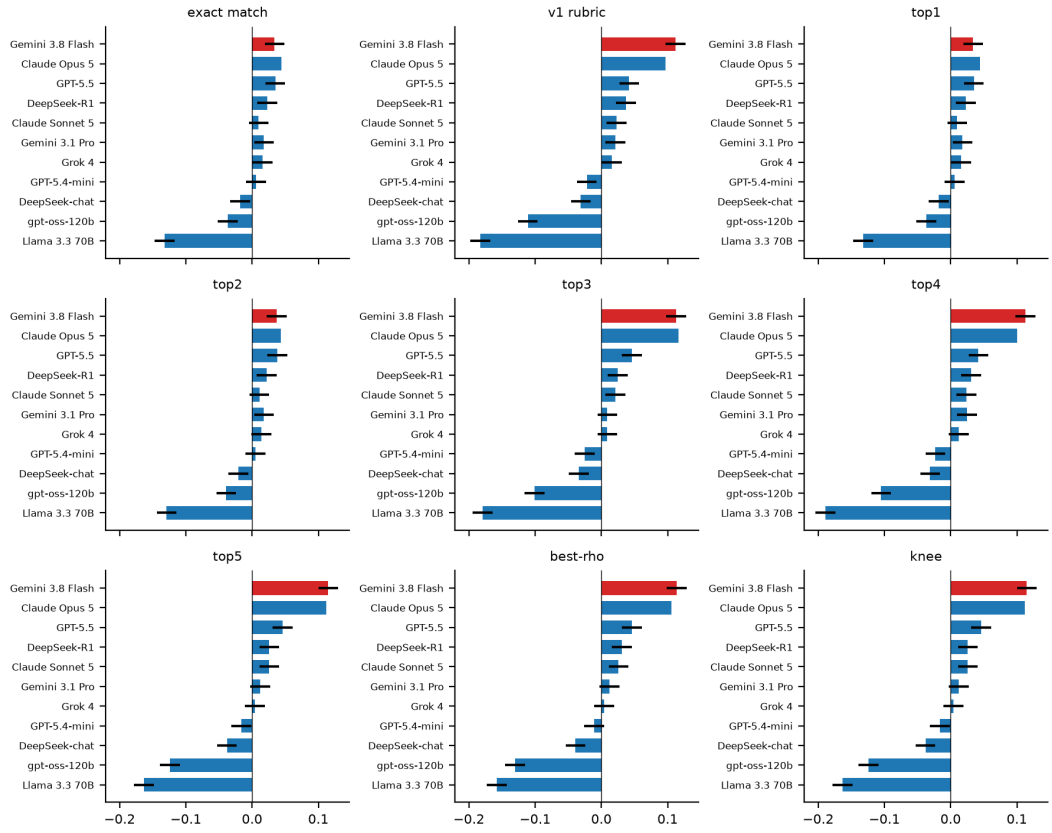


Figure 12: Bradley–Terry strengths (± SE vs reference) under every reported rule; models fixed in the headline order.

2.6 Comparability and diagnostics

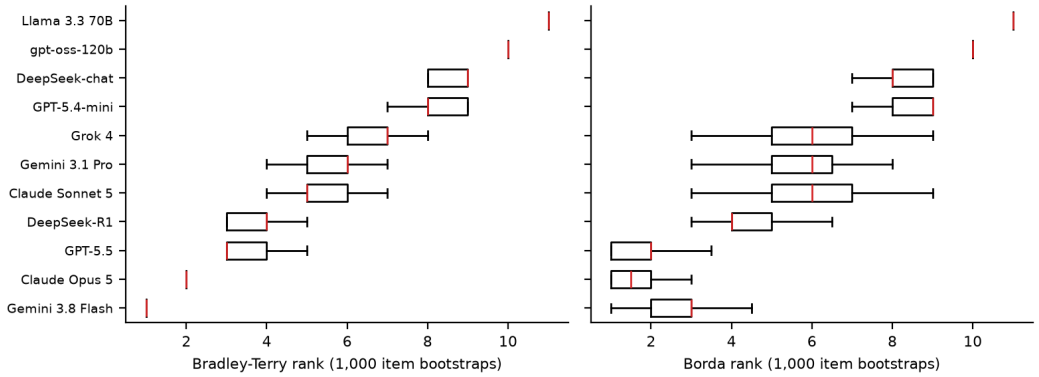


Figure 13: Bootstrap distribution of each model's rank under the headline rule (v1 rubric).

Table 7: Official-split estimates under the headline rule averaged over the ten models.

dataset	suite_mean	official_ estimate	suite_f1_ pos	official_precision_ pos	official_recall_ pos	official_f1_ pos	official_positive_ share
abt_buy	0.971	0.987	0.969	0.933	0.950	0.940	0.108
alaska_camera	0.945	0.945	0.939	1.000	0.895	0.939	0.500
alaska_schema	0.929	0.929	–	–	–	–	–
amazon_google	0.935	0.932	0.928	0.609	0.941	0.738	0.102
bird	0.191	0.218	–	–	–	–	–
dblp_acm	0.987	0.992	0.986	0.981	0.980	0.978	0.180
dblp_scholar	0.948	0.966	0.947	0.901	0.922	0.909	0.186
fetaqa	0.387	0.387	–	–	–	–	–
fintagging	0.884	0.882	–	–	–	–	–
hitab	0.772	0.776	–	–	–	–	–
machamp	0.887	0.891	0.892	0.649	0.852	0.735	0.175
magneto_gdc	0.937	0.940	–	–	–	–	–
mmtu	0.874	0.837	–	–	–	–	–
officeqa	0.135	0.135	–	–	–	–	–
officeqa_pro_v2	0.151	0.151	–	–	–	–	–
opensanctions_ pairs	0.835	0.748	0.790	0.993	0.677	0.795	0.769
papadakis_dn	0.940	0.987	0.831	0.527	0.967	0.662	0.018
realhitbench	0.677	0.676	–	–	–	–	–
smat	0.916	0.914	–	–	–	–	–
suc	0.917	0.915	–	–	–	–	–
tab_fact	0.802	0.792	0.761	0.903	0.674	0.767	0.500
tabis	0.918	0.923	–	–	–	–	–
tablebench	0.824	0.822	–	–	–	–	–
tableeval	0.755	0.758	–	–	–	–	–
tat_qa	0.804	0.759	–	–	–	–	–
tpcdi_cells	0.456	0.462	–	–	–	–	–
valentine	0.985	0.981	–	–	–	–	–
walmart_amazon	0.959	0.974	0.949	0.824	0.931	0.872	0.094
wdc_lspc	0.982	0.984	0.983	0.968	0.978	0.972	0.264
wdc_products	0.978	0.986	0.979	0.982	0.968	0.975	0.280
wikitablequestions	0.848	0.846	–	–	–	–	–

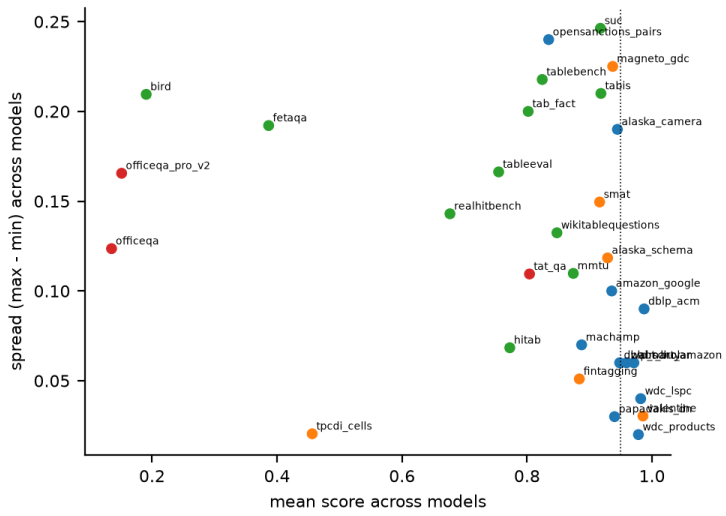


Figure 14: Dataset diagnostics: mean score against spread across models; dotted line marks the ceiling flag.

A All figures

Every figure that can be drawn from the saved outputs; source data in `data/`.

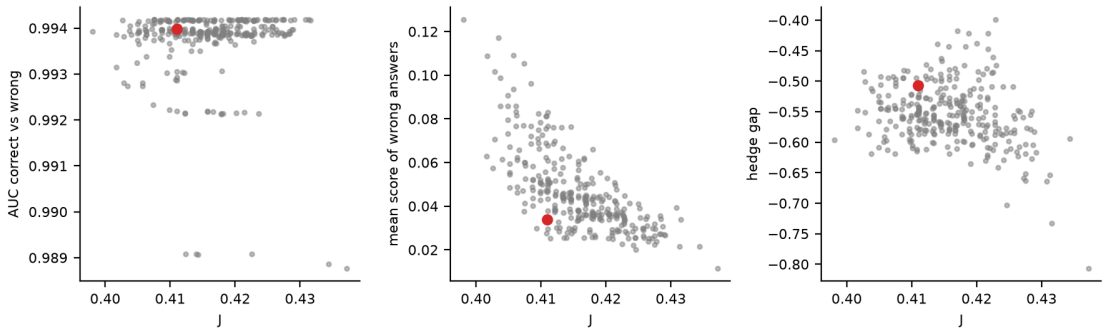


Figure 15: Three further anchor diagnostics against J for every candidate (red = v1 rubric).

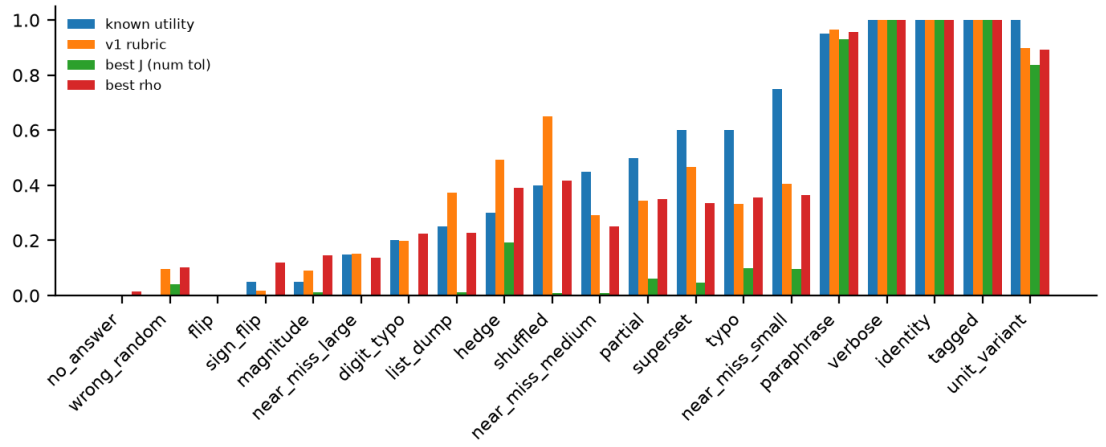


Figure 16: Anchors: known utility of each operator against the mean score it receives under the v1 rubric, the best-J and the best-ρ candidates.

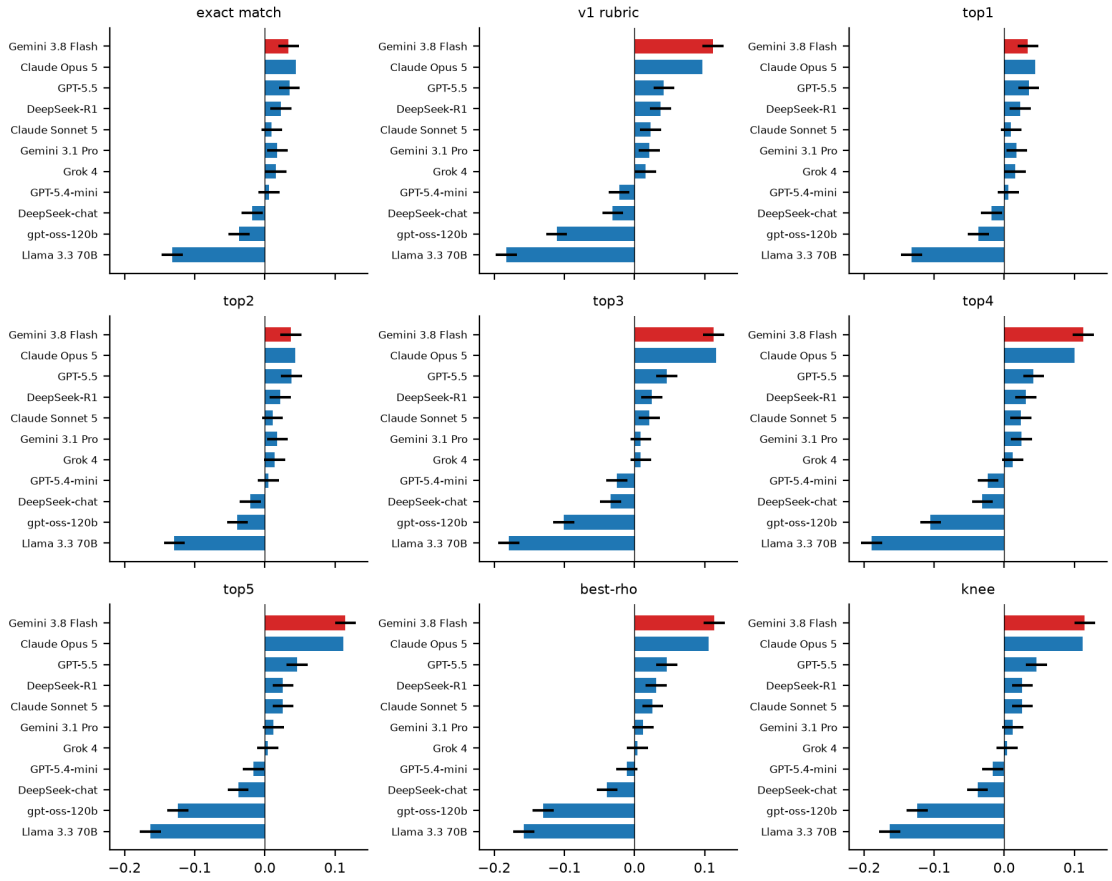


Figure 17: Bradley–Terry strengths (± SE vs reference) under every reported rule; models fixed in the headline order.

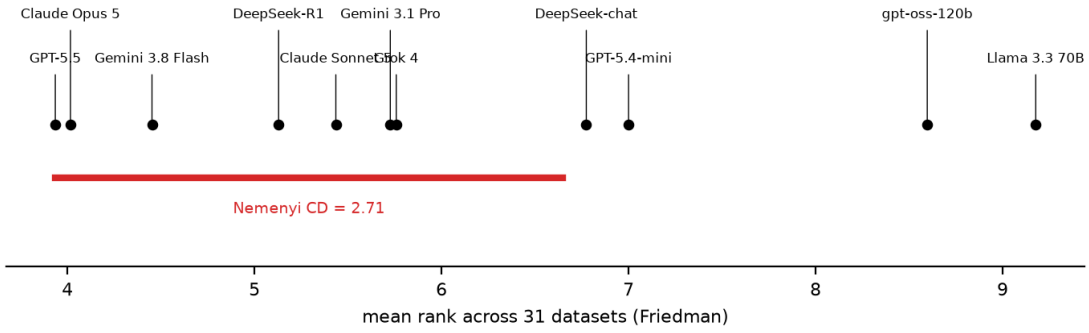


Figure 18: Critical-difference view of the Friedman test under the headline rule: mean rank per model and the Nemenyi critical difference at $\alpha = .05$.

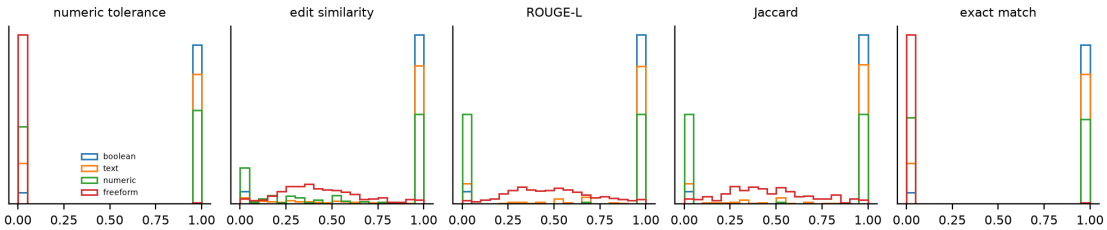


Figure 19: Distributions of the five components not in v1, by answer type.

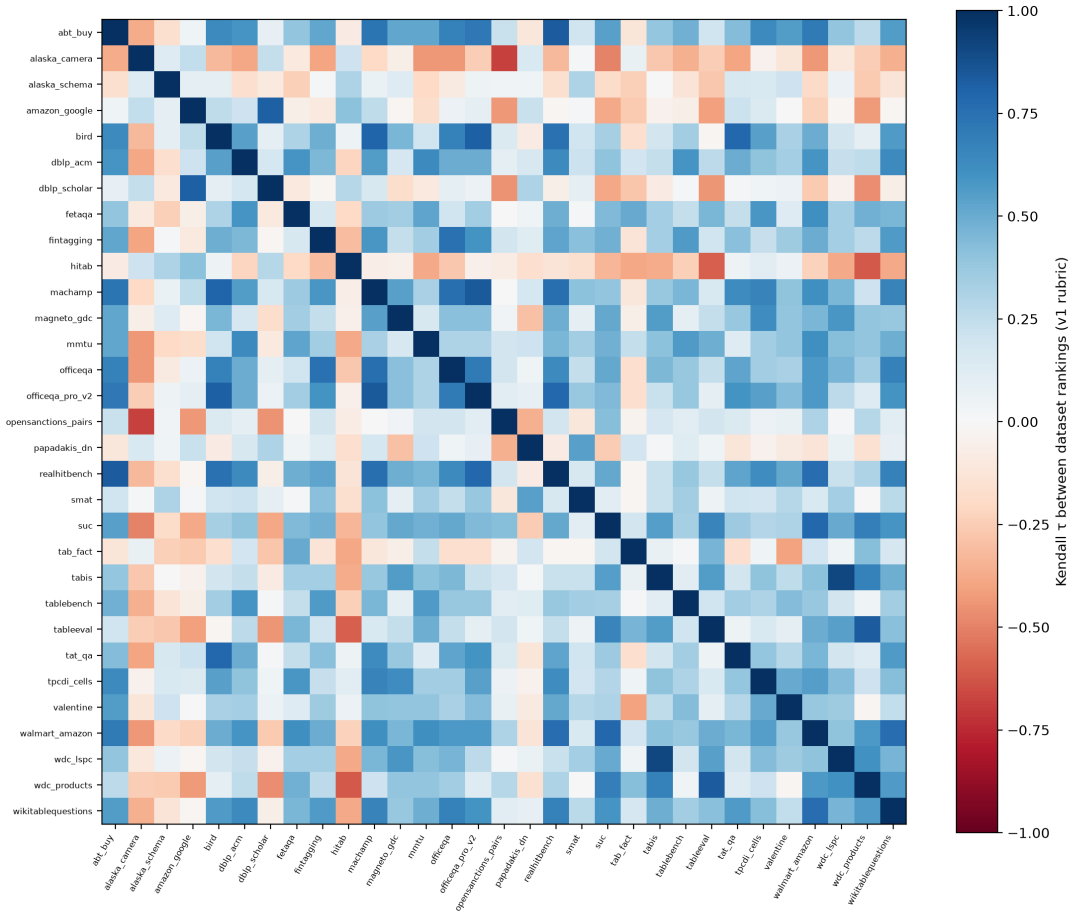


Figure 20: Kendall τ between the model rankings of each pair of datasets under the headline rule.

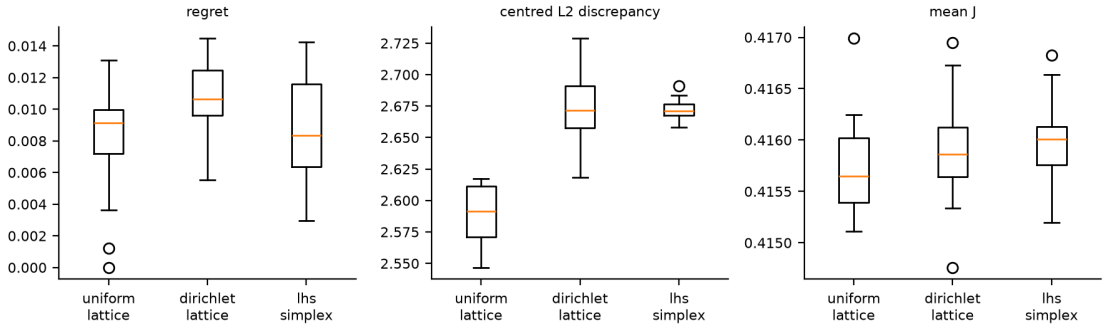


Figure 21: Sampled designs over 20 seeds: regret against the best over all seeds and designs, centred L_2 discrepancy and mean J of the design.

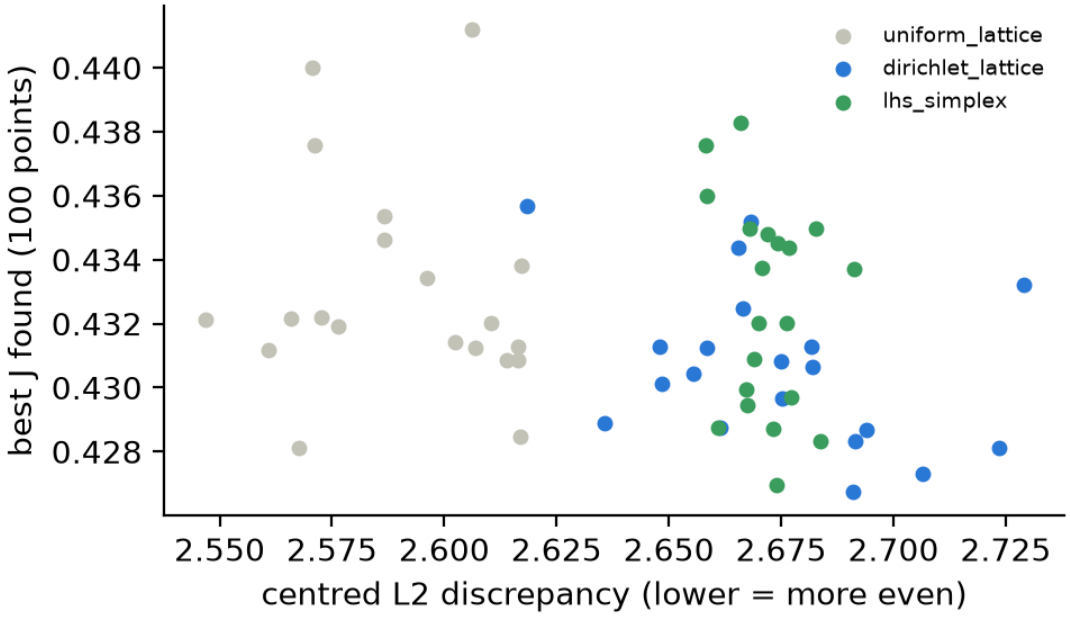


Figure 22: Sampled designs: best J found per seed.

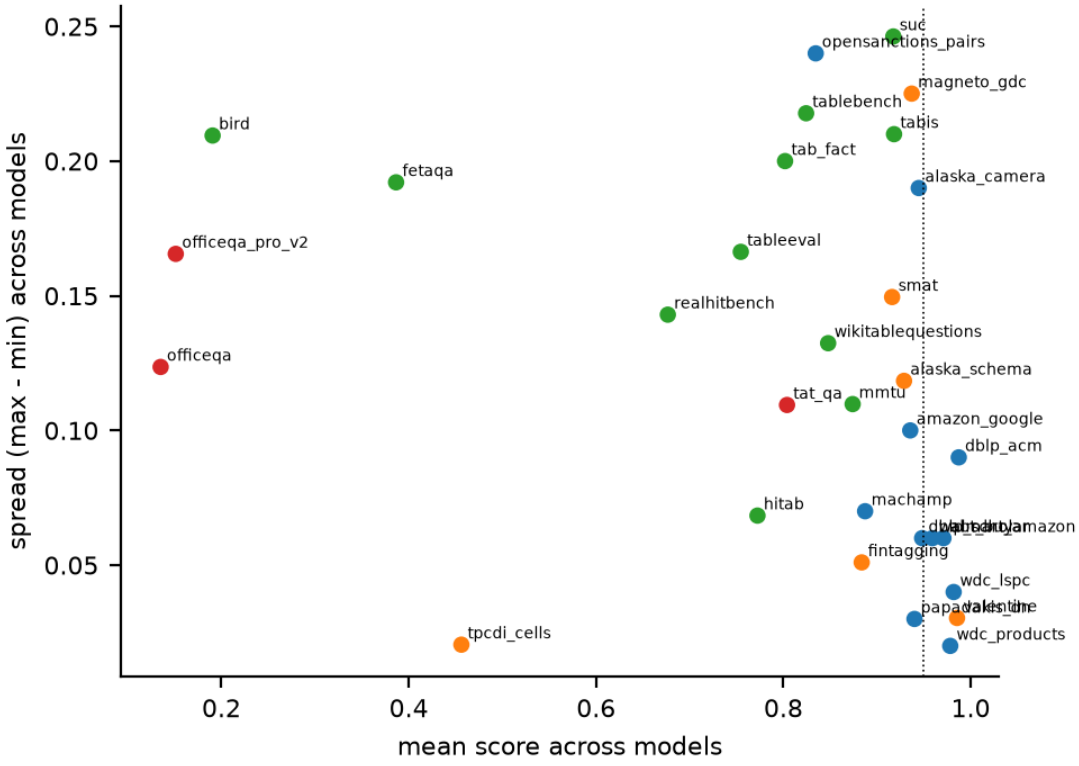


Figure 23: Dataset diagnostics: mean score against spread across models; dotted line marks the ceiling flag.

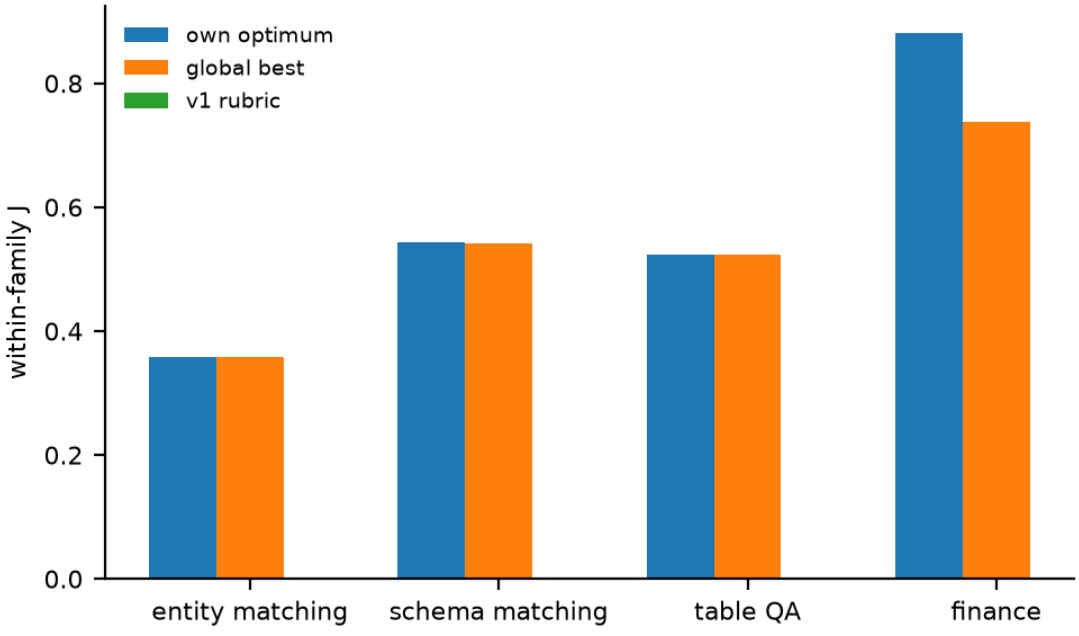


Figure 24: Within-family J under each family's own optimum, the global best and the v1 rubric.

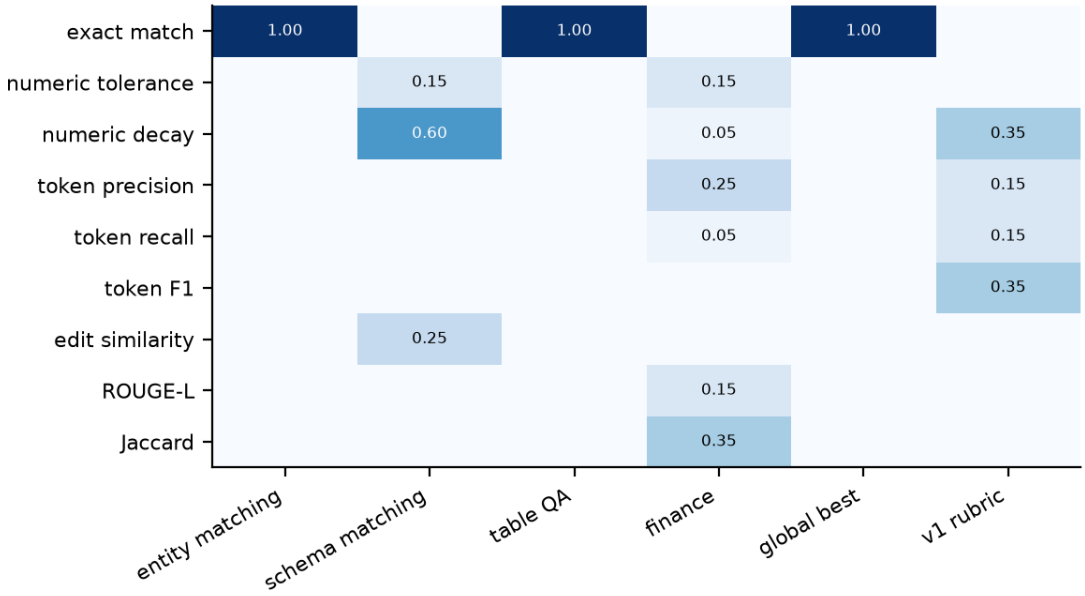


Figure 25: Per-family optimum, global optimum and the v1 rubric over the nine components.

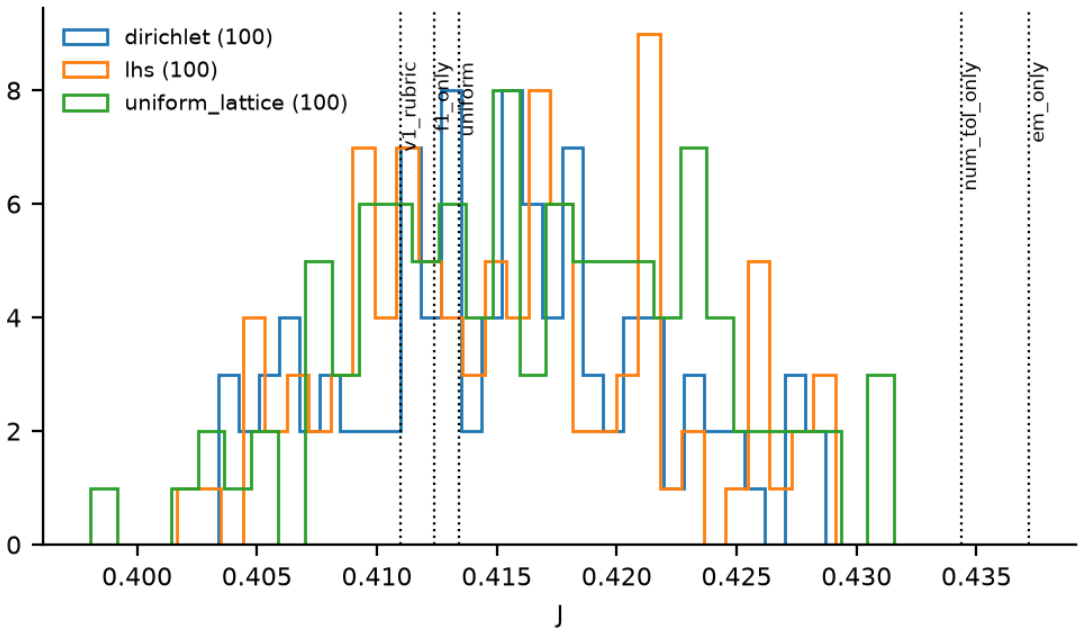


Figure 26: Distribution of J over the 100 candidates of each design, with the five baselines marked.

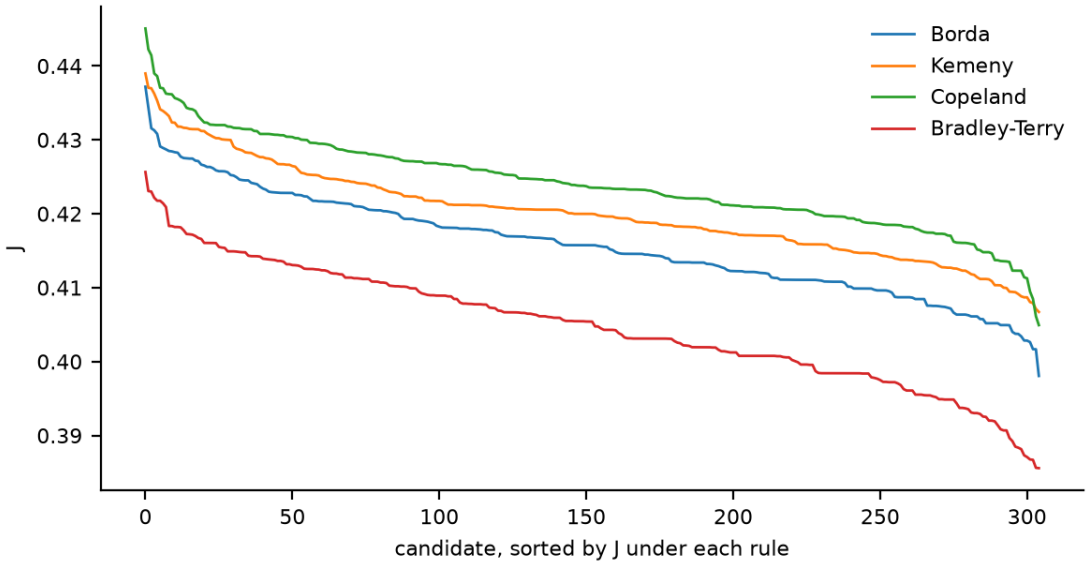


Figure 27: Objective over all candidates under four pooling rules.

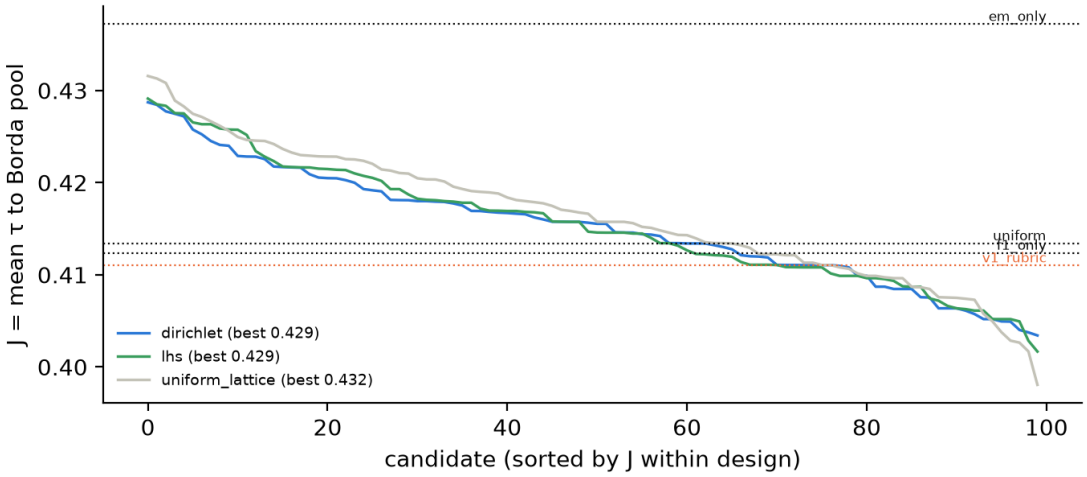


Figure 28: All 305 candidates sorted by J, with the baselines marked.

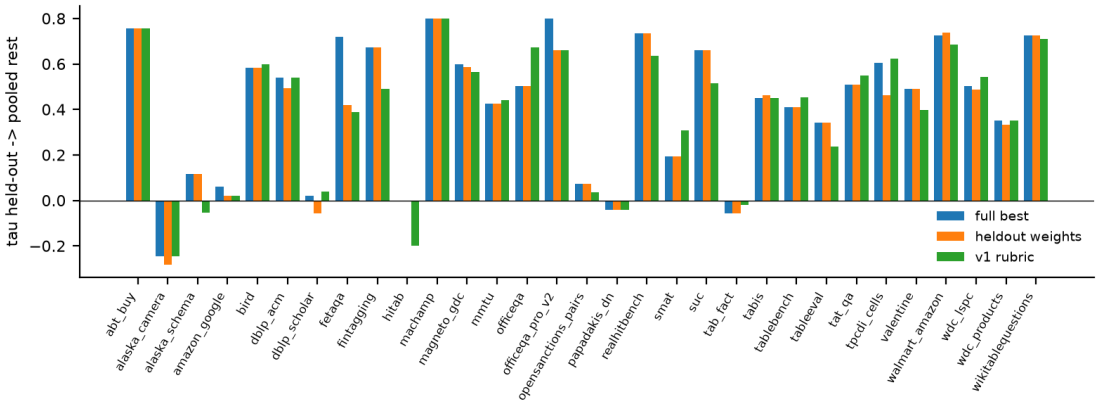


Figure 29: Leave-one-dataset-out: agreement of the held-out dataset with the pooled rest under re-fitted weights, the full best and the v1 rubric.

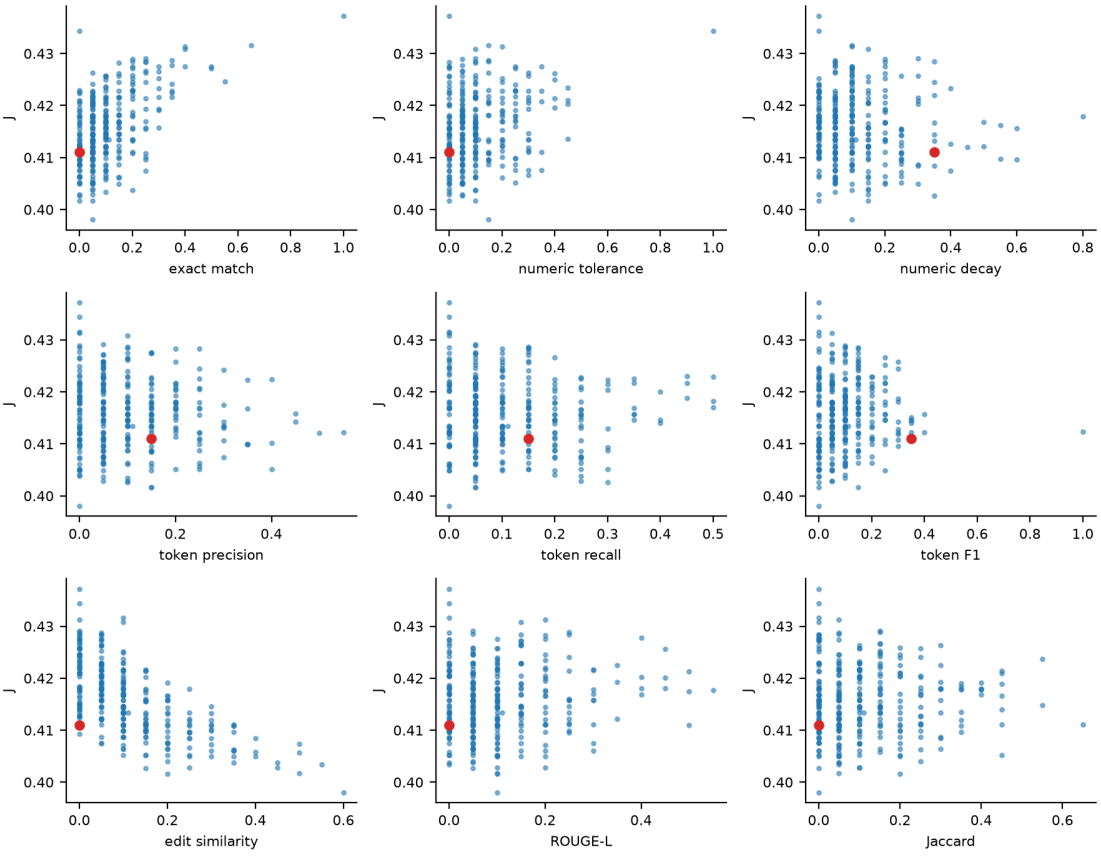


Figure 30: J against each component weight over all candidates; red = v1 rubric.

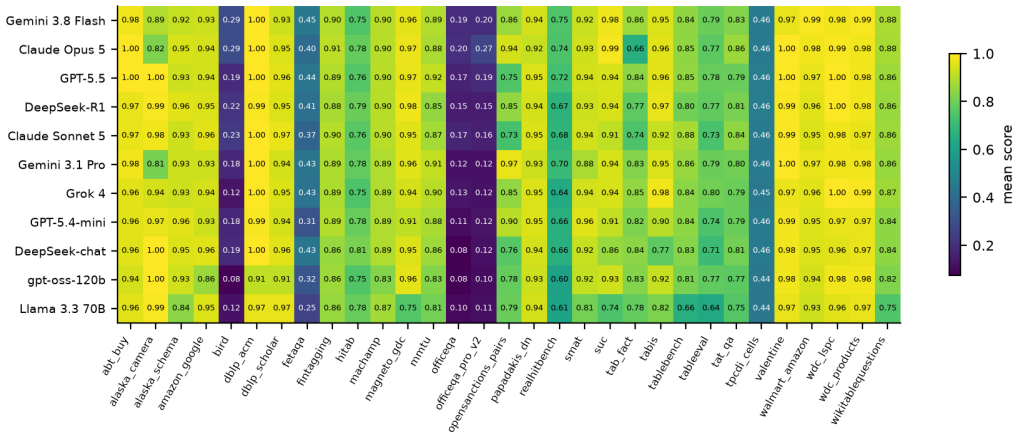


Figure 31: Mean score per model and dataset under the headline rule.

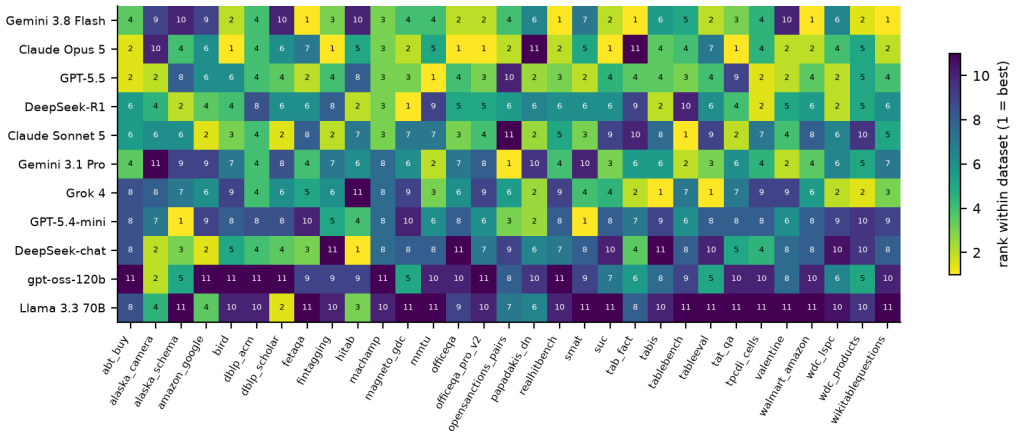


Figure 32: Rank per model and dataset under the headline rule.

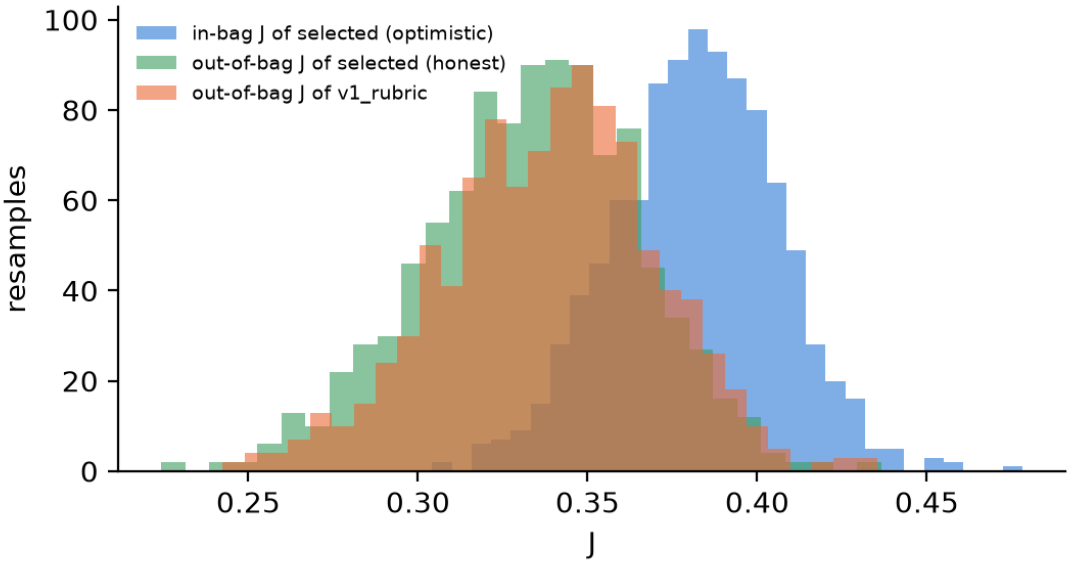


Figure 33: In-bag vs out-of-bag J of the selected candidate against the v1 rubric.

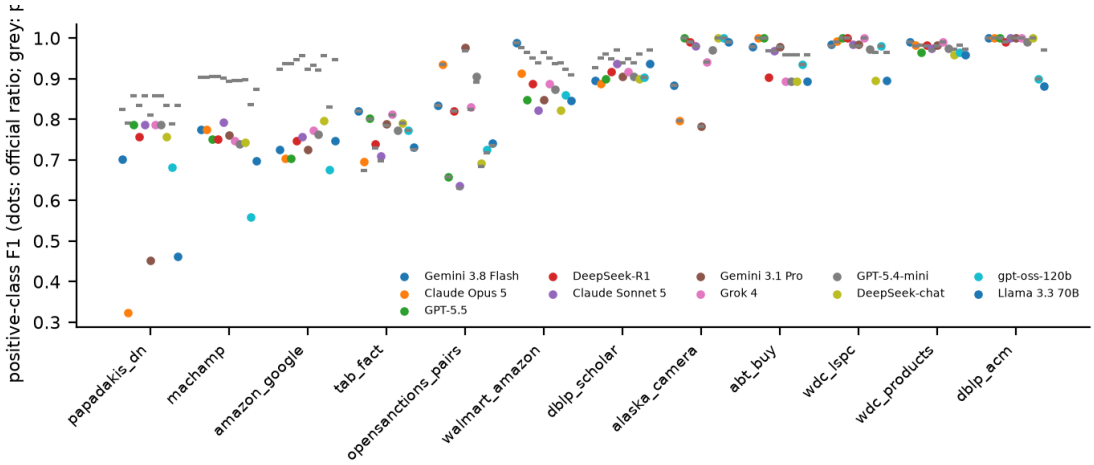


Figure 34: Positive-class F1 on the balanced pool (grey) and at the official match ratio (coloured) per model and matching dataset.

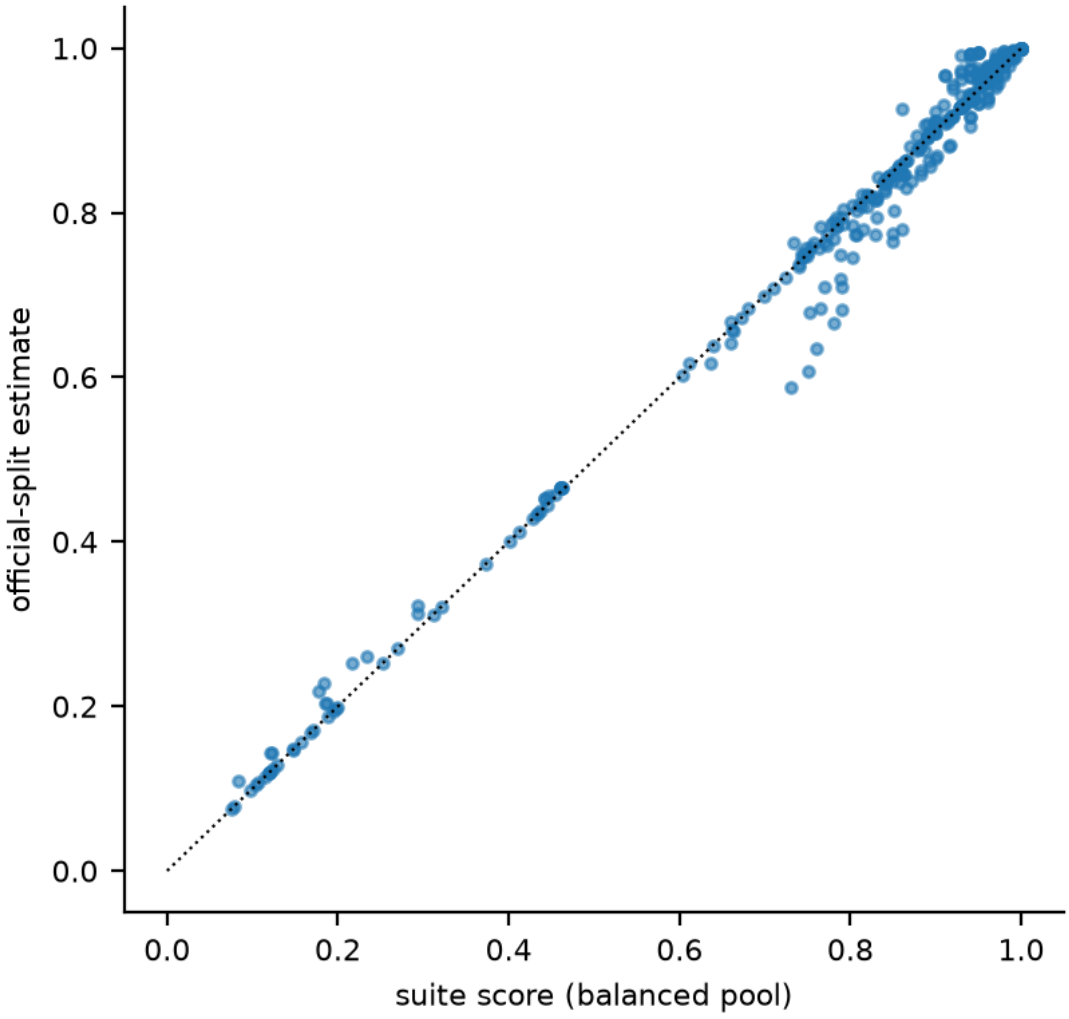


Figure 35: Suite score against post-stratified official-split estimate for every (dataset, model) pair under the headline rule.

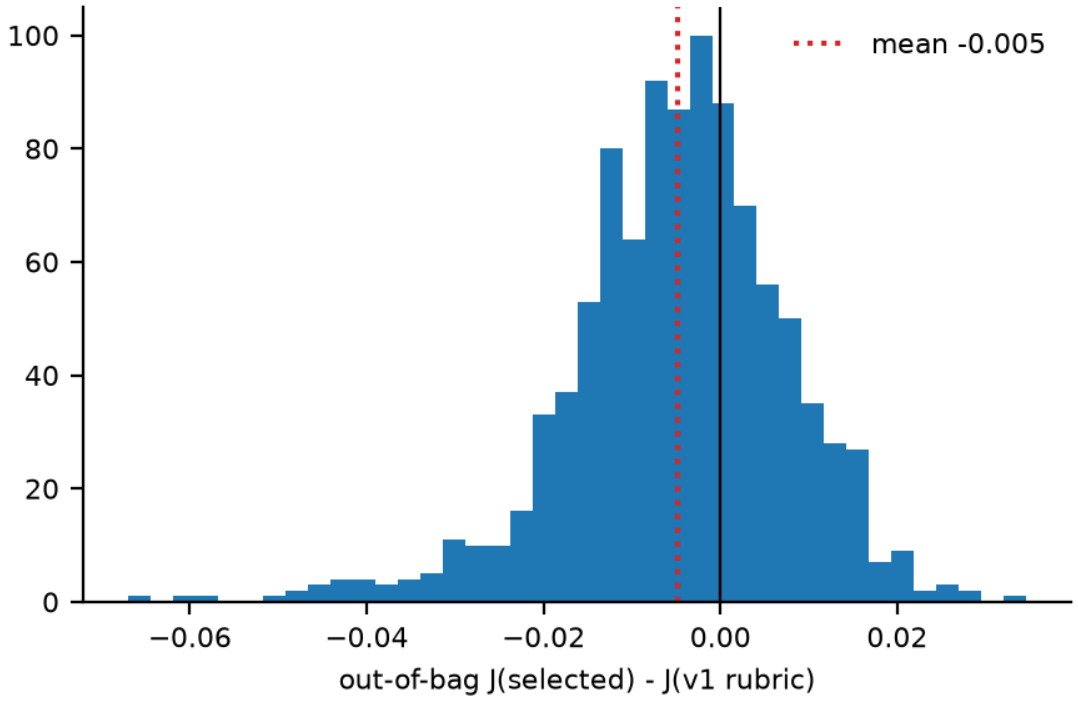


Figure 36: Paired out-of-bag difference between the selected candidate and the v1 rubric over 1,000 resamples.

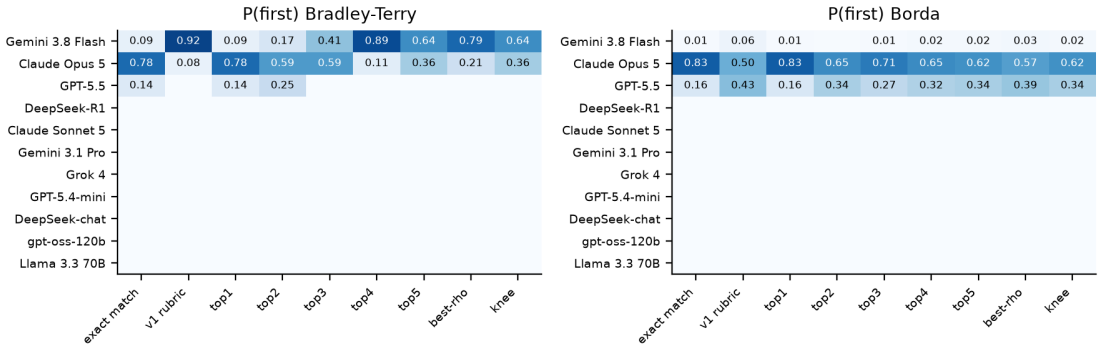


Figure 37: Bootstrap probability of first place under every rule, by pooling method.

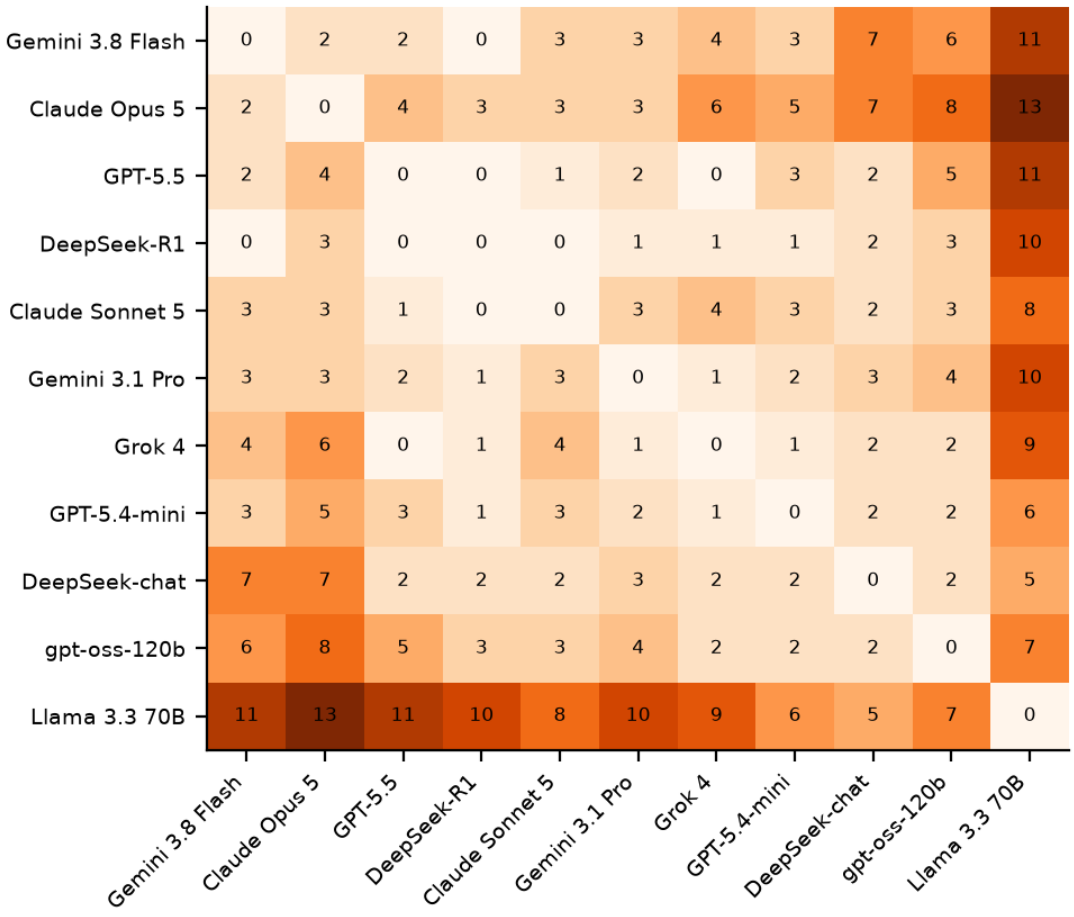


Figure 38: Number of datasets (of 31) in which the per-item paired test separates each pair of models under the headline rule (Holm within dataset).

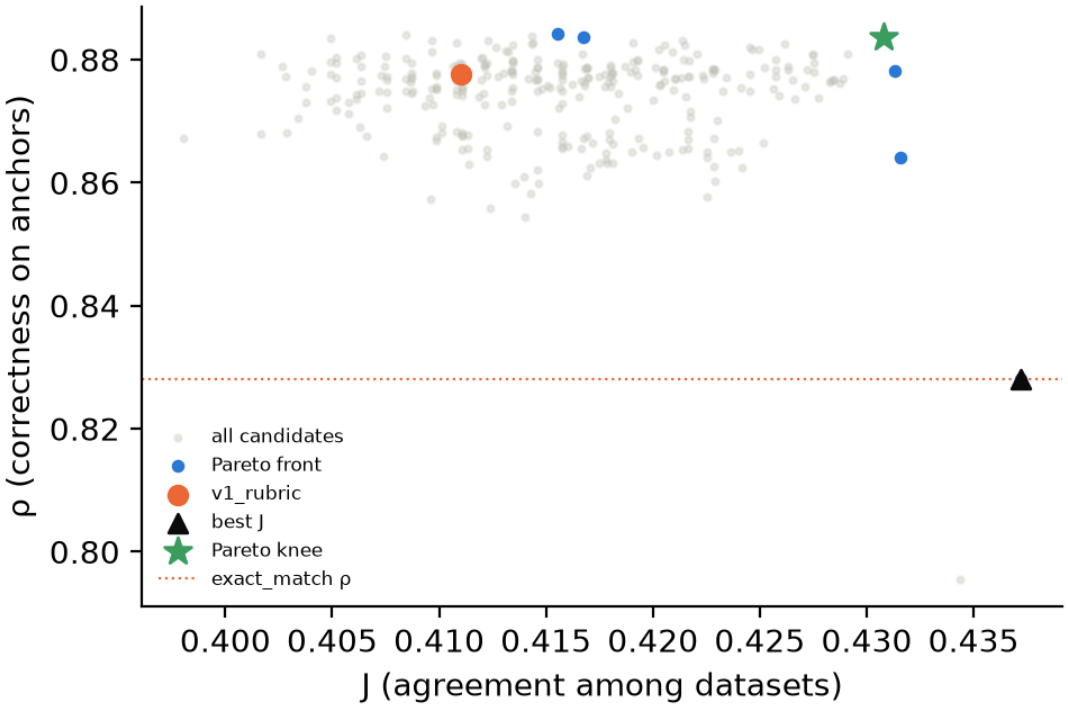


Figure 39: Agreement J against anchor correctness ρ for every candidate.

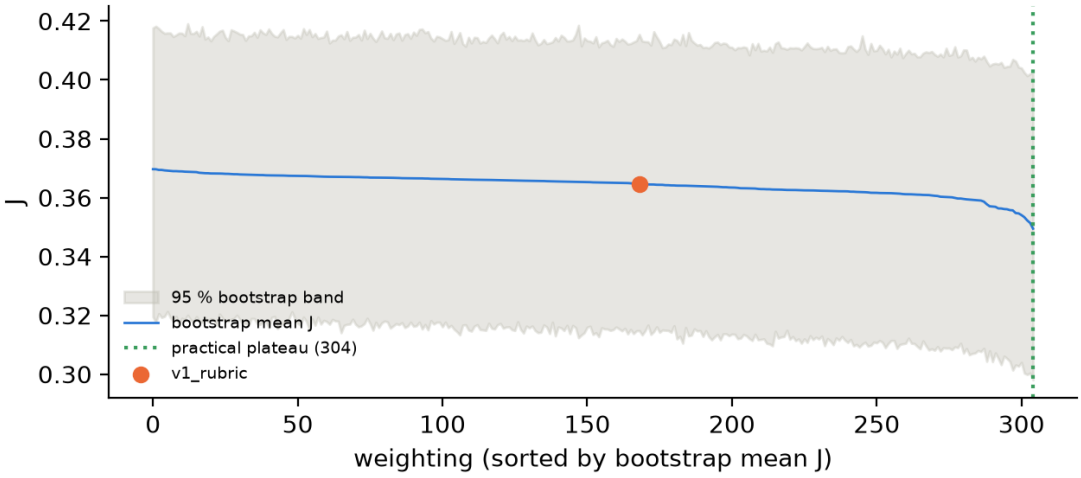


Figure 40: Bootstrap band over sorted candidates and the plateau boundary.

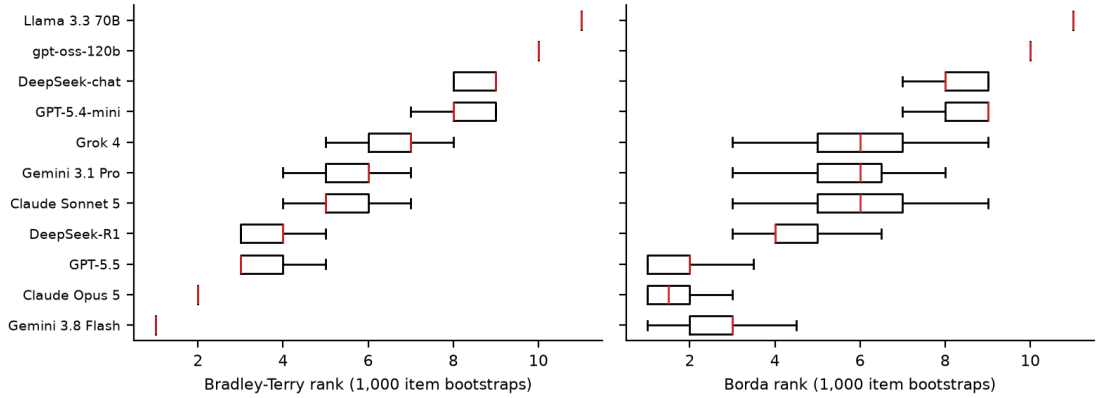


Figure 41: Bootstrap distribution of each model's rank under the headline rule (v1 rubric).

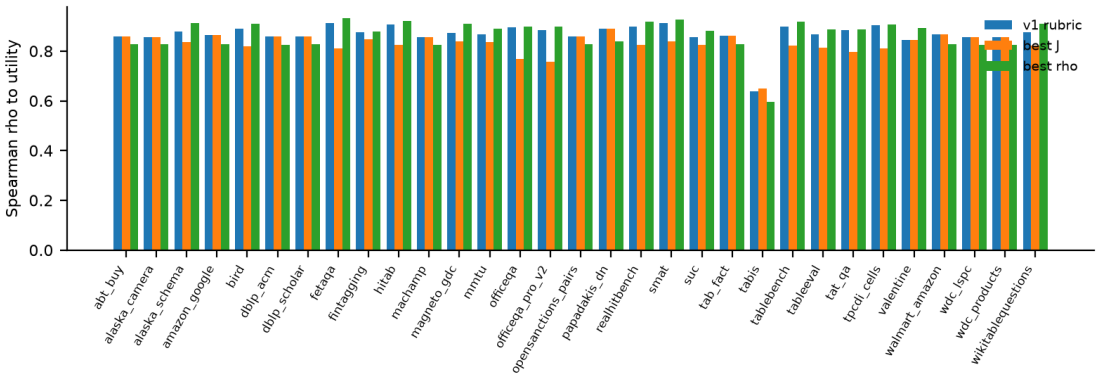


Figure 42: Anchor correctness per dataset under the v1 rubric, the best-J and the best-ρ candidates.

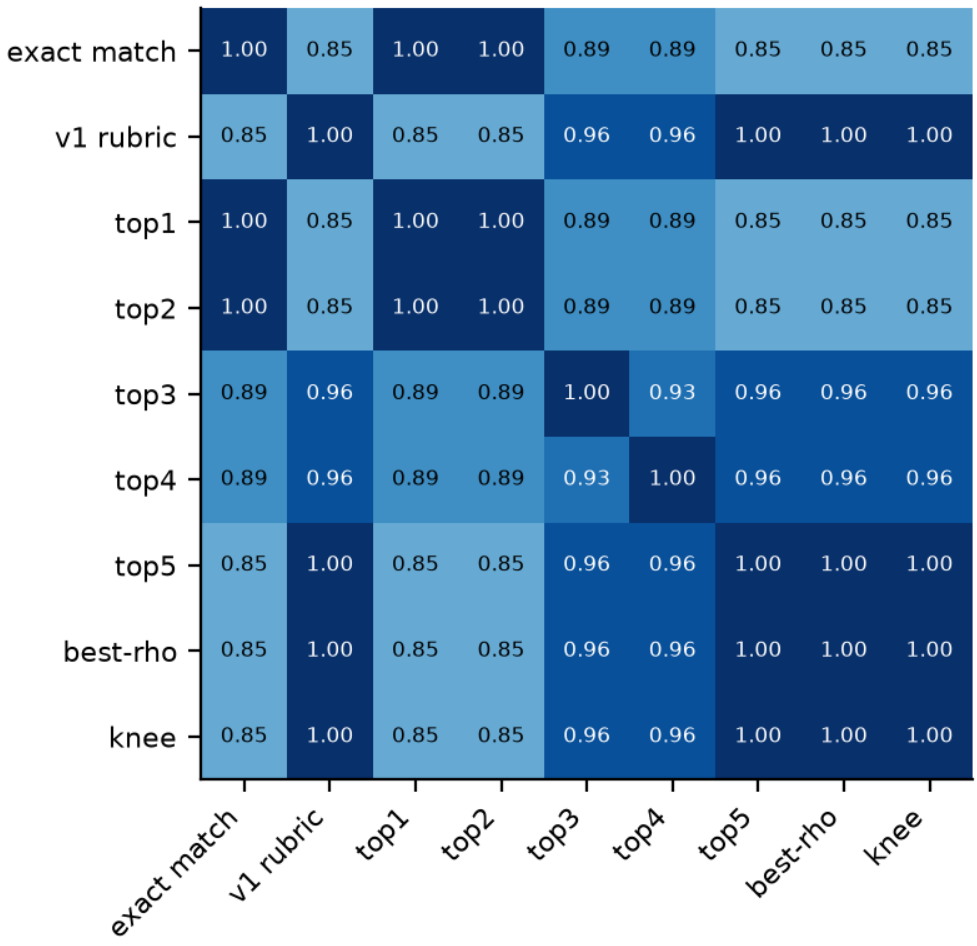


Figure 43: Kendall τ between the Bradley–Terry rankings of each pair of rules.

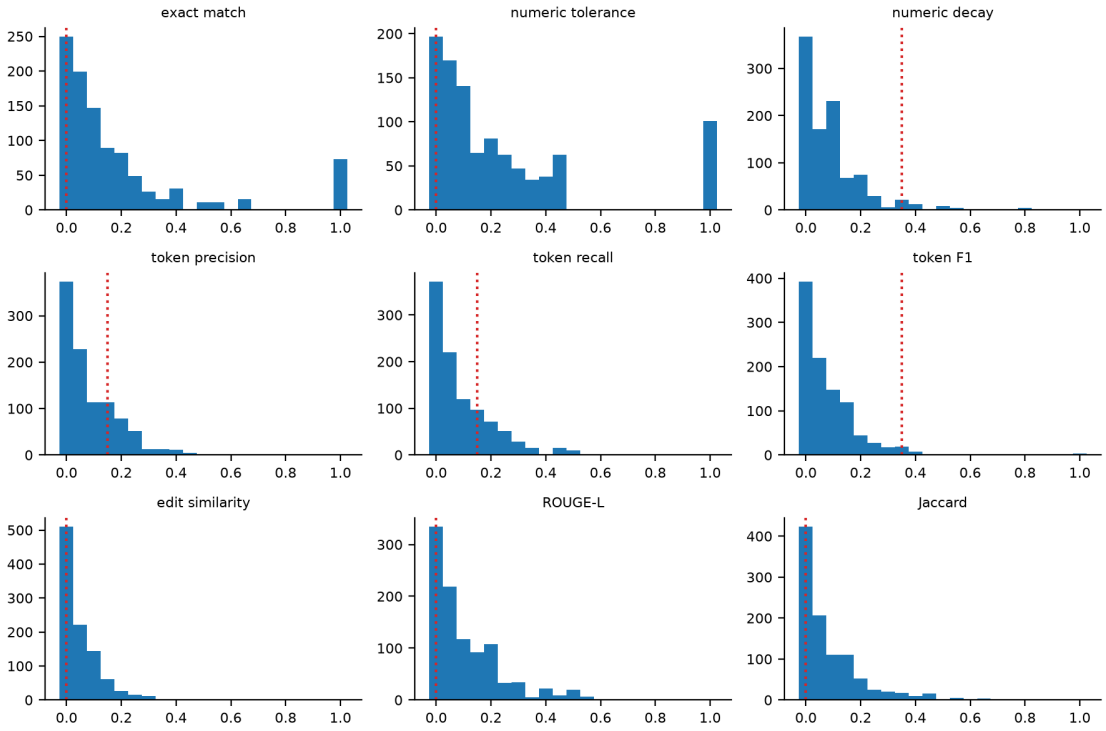


Figure 44: Weights selected as in-bag best across 1,000 bootstrap resamples; dotted = v1 rubric.

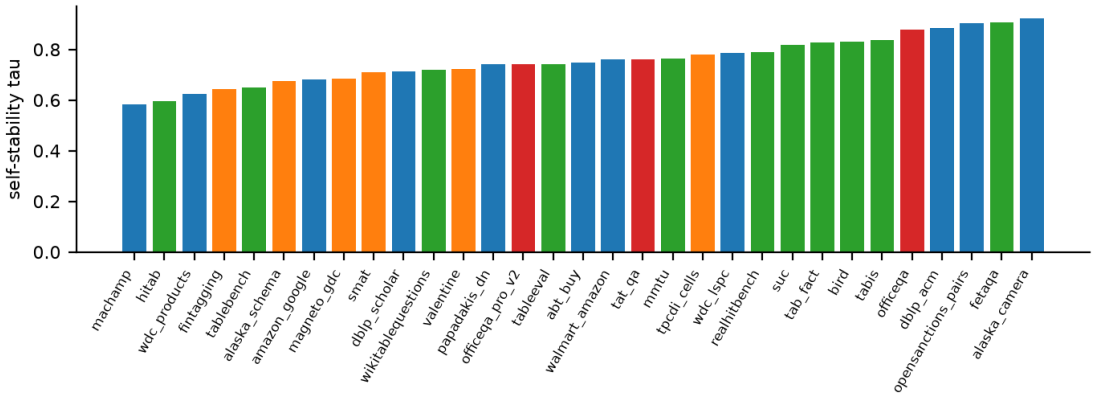


Figure 45: Self-stability of each dataset's model ranking (Kendall τ between bootstrap halves).

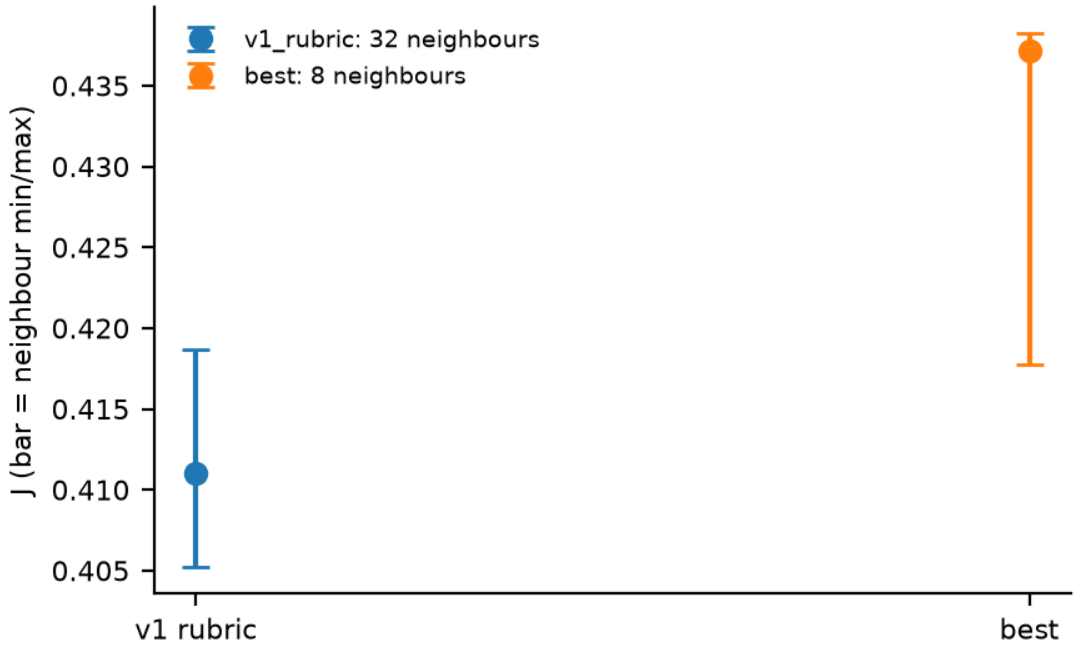


Figure 46: One-step sensitivity around the v1 rubric and the best candidate.

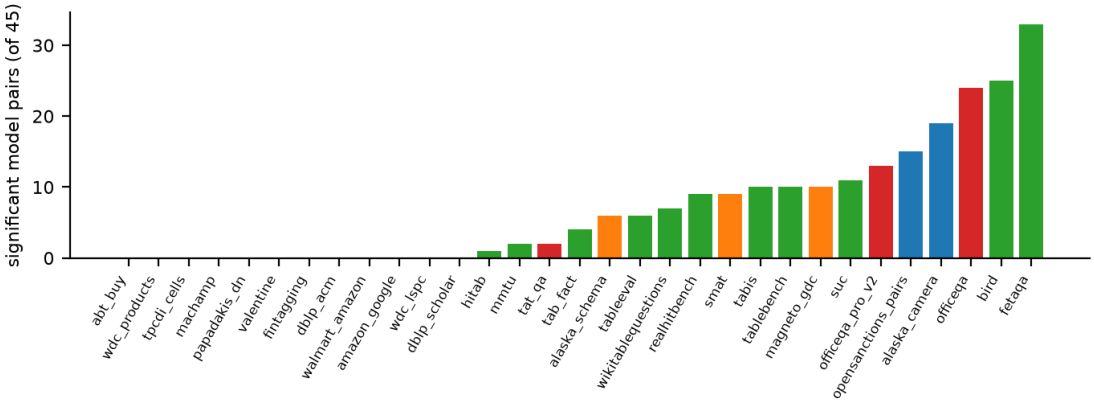


Figure 47: How many of the 45 model pairs each dataset separates at the item level under the headline rule (colour = family).

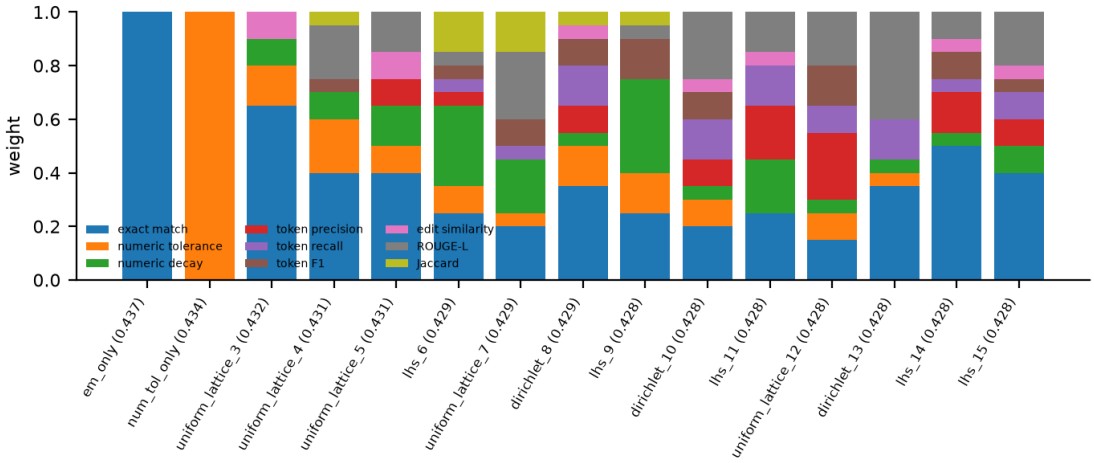


Figure 48: Component weights of the fifteen best candidates.

B All tables

Every saved table, one file each under `tables/`; include any in the main text with `\input`.

Table 8: Anchor operators: utility and mean score under three rules.

op	utility	v1	bestJ	bestrho	em	n
no_answer	0.000	0.001	0.000	0.014	0.000	3083
wrong_random	0.000	0.097	0.041	0.101	0.041	1845
flip	0.000	0.000	0.000	0.000	0.000	1238
sign_flip	0.050	0.018	0.003	0.119	0.003	632
magnitude	0.050	0.091	0.013	0.146	0.013	632
near_miss_large	0.150	0.150	0.000	0.137	0.000	632
digit_typo	0.200	0.199	0.000	0.224	0.000	632
list_dump	0.250	0.373	0.013	0.226	0.013	1845
hedge	0.300	0.493	0.192	0.392	0.192	2974
shuffled	0.400	0.652	0.009	0.418	0.009	109
near_miss_medium	0.450	0.293	0.008	0.252	0.008	632
partial	0.500	0.345	0.061	0.350	0.061	1213
superset	0.600	0.467	0.045	0.335	0.045	1213
typo	0.600	0.334	0.101	0.355	0.101	1104
near_miss_small	0.750	0.405	0.097	0.366	0.097	632
paraphrase	0.950	0.965	0.930	0.958	0.930	3083
verbose	1.000	1.000	0.999	1.000	0.999	3083
identity	1.000	1.000	0.999	1.000	0.999	3083
tagged	1.000	1.000	0.999	1.000	0.999	3083
unit_variant	1.000	0.898	0.837	0.891	0.837	632

Table 9: Anchor correctness for the labelled candidates.

label	w_em	w_num_tol	w_num_decay	w_tok_prec	w_tok_rec	w_tok_f1	w_edit_sim	w_rouge_l	w_jaccard	J	rho	rho_ds	auc	wrong_mean	hedge_gap
em_only	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.437	0.828	0.830	0.989	0.011	-0.807
num_tol_only	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.434	0.796	0.808	0.989	0.022	-0.595
uniform	0.111	0.111	0.111	0.111	0.111	0.111	0.111	0.111	0.111	0.413	0.878	0.861	0.994	0.046	-0.555
f1_only	0.000	0.000	0.000	0.000	0.000	1.000	0.000	0.000	0.000	0.412	0.856	0.846	0.989	0.030	-0.496
v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.411	0.878	0.868	0.994	0.034	-0.506

Table 10: The five fixed baselines on the candidate set.

rank	label	w_em	w_num_tol	w_num_decay	w_tok_prec	w_tok_rec	w_tok_f1	w_edit_sim	w_rouge_l	w_jaccard	J	J_kemeny	J_copeland	J_bt	tau_native
1	em_only	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.437	0.439	0.439	0.422	0.916
2	num_tol_only	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.434	0.436	0.436	0.426	0.973
187	uniform	0.111	0.111	0.111	0.111	0.111	0.111	0.111	0.111	0.111	0.413	0.421	0.427	0.403	0.916
199	f1_only	0.000	0.000	0.000	0.000	0.000	1.000	0.000	0.000	0.000	0.412	0.415	0.422	0.388	0.877
230	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.411	0.415	0.424	0.403	0.915

Table 11: Sampled designs: summary over 20 seeds.

design	best_J_mean	best_J_sd	best_J_max	mean_J	regret_mean	regret_max	min_pairwise_L1	mean_nearest_L1	discrepancy	zero_share
dirichlet_lattice	0.4307	0.0025	0.4357	0.4159	0.0105	0.0145	0.1950	0.4027	2.6738	0.8820
lhs_simplex	0.4325	0.0033	0.4383	0.4160	0.0087	0.0143	0.2150	0.4096	2.6718	0.8780
uniform_lattice	0.4330	0.0034	0.4412	0.4158	0.0082	0.0131	0.2350	0.4650	2.5904	0.9705

Table 12: Dataset diagnostics under the headline rule.

dataset	family	mean_score	spread_ across_models	floor	ceiling	share_ constant_ bootstrap_ rankings	ranking_self_ stability_tau	tau_to_ others_mean	n_items	exclude_ from_ objective
abt_buy	EM	0.971	0.060	no	no	0.000	0.748	0.367	100	no
alaska_camera	EM	0.945	0.190	no	no	0.000	0.924	-0.199	100	no
alaska_schema	SM	0.929	0.118	no	no	0.000	0.676	-0.022	93	no
amazon_google	EM	0.935	0.100	no	no	0.000	0.683	0.015	100	no
bird	TQA	0.191	0.209	no	no	0.000	0.830	0.325	100	no
dblp_acm	EM	0.987	0.090	no	no	0.000	0.883	0.328	100	no
dblp_scholar	EM	0.948	0.060	no	no	0.000	0.713	-0.015	100	no
fetaqa	TQA	0.387	0.192	no	no	0.000	0.906	0.264	100	no
fintagging	SM	0.884	0.051	no	no	0.000	0.644	0.290	100	no
hitab	TQA	0.772	0.068	no	no	0.000	0.597	-0.140	100	no
machamp	EM	0.887	0.070	no	no	0.000	0.584	0.399	100	no
magneto_gdc	SM	0.937	0.225	no	no	0.000	0.684	0.266	100	no
mmtu	TQA	0.874	0.110	no	no	0.000	0.765	0.274	100	no
officeqa	FIN	0.135	0.124	no	no	0.000	0.877	0.337	100	no
officeqa_pro_v2	FIN	0.151	0.166	no	no	0.000	0.741	0.354	90	no
opensanctions_pairs	EM	0.835	0.240	no	no	0.000	0.905	0.038	100	no
papadakis_dn	EM	0.940	0.030	no	no	0.002	0.741	0.025	100	no
realhitbench	TQA	0.677	0.143	no	no	0.000	0.788	0.363	100	no
smat	SM	0.916	0.150	no	no	0.000	0.710	0.185	100	no
suc	TQA	0.917	0.246	no	no	0.000	0.818	0.294	100	no
tab_fact	TQA	0.802	0.200	no	no	0.000	0.827	0.003	100	no
tabis	TQA	0.918	0.210	no	no	0.000	0.836	0.267	100	no
tablebench	TQA	0.824	0.218	no	no	0.000	0.649	0.233	100	no
tableeval	TQA	0.755	0.166	no	no	0.000	0.742	0.177	100	no
tat_qa	FIN	0.804	0.109	no	no	0.000	0.762	0.274	100	no
tpcdi_cells	SM	0.456	0.020	no	no	0.000	0.779	0.330	100	no
valentine	SM	0.985	0.030	no	yes	0.000	0.722	0.236	100	yes
walmart_amazon	EM	0.959	0.060	no	no	0.000	0.761	0.362	100	no
wdc_lspc	EM	0.982	0.040	no	yes	0.004	0.788	0.282	100	yes
wdc_products	EM	0.978	0.020	no	yes	0.006	0.623	0.199	100	yes
wikitablequestions	TQA	0.848	0.132	no	no	0.000	0.719	0.361	100	no

Table 13: Mean v1-rubric score per model within each family.

model	entity matching	finance	schema matching	table QA
Gemini 3.8 Flash	0.945	0.414	0.854	0.766
Claude Opus 5	0.947	0.449	0.869	0.745
GPT-5.5	0.950	0.388	0.864	0.751
DeepSeek-R1	0.953	0.378	0.865	0.733
Claude Sonnet 5	0.942	0.399	0.862	0.724
Gemini 3.1 Pro	0.944	0.355	0.852	0.749
Grok 4	0.948	0.354	0.851	0.738
GPT-5.4-mini	0.947	0.347	0.859	0.716
DeepSeek-chat	0.941	0.342	0.852	0.711
gpt-oss-120b	0.915	0.324	0.848	0.698
Llama 3.3 70B	0.936	0.328	0.777	0.633

Table 14: The 25 best candidates with the objective under the other pooling rules.

rank	label	w_em	w_num_ tol	w_num_ decay	w_tok_ prec	w_tok_ rec	w_tok_ f1	w_edit_ sim	w_ rouge_l	w_ jaccard	J	J_ kemeny	J_ copeland	J_bt	tau_ native
1	em_only	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.437	0.439	0.439	0.422	0.916
2	num_ tol_only	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.434	0.436	0.436	0.426	0.973
3	uniform_ lattice_3	0.650	0.150	0.100	0.000	0.000	0.000	0.100	0.000	0.000	0.432	0.437	0.437	0.423	0.908
4	uniform_ lattice_4	0.400	0.200	0.100	0.000	0.000	0.050	0.000	0.200	0.050	0.431	0.434	0.445	0.417	0.928
5	uniform_ lattice_5	0.400	0.100	0.150	0.100	0.000	0.000	0.100	0.150	0.000	0.431	0.437	0.437	0.415	0.918
6	lhs_6	0.250	0.100	0.300	0.050	0.050	0.050	0.000	0.050	0.150	0.429	0.434	0.434	0.418	0.928
7	uniform_ lattice_7	0.200	0.050	0.200	0.000	0.050	0.100	0.000	0.250	0.150	0.429	0.431	0.431	0.414	0.918
8	dirichlet_ 8	0.350	0.150	0.050	0.100	0.150	0.100	0.050	0.000	0.050	0.429	0.430	0.430	0.416	0.921
9	lhs_9	0.250	0.150	0.350	0.000	0.000	0.150	0.000	0.050	0.050	0.428	0.432	0.432	0.417	0.933
10	dirichlet_ 10	0.200	0.100	0.050	0.100	0.150	0.100	0.050	0.250	0.000	0.428	0.428	0.428	0.409	0.911
11	lhs_11	0.250	0.000	0.200	0.200	0.150	0.000	0.050	0.150	0.000	0.428	0.432	0.432	0.415	0.913
12	uniform_ lattice_ 12	0.150	0.100	0.050	0.250	0.100	0.150	0.000	0.200	0.000	0.428	0.428	0.428	0.411	0.906
13	dirichlet_ 13	0.350	0.050	0.050	0.000	0.150	0.000	0.000	0.400	0.000	0.428	0.429	0.429	0.414	0.913
14	lhs_14	0.500	0.000	0.050	0.150	0.050	0.100	0.050	0.100	0.000	0.428	0.432	0.432	0.417	0.911
15	lhs_15	0.400	0.000	0.100	0.100	0.100	0.050	0.050	0.200	0.000	0.428	0.432	0.432	0.417	0.913
16	dirichlet_ 16	0.200	0.250	0.200	0.150	0.050	0.150	0.000	0.000	0.000	0.427	0.434	0.434	0.421	0.933
17	uniform_ lattice_ 17	0.300	0.350	0.100	0.150	0.000	0.050	0.000	0.000	0.050	0.427	0.433	0.433	0.414	0.934
18	dirichlet_ 18	0.500	0.000	0.100	0.050	0.050	0.100	0.050	0.150	0.000	0.427	0.431	0.431	0.414	0.918
19	uniform_ lattice_ 19	0.200	0.250	0.100	0.050	0.150	0.100	0.000	0.150	0.000	0.427	0.426	0.426	0.413	0.928
20	uniform_ lattice_ 20	0.250	0.050	0.100	0.000	0.200	0.250	0.000	0.000	0.150	0.427	0.429	0.429	0.413	0.912
21	lhs_21	0.300	0.050	0.050	0.100	0.050	0.150	0.000	0.150	0.150	0.427	0.428	0.428	0.412	0.917
22	lhs_22	0.250	0.150	0.050	0.000	0.000	0.050	0.050	0.150	0.300	0.426	0.432	0.432	0.416	0.921
23	lhs_23	0.050	0.300	0.100	0.100	0.100	0.050	0.050	0.150	0.100	0.426	0.431	0.442	0.416	0.930
24	uniform_ lattice_ 24	0.150	0.400	0.200	0.100	0.000	0.100	0.050	0.000	0.000	0.426	0.432	0.432	0.414	0.939
25	lhs_25	0.200	0.100	0.100	0.050	0.150	0.150	0.000	0.050	0.200	0.426	0.423	0.423	0.406	0.917

Table 15: Full leaderboard under rule best-rho; weights: em 0.20, num_decay 0.25, tok_prec 0.25, tok_rec 0.05, tok_f1 0.05, edit_sim 0.15, rouge_l 0.05.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Gemini 3.8 Flash	0.114	0.015	1.000	1.000	2.000	0.786	3.000	1.000	4.000	0.028	3.000	3.000	3.000	3.000	2.000	0.806
Claude Opus 5	0.106	0.000	2.000	1.000	2.000	0.214	2.000	1.000	3.000	0.568	1.000	1.000	1.000	1.000	1.000	0.807
GPT-5.5	0.046	0.015	3.000	3.000	5.000	0.000	1.000	1.000	3.000	0.393	2.000	2.000	2.000	2.000	3.000	0.803
DeepSeek-R1	0.031	0.015	4.000	3.000	6.000	0.000	4.000	3.000	7.000	0.000	5.000	5.000	4.000	4.000	5.000	0.796
Claude Sonnet 5	0.026	0.015	5.000	3.000	7.000	0.000	5.000	4.000	7.000	0.000	4.000	4.000	5.000	5.000	4.000	0.791
Gemini 3.1 Pro	0.012	0.015	6.000	4.000	7.025	0.000	6.000	4.000	7.000	0.000	7.000	7.000	6.000	6.000	6.000	0.794
Grok 4	0.004	0.015	7.000	5.000	8.000	0.000	7.000	4.000	8.000	0.000	6.000	6.000	7.000	7.000	7.000	0.792
GPT-5.4-mini	-0.011	0.015	8.000	6.975	8.025	0.000	8.000	7.000	9.000	0.000	8.000	8.000	8.000	8.000	8.000	0.787
DeepSeek-chat	-0.039	0.015	9.000	8.000	9.000	0.000	9.000	8.000	9.000	0.000	9.000	9.000	9.000	9.000	9.000	0.778
gpt-oss-120b	-0.130	0.015	10.000	10.000	11.000	0.000	10.000	10.000	11.000	0.000	10.000	10.000	10.000	10.000	10.000	0.762
Llama 3.3 70B	-0.158	0.015	11.000	10.000	11.000	0.000	11.000	10.000	11.000	0.000	11.000	11.000	11.000	11.000	11.000	0.733

Table 16: Full leaderboard under rule exact match; weights: em 1.00.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Claude Opus 5	0.044	0.000	1.000	1.000	3.000	0.777	1.000	1.000	2.000	0.826	1.000	1.000	1.000	1.000	1.000	0.768
GPT-5.5	0.035	0.015	2.000	1.000	4.000	0.136	2.000	1.000	4.000	0.156	2.000	2.000	2.000	2.000	2.000	0.764
Gemini 3.8 Flash	0.034	0.015	3.000	1.000	4.000	0.085	3.000	2.000	5.512	0.009	3.000	3.000	3.000	3.000	4.000	0.763
DeepSeek-R1	0.023	0.015	4.000	2.000	7.000	0.002	4.000	2.000	7.000	0.001	4.000	4.000	4.000	4.000	3.000	0.758
Gemini 3.1 Pro	0.018	0.015	5.000	3.000	8.000	0.000	7.000	4.000	8.000	0.000	7.000	7.000	6.000	7.000	6.000	0.755
Grok 4	0.016	0.015	6.000	4.000	8.000	0.000	6.000	4.000	8.000	0.000	6.000	6.000	7.000	6.000	7.000	0.754
Claude Sonnet 5	0.010	0.015	7.000	4.000	8.000	0.000	5.000	3.000	7.000	0.000	5.000	5.000	5.000	5.000	5.000	0.751
GPT-5.4-mini	0.006	0.015	8.000	5.000	8.000	0.000	9.000	7.988	9.000	0.000	8.000	8.000	8.000	9.000	9.000	0.749
DeepSeek-chat	-0.018	0.015	9.000	9.000	10.000	0.000	8.000	7.000	9.000	0.000	9.000	9.000	9.000	8.000	8.000	0.737
gpt-oss-120b	-0.037	0.015	10.000	9.000	10.000	0.000	10.000	10.000	10.000	0.000	10.000	10.000	10.000	10.000	10.000	0.728
Llama 3.3 70B	-0.132	0.015	11.000	11.000	11.000	0.000	11.000	11.000	11.000	0.000	11.000	11.000	11.000	11.000	11.000	0.680

Table 17: Leaderboard under the headline rule (v1 rubric).

rank	model	BT strength	SE	Borda rank	Borda 95%	P(first) BT	mean score
1	Gemini 3.8 Flash	0.112	0.015	3	[1, 4]	0.920	0.812
2	Claude Opus 5	0.097	0.000	2	[1, 3]	0.080	0.812
3	GPT-5.5	0.042	0.015	1	[1, 3]	0.000	0.808
4	DeepSeek-R1	0.037	0.015	4	[3, 7]	0.000	0.801
5	Claude Sonnet 5	0.023	0.015	5	[4, 7]	0.000	0.796
6	Gemini 3.1 Pro	0.021	0.015	6	[4, 7]	0.000	0.799
7	Grok 4	0.016	0.015	7	[4, 7]	0.000	0.797
8	GPT-5.4-mini	-0.022	0.015	9	[7, 9]	0.000	0.790
9	DeepSeek-chat	-0.031	0.015	8	[7, 9]	0.000	0.784
10	gpt-oss-120b	-0.111	0.015	10	[10, 11]	0.000	0.767
11	Llama 3.3 70B	-0.183	0.015	11	[10, 11]	0.000	0.738

Table 18: Full leaderboard under rule knee; weights: em 0.40, num_tol 0.10, num_decay 0.15, tok_prec 0.10, edit_sim 0.10, rouge_l 0.15.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Gemini 3.8 Flash	0.115	0.015	1.000	1.000	2.000	0.643	3.000	2.000	4.000	0.017	3.000	3.000	3.000	3.000	3.000	0.791
Claude Opus 5	0.112	0.000	2.000	1.000	2.000	0.357	2.000	1.000	3.000	0.625	1.000	1.000	1.000	1.000	1.000	0.793
GPT-5.5	0.046	0.015	3.000	3.000	5.000	0.000	1.000	1.000	3.000	0.343	2.000	2.000	2.000	2.000	2.000	0.788
DeepSeek-R1	0.026	0.015	4.000	3.000	7.000	0.000	4.000	3.000	7.000	0.000	5.000	5.000	4.000	4.000	5.000	0.782
Claude Sonnet 5	0.026	0.015	5.000	3.000	7.000	0.000	5.000	4.000	7.000	0.000	4.000	4.000	5.000	5.000	4.000	0.776
Gemini 3.1 Pro	0.012	0.015	6.000	4.000	7.000	0.000	6.000	4.000	7.000	0.000	7.000	7.000	6.000	6.000	6.000	0.779
Grok 4	0.004	0.015	7.000	5.000	8.000	0.000	7.000	4.000	7.025	0.000	6.000	6.000	7.000	7.000	7.000	0.778
GPT-5.4-mini	-0.016	0.015	8.000	7.000	9.000	0.000	8.000	7.000	9.000	0.000	8.000	8.000	8.000	8.000	8.000	0.773
DeepSeek-chat	-0.038	0.015	9.000	8.000	9.000	0.000	9.000	8.000	9.000	0.000	9.000	9.000	9.000	9.000	9.000	0.763
gpt-oss-120b	-0.124	0.015	10.000	10.000	10.000	0.000	10.000	10.000	11.000	0.000	10.000	10.000	10.000	10.000	10.000	0.750
Llama 3.3 70B	-0.163	0.015	11.000	11.000	11.000	0.000	11.000	10.000	11.000	0.000	11.000	11.000	11.000	11.000	11.000	0.713

Table 19: Full leaderboard under rule top1; weights: em 1.00.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Claude Opus 5	0.044	0.000	1.000	1.000	3.000	0.777	1.000	1.000	2.000	0.826	1.000	1.000	1.000	1.000	1.000	0.768
GPT-5.5	0.035	0.015	2.000	1.000	4.000	0.136	2.000	1.000	4.000	0.156	2.000	2.000	2.000	2.000	2.000	0.764
Gemini 3.8 Flash	0.034	0.015	3.000	1.000	4.000	0.085	3.000	2.000	5.512	0.009	3.000	3.000	3.000	3.000	4.000	0.763
DeepSeek-R1	0.023	0.015	4.000	2.000	7.000	0.002	4.000	2.000	7.000	0.001	4.000	4.000	4.000	4.000	3.000	0.758
Gemini 3.1 Pro	0.018	0.015	5.000	3.000	8.000	0.000	7.000	4.000	8.000	0.000	7.000	7.000	6.000	7.000	6.000	0.755
Grok 4	0.016	0.015	6.000	4.000	8.000	0.000	6.000	4.000	8.000	0.000	6.000	6.000	7.000	6.000	7.000	0.754
Claude Sonnet 5	0.010	0.015	7.000	4.000	8.000	0.000	5.000	3.000	7.000	0.000	5.000	5.000	5.000	5.000	5.000	0.751
GPT-5.4-mini	0.006	0.015	8.000	5.000	8.000	0.000	9.000	7.988	9.000	0.000	8.000	8.000	8.000	9.000	9.000	0.749
DeepSeek-chat	-0.018	0.015	9.000	9.000	10.000	0.000	8.000	7.000	9.000	0.000	9.000	9.000	9.000	8.000	8.000	0.737
gpt-oss-120b	-0.037	0.015	10.000	9.000	10.000	0.000	10.000	10.000	10.000	0.000	10.000	10.000	10.000	10.000	10.000	0.728
Llama 3.3 70B	-0.132	0.015	11.000	11.000	11.000	0.000	11.000	11.000	11.000	0.000	11.000	11.000	11.000	11.000	11.000	0.680

Table 20: Full leaderboard under rule top2; weights: num_tol 1.00.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Claude Opus 5	0.043	0.000	1.000	1.000	3.000	0.586	1.000	1.000	2.000	0.649	1.000	1.000	1.000	1.000	1.000	0.779
GPT-5.5	0.038	0.015	2.000	1.000	3.025	0.246	2.000	1.000	3.000	0.341	2.000	2.000	2.000	2.000	2.000	0.776
Gemini 3.8 Flash	0.037	0.015	3.000	1.000	4.000	0.166	3.000	2.000	5.000	0.005	3.000	3.000	3.000	3.000	3.000	0.776
DeepSeek-R1	0.022	0.015	4.000	3.000	7.000	0.002	4.000	3.000	7.000	0.000	4.000	4.000	4.000	4.000	4.000	0.768
Gemini 3.1 Pro	0.018	0.015	5.000	4.000	8.000	0.000	7.000	4.000	7.000	0.000	7.000	7.000	6.000	7.000	6.000	0.766
Grok 4	0.014	0.015	6.000	4.000	8.000	0.000	6.000	4.000	7.000	0.000	6.000	6.000	7.000	6.000	7.000	0.764
Claude Sonnet 5	0.011	0.015	7.000	4.000	8.000	0.000	5.000	3.000	7.000	0.000	5.000	5.000	5.000	5.000	5.000	0.763
GPT-5.4-mini	0.005	0.015	8.000	5.000	8.000	0.000	9.000	7.000	9.000	0.000	8.000	8.000	8.000	9.000	8.000	0.759
DeepSeek-chat	-0.021	0.015	9.000	9.000	10.000	0.000	8.000	8.000	9.000	0.000	9.000	9.000	9.000	8.000	9.000	0.747
gpt-oss-120b	-0.039	0.015	10.000	9.000	10.000	0.000	10.000	10.000	10.000	0.000	10.000	10.000	10.000	10.000	10.000	0.737
Llama 3.3 70B	-0.129	0.015	11.000	11.000	11.000	0.000	11.000	11.000	11.000	0.000	11.000	11.000	11.000	11.000	11.000	0.693

Table 21: Full leaderboard under rule top3; weights: em 0.65, num_tol 0.15, num_decay 0.10, edit_sim 0.10.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Claude Opus 5	0.116	0.000	1.000	1.000	2.000	0.586	1.000	1.000	2.000	0.713	1.000	1.000	1.000	1.000	1.000	0.781
Gemini 3.8 Flash	0.113	0.015	2.000	1.000	2.000	0.414	3.000	2.000	5.000	0.009	3.000	3.000	3.000	3.000	3.000	0.777
GPT-5.5	0.046	0.015	3.000	3.000	4.000	0.000	2.000	1.000	3.000	0.272	2.000	2.000	2.000	2.000	2.000	0.776
DeepSeek-R1	0.025	0.015	4.000	3.000	7.000	0.000	4.000	3.000	7.000	0.000	5.000	5.000	4.000	4.000	5.000	0.769
Claude Sonnet 5	0.021	0.015	5.000	4.000	7.000	0.000	5.000	4.000	7.000	0.000	4.000	4.000	5.000	5.000	4.000	0.764
Gemini 3.1 Pro	0.009	0.015	6.000	4.000	7.000	0.000	6.000	4.000	7.000	0.000	7.000	7.000	6.000	6.000	6.000	0.766
Grok 4	0.009	0.015	7.000	4.000	7.000	0.000	7.000	4.000	8.000	0.000	6.000	6.000	7.000	7.000	7.000	0.766
GPT-5.4-mini	-0.025	0.015	8.000	8.000	9.000	0.000	8.000	7.000	9.000	0.000	8.000	8.000	8.000	8.000	8.000	0.761
DeepSeek-chat	-0.034	0.015	9.000	8.000	9.000	0.000	9.000	7.000	9.000	0.000	9.000	9.000	9.000	9.000	9.000	0.749
gpt-oss-120b	-0.101	0.015	10.000	10.000	10.000	0.000	10.000	10.000	11.000	0.000	10.000	10.000	10.000	10.000	10.000	0.738
Llama 3.3 70B	-0.179	0.015	11.000	11.000	11.000	0.000	11.000	10.000	11.000	0.000	11.000	11.000	11.000	11.000	11.000	0.695

Table 22: Full leaderboard under rule top4; weights: em 0.40, num_tol 0.20, num_decay 0.10, tok_f1 0.05, rouge_l 0.20, jaccard 0.05.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Gemini 3.8 Flash	0.113	0.015	1.000	1.000	2.000	0.894	3.000	2.000	4.000	0.019	3.000	3.000	3.000	3.000	3.000	0.783
Claude Opus 5	0.100	0.000	2.000	1.000	2.000	0.106	1.000	1.000	3.000	0.652	1.000	1.000	1.000	1.000	1.000	0.786
GPT-5.5	0.042	0.015	3.000	3.000	5.000	0.000	2.000	1.000	3.000	0.324	2.000	2.000	2.000	2.000	2.000	0.782
DeepSeek-R1	0.031	0.015	4.000	3.000	7.000	0.000	4.000	3.000	7.000	0.000	5.000	6.000	4.000	4.000	5.000	0.775
Gemini 3.1 Pro	0.025	0.015	5.000	3.000	7.000	0.000	6.000	4.000	7.000	0.000	6.000	6.000	6.000	6.000	6.000	0.774
Claude Sonnet 5	0.024	0.015	6.000	3.000	7.000	0.000	5.000	4.000	7.000	0.000	4.000	4.000	5.000	5.000	4.000	0.770
Grok 4	0.012	0.015	7.000	4.975	7.000	0.000	7.000	4.000	7.000	0.000	7.000	6.000	7.000	7.000	7.000	0.772
GPT-5.4-mini	-0.023	0.015	8.000	8.000	9.000	0.000	9.000	8.000	9.000	0.000	8.000	8.000	8.000	9.000	8.000	0.765
DeepSeek-chat	-0.031	0.015	9.000	8.000	9.000	0.000	8.000	7.488	9.000	0.000	9.000	9.000	9.000	8.000	9.000	0.756
gpt-oss-120b	-0.105	0.015	10.000	10.000	10.000	0.000	10.000	10.000	10.025	0.000	10.000	10.000	10.000	10.000	10.000	0.744
Llama 3.3 70B	-0.189	0.015	11.000	11.000	11.000	0.000	11.000	10.975	11.000	0.000	11.000	11.000	11.000	11.000	11.000	0.704

Table 23: Full leaderboard under rule top5; weights: em 0.40, num_tol 0.10, num_decay 0.15, tok_prec 0.10, edit_sim 0.10, rouge_l 0.15.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Gemini 3.8 Flash	0.115	0.015	1.000	1.000	2.000	0.643	3.000	2.000	4.000	0.017	3.000	3.000	3.000	3.000	3.000	0.791
Claude Opus 5	0.112	0.000	2.000	1.000	2.000	0.357	2.000	1.000	3.000	0.625	1.000	1.000	1.000	1.000	1.000	0.793
GPT-5.5	0.046	0.015	3.000	3.000	5.000	0.000	1.000	1.000	3.000	0.343	2.000	2.000	2.000	2.000	2.000	0.788
DeepSeek-R1	0.026	0.015	4.000	3.000	7.000	0.000	4.000	3.000	7.000	0.000	5.000	5.000	4.000	4.000	5.000	0.782
Claude Sonnet 5	0.026	0.015	5.000	3.000	7.000	0.000	5.000	4.000	7.000	0.000	4.000	4.000	5.000	5.000	4.000	0.776
Gemini 3.1 Pro	0.012	0.015	6.000	4.000	7.000	0.000	6.000	4.000	7.000	0.000	7.000	7.000	6.000	6.000	6.000	0.779
Grok 4	0.004	0.015	7.000	5.000	8.000	0.000	7.000	4.000	7.025	0.000	6.000	6.000	7.000	7.000	7.000	0.778
GPT-5.4-mini	-0.016	0.015	8.000	7.000	9.000	0.000	8.000	7.000	9.000	0.000	8.000	8.000	8.000	8.000	8.000	0.773
DeepSeek-chat	-0.038	0.015	9.000	8.000	9.000	0.000	9.000	8.000	9.000	0.000	9.000	9.000	9.000	9.000	9.000	0.763
gpt-oss-120b	-0.124	0.015	10.000	10.000	10.000	0.000	10.000	10.000	11.000	0.000	10.000	10.000	10.000	10.000	10.000	0.750
Llama 3.3 70B	-0.163	0.015	11.000	11.000	11.000	0.000	11.000	10.000	11.000	0.000	11.000	11.000	11.000	11.000	11.000	0.713

Table 24: Full leaderboard under rule v1 rubric; weights: num_decay 0.35, tok_prec 0.15, tok_rec 0.15, tok_f1 0.35.

model	bt_strength	bt_se_vs_ref	rank_bt	bt_rank_lo95	bt_rank_hi95	p_first_bt	rank_borda	borda_rank_lo95	borda_rank_hi95	p_first_borda	rank_kemeny	rank_copeland	rank_mean_z	rank_rrf	rank_family_balanced_z	mean_score
Gemini 3.8 Flash	0.112	0.015	1.000	1.000	2.000	0.922	3.000	1.000	4.000	0.056	3.000	3.000	3.000	3.000	2.000	0.812
Claude Opus 5	0.097	0.000	2.000	1.000	2.000	0.078	2.000	1.000	3.000	0.499	1.000	1.000	1.000	2.000	1.000	0.812
GPT-5.5	0.042	0.015	3.000	3.000	5.025	0.000	1.000	1.000	3.000	0.431	2.000	2.000	2.000	1.000	3.000	0.808
DeepSeek-R1	0.037	0.015	4.000	3.000	6.000	0.000	4.000	3.000	7.000	0.000	5.000	5.000	4.000	4.000	5.000	0.801
Claude Sonnet 5	0.023	0.015	5.000	3.000	7.000	0.000	5.000	4.000	7.000	0.000	4.000	5.000	5.000	5.000	4.000	0.796
Gemini 3.1 Pro	0.021	0.015	6.000	4.000	7.000	0.000	6.000	4.000	7.000	0.000	6.000	5.000	6.000	6.000	6.000	0.799
Grok 4	0.016	0.015	7.000	4.000	7.000	0.000	7.000	4.000	7.000	0.000	7.000	7.000	7.000	7.000	7.000	0.797
GPT-5.4-mini	-0.022	0.015	8.000	8.000	9.000	0.000	9.000	7.000	9.000	0.000	8.000	8.000	8.000	9.000	8.000	0.790
DeepSeek-chat	-0.031	0.015	9.000	8.000	9.000	0.000	8.000	7.000	9.000	0.000	9.000	9.000	9.000	8.000	9.000	0.784
gpt-oss-120b	-0.111	0.015	10.000	10.000	10.000	0.000	10.000	10.000	11.000	0.000	10.000	10.000	10.000	10.000	10.000	0.767
Llama 3.3 70B	-0.183	0.015	11.000	11.000	11.000	0.000	11.000	10.000	11.000	0.000	11.000	11.000	11.000	11.000	11.000	0.738

Table 25: Leave-one-dataset-out, every dataset.

kind	held_out	weights_from	w_em	w_num_tol	w_num_decay	w_tok_prec	w_tok_rec	w_tok_f1	w_edit_sim	w_rouge_l	w_jaccard	J_rest	tau_heldout_mean	n_heldout
dataset	abt_buy	heldout_weights	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.427	0.755	1
dataset	abt_buy	full_best	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.427	0.755	1
dataset	abt_buy	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.399	0.755	1
dataset	alaska_camera	heldout_weights	0.400	0.100	0.150	0.100	0.000	0.000	0.100	0.150	0.000	0.458	-0.283	1
dataset	alaska_camera	full_best	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.456	-0.245	1
dataset	alaska_camera	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.435	-0.245	1
dataset	alaska_schema	heldout_weights	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.448	0.117	1
dataset	alaska_schema	full_best	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.448	0.117	1
dataset	alaska_schema	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.427	-0.055	1
dataset	amazon_google	heldout_weights	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.447	0.020	1
dataset	amazon_google	full_best	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.446	0.059	1
dataset	amazon_google	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.425	0.020	1
dataset	bird	heldout_weights	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.432	0.584	1
dataset	bird	full_best	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.432	0.584	1
dataset	bird	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.404	0.600	1
dataset	dblp_acm	heldout_weights	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.434	0.493	1
dataset	dblp_acm	full_best	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.434	0.540	1
dataset	dblp_acm	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.407	0.540	1
dataset	dblp_scholar	heldout_weights	0.400	0.100	0.150	0.100	0.000	0.000	0.100	0.150	0.000	0.449	-0.058	1
dataset	dblp_scholar	full_best	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.447	0.019	1
dataset	dblp_scholar	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.428	0.039	1
dataset	fetaqa	heldout_weights	0.550	0.050	0.150	0.050	0.000	0.000	0.000	0.050	0.150	0.435	0.418	1
dataset	fetaqa	full_best	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.428	0.719	1
dataset	fetaqa	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.422	0.389	1
dataset	fintagging	heldout_weights	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.431	0.674	1
dataset	fintagging	full_best	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.431	0.674	1
dataset	fintagging	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.408	0.491	1
dataset	hitab	heldout_weights	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.452	0.000	1
dataset	hitab	full_best	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.452	0.000	1
dataset	hitab	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.431	-0.200	1
dataset	machamp	heldout_weights	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.425	0.799	1
dataset	machamp	full_best	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.425	0.799	1
dataset	machamp	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.398	0.799	1
dataset	magneto_gdc	heldout_weights	0.650	0.150	0.100	0.000	0.000	0.000	0.100	0.000	0.000	0.433	0.587	1
dataset	magneto_gdc	full_best	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.432	0.598	1
dataset	magneto_gdc	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.405	0.564	1
dataset	mmtu	heldout_weights	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.438	0.426	1

dataset	mmtu	full_best	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.438	0.426	1
dataset	mmtu	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.416	0.440	1
dataset	officeqa	heldout_weights	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.436	0.502	1
dataset	officeqa	full_best	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.436	0.502	1
dataset	officeqa	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.402	0.673	1
dataset	officeqa_pro_v2	heldout_weights	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.427	0.662	1
dataset	officeqa_pro_v2	full_best	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.425	0.801	1
dataset	officeqa_pro_v2	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.406	0.661	1
dataset	opensanctions_pairs	heldout_weights	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.449	0.073	1
dataset	opensanctions_pairs	full_best	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.449	0.073	1
dataset	opensanctions_pairs	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.423	0.037	1
dataset	papadakis_dn	heldout_weights	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.449	-0.042	1
dataset	papadakis_dn	full_best	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.449	-0.042	1
dataset	papadakis_dn	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.428	-0.042	1
dataset	realhitbench	heldout_weights	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.427	0.734	1
dataset	realhitbench	full_best	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.427	0.734	1
dataset	realhitbench	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.403	0.636	1
dataset	smat	heldout_weights	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.441	0.195	1
dataset	smat	full_best	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.441	0.195	1
dataset	smat	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.416	0.309	1
dataset	suc	heldout_weights	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.430	0.662	1
dataset	suc	full_best	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.430	0.662	1
dataset	suc	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.407	0.514	1
dataset	tab_fact	heldout_weights	0.400	0.100	0.150	0.100	0.000	0.000	0.100	0.150	0.000	0.450	-0.056	1
dataset	tab_fact	full_best	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.446	-0.056	1
dataset	tab_fact	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.427	-0.019	1
dataset	tabis	heldout_weights	0.400	0.100	0.150	0.100	0.000	0.000	0.100	0.150	0.000	0.435	0.463	1
dataset	tabis	full_best	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.433	0.449	1
dataset	tabis	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.410	0.449	1
dataset	tablebench	heldout_weights	0.000	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.438	0.411	1
dataset	tablebench	full_best	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.438	0.411	1
dataset	tablebench	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.409	0.455	1
dataset	tableeval	heldout_weights	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.443	0.343	1
dataset	tableeval	full_best	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.443	0.343	1
dataset	tableeval	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.419	0.236	1
dataset	tat_qa	heldout_weights	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.437	0.510	1
dataset	tat_qa	full_best	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.437	0.510	1
dataset	tat_qa	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.410	0.550	1
dataset	tpcdi_cells	heldout_weights	0.650	0.150	0.100	0.000	0.000	0.000	0.100	0.000	0.000	0.435	0.463	1
dataset	tpcdi_cells	full_best	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.432	0.605	1
dataset	tpcdi_cells	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.405	0.623	1
dataset	valentine	heldout_weights	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.435	0.492	1
dataset	valentine	full_best	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.435	0.492	1
dataset	valentine	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.411	0.397	1

dataset	walmart_amazon	heldout_weights	0.650	0.150	0.100	0.000	0.000	0.000	0.100	0.000	0.000	0.430	0.738	1
dataset	walmart_amazon	full_best	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.428	0.725	1
dataset	walmart_amazon	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.401	0.686	1
dataset	wdc_lspc	heldout_weights	0.200	0.050	0.200	0.000	0.050	0.100	0.000	0.250	0.150	0.432	0.487	1
dataset	wdc_lspc	full_best	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.431	0.503	1
dataset	wdc_lspc	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.407	0.543	1
dataset	wdc_products	heldout_weights	0.200	0.050	0.200	0.000	0.050	0.100	0.000	0.250	0.150	0.437	0.331	1
dataset	wdc_products	full_best	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.436	0.350	1
dataset	wdc_products	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.413	0.350	1
dataset	wikitablequestions	heldout_weights	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.424	0.725	1
dataset	wikitablequestions	full_best	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.424	0.725	1
dataset	wikitablequestions	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.401	0.709	1

Table 26: Leave-one-family-out.

kind	held_out	weights_from	w_em	w_num_tol	w_num_decay	w_tok_prec	w_tok_rec	w_tok_f1	w_edit_sim	w_rouge_l	w_jaccard	J_rest	tau_heldout_mean	n_heldout
family	EM	heldout_weights	0.400	0.200	0.100	0.000	0.000	0.050	0.000	0.200	0.050	0.498	0.312	11
family	EM	full_best	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.494	0.328	11
family	EM	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.465	0.324	11
family	SM	heldout_weights	0.250	0.050	0.100	0.000	0.200	0.250	0.000	0.000	0.150	0.432	0.311	6
family	SM	full_best	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.427	0.422	6
family	SM	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.409	0.277	6
family	TQA	heldout_weights	0.100	0.450	0.100	0.050	0.050	0.000	0.000	0.200	0.050	0.444	0.378	11
family	TQA	full_best	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.435	0.415	11
family	TQA	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.420	0.357	11
family	FIN	heldout_weights	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.420	0.524	3
family	FIN	full_best	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.420	0.524	3
family	FIN	v1_rubric	0.000	0.000	0.350	0.150	0.150	0.350	0.000	0.000	0.000	0.391	0.563	3

Table 27: Every quantity saved by the nested item bootstrap.

quantity	value
B	1000
n_candidates	305
reference	v1_rubric
in_bag_J_of_selected_mean	0.382
out_of_bag_J_of_selected_mean	0.334
optimism_gap	0.048
out_of_bag_J_ref_mean	0.339
out_of_bag_J_full_best_mean	0.332
selected_vs_ref_out_of_bag	"mean_diff": -0.00486262109875679, "ci95_lo": -0.03271459862589836, "ci95_hi": 0.016647919267415996, "p_boot_one_sided": 0.659, "share_a_ahead": 0.341
full_best_vs_ref_out_of_bag	"mean_diff": -0.006209747359156609, "ci95_lo": -0.042142198979854585, "ci95_hi": 0.027577679604291916, "p_boot_one_sided": 0.631, "share_a_ahead": 0.369
full_best_vs_ref_in_bag	"mean_diff": -0.0006272097527980805, "ci95_lo": -0.03069242984056473, "ci95_hi": 0.026698747277259825, "p_boot_one_sided": 0.498, "share_a_ahead": 0.502
share_resamples_ref_in_top_5pct	0.046
selected_weights_mean	"em": 0.187, "num_tol": 0.235, "num_decay": 0.089, "tok_prec": 0.083, "tok_rec": 0.091, "tok_f1": 0.077, "edit_sim": 0.05, "rouge_l": 0.104, "jaccard": 0.083
selected_weights_sd	"em": 0.265, "num_tol": 0.289, "num_decay": 0.113, "tok_prec": 0.096, "tok_rec": 0.109, "tok_f1": 0.102, "edit_sim": 0.071, "rouge_l": 0.123, "jaccard": 0.113
n_distinct_selected	230
plateau_size	305
plateau_share	1.000
ref_in_plateau	yes
ref_p_vs_best	0.498
practical_plateau_size	304
practical_margin	0.020
ref_in_practical_plateau	yes
practical_plateau_weight_ranges	"em": [0.0, 1.0], "num_tol": [0.0, 1.0], "num_decay": [0.0, 0.8], "tok_prec": [0.0, 0.55], "tok_rec": [0.0, 0.5], "tok_f1": [0.0, 1.0], "edit_sim": [0.0, 0.6], "rouge_l": [0.0, 0.55], "jaccard": [0.0, 0.65]

Table 28: Official-split estimates under the headline rule for every dataset and model.

dataset	model	suite_mean	official_ estimate	official_lo95	official_hi95	effective_n	suite_f1_pos	official_f1_ pos	official_f1_ lo95	official_f1_ hi95	official_ positive_ share
abt_buy	Claude Opus 5	1.000	1.000	1.000	1.000	63.075	1.000	1.000	1.000	1.000	0.108
abt_buy	Claude Sonnet 5	0.970	0.993	0.985	1.000	63.075	0.968	0.968	0.925	1.000	0.108
abt_buy	DeepSeek-chat	0.960	0.976	0.933	0.998	63.075	0.958	0.893	0.735	0.990	0.108
abt_buy	DeepSeek-R1	0.970	0.978	0.931	1.000	63.075	0.969	0.904	0.755	1.000	0.108
abt_buy	Gemini 3.1 Pro	0.980	0.996	0.988	1.000	63.075	0.979	0.979	0.945	1.000	0.108
abt_buy	Gemini 3.8 Flash	0.980	0.996	0.988	1.000	63.075	0.979	0.979	0.944	1.000	0.108
abt_buy	GPT-5.4-mini	0.960	0.976	0.937	0.998	63.075	0.958	0.893	0.736	0.991	0.108
abt_buy	GPT-5.5	1.000	1.000	1.000	1.000	63.075	1.000	1.000	1.000	1.000	0.108
abt_buy	gpt-oss-120b	0.940	0.987	0.974	0.996	63.075	0.935	0.935	0.871	0.978	0.108
abt_buy	Grok 4	0.960	0.976	0.935	0.998	63.075	0.958	0.893	0.739	0.990	0.108
abt_buy	Llama 3.3 70B	0.960	0.976	0.937	0.998	63.075	0.958	0.893	0.737	0.990	0.108
alaska_camera	Claude Opus 5	0.820	0.820	0.740	0.890	100.000	0.795	0.795	0.700	0.882	0.500
alaska_camera	Claude Sonnet 5	0.980	0.980	0.950	1.000	100.000	0.981	0.981	0.948	1.000	0.500
alaska_camera	DeepSeek-chat	1.000	1.000	1.000	1.000	100.000	1.000	1.000	1.000	1.000	0.500
alaska_camera	DeepSeek-R1	0.990	0.990	0.970	1.000	100.000	0.990	0.990	0.968	1.000	0.500
alaska_camera	Gemini 3.1 Pro	0.810	0.810	0.730	0.880	100.000	0.782	0.782	0.667	0.869	0.500
alaska_camera	Gemini 3.8 Flash	0.890	0.890	0.830	0.950	100.000	0.884	0.884	0.804	0.945	0.500
alaska_camera	GPT-5.4-mini	0.970	0.970	0.930	1.000	100.000	0.971	0.971	0.932	1.000	0.500
alaska_camera	GPT-5.5	1.000	1.000	1.000	1.000	100.000	1.000	1.000	1.000	1.000	0.500
alaska_camera	gpt-oss-120b	1.000	1.000	1.000	1.000	100.000	1.000	1.000	1.000	1.000	0.500
alaska_camera	Grok 4	0.940	0.940	0.890	0.980	100.000	0.940	0.940	0.889	0.981	0.500
alaska_camera	Llama 3.3 70B	0.990	0.990	0.970	1.000	100.000	0.990	0.990	0.968	1.000	0.500
alaska_schema	Claude Opus 5	0.945	0.945	0.903	0.982	93.000	–	–	–	–	–
alaska_schema	Claude Sonnet 5	0.929	0.929	0.880	0.972	93.000	–	–	–	–	–
alaska_schema	DeepSeek-chat	0.951	0.951	0.910	0.985	93.000	–	–	–	–	–
alaska_schema	DeepSeek-R1	0.957	0.957	0.923	0.988	93.000	–	–	–	–	–
alaska_schema	Gemini 3.1 Pro	0.928	0.928	0.875	0.970	93.000	–	–	–	–	–
alaska_schema	Gemini 3.8 Flash	0.918	0.918	0.863	0.962	93.000	–	–	–	–	–
alaska_schema	GPT-5.4-mini	0.958	0.958	0.920	0.989	93.000	–	–	–	–	–
alaska_schema	GPT-5.5	0.928	0.928	0.880	0.972	93.000	–	–	–	–	–
alaska_schema	gpt-oss-120b	0.934	0.934	0.889	0.972	93.000	–	–	–	–	–
alaska_schema	Grok 4	0.929	0.929	0.880	0.973	93.000	–	–	–	–	–
alaska_schema	Llama 3.3 70B	0.840	0.840	0.780	0.894	93.000	–	–	–	–	–
amazon_google	Claude Opus 5	0.940	0.916	0.839	0.981	67.152	0.936	0.704	0.526	0.919	0.102
amazon_google	Claude Sonnet 5	0.960	0.935	0.868	0.984	67.152	0.957	0.758	0.588	0.935	0.102
amazon_google	DeepSeek-chat	0.960	0.949	0.893	0.998	67.152	0.957	0.796	0.614	0.990	0.102

amazon_google	DeepSeek-R1	0.950	0.932	0.866	0.984	67.152	0.946	0.747	0.579	0.932	0.102
amazon_google	Gemini 3.1 Pro	0.930	0.928	0.862	0.981	67.152	0.923	0.725	0.551	0.914	0.102
amazon_google	Gemini 3.8 Flash	0.930	0.928	0.859	0.981	67.152	0.923	0.725	0.548	0.913	0.102
amazon_google	GPT-5.4-mini	0.930	0.942	0.880	0.990	67.152	0.921	0.762	0.583	0.959	0.102
amazon_google	GPT-5.5	0.940	0.916	0.844	0.982	67.152	0.936	0.704	0.535	0.915	0.102
amazon_google	gpt-oss-120b	0.860	0.926	0.861	0.973	67.152	0.829	0.676	0.507	0.874	0.102
amazon_google	Grok 4	0.940	0.944	0.875	0.992	67.152	0.933	0.773	0.607	0.966	0.102
amazon_google	Llama 3.3 70B	0.950	0.932	0.861	0.982	67.152	0.946	0.747	0.561	0.941	0.102
bird	Claude Opus 5	0.293	0.323	0.233	0.419	74.082	–	–	–	–	–
bird	Claude Sonnet 5	0.233	0.260	0.174	0.349	74.082	–	–	–	–	–
bird	DeepSeek-chat	0.187	0.203	0.132	0.289	74.082	–	–	–	–	–
bird	DeepSeek-R1	0.216	0.253	0.170	0.343	74.082	–	–	–	–	–
bird	Gemini 3.1 Pro	0.183	0.228	0.143	0.315	74.082	–	–	–	–	–
bird	Gemini 3.8 Flash	0.293	0.313	0.227	0.402	74.082	–	–	–	–	–
bird	GPT-5.4-mini	0.178	0.218	0.131	0.308	74.082	–	–	–	–	–
bird	GPT-5.5	0.185	0.204	0.118	0.293	74.082	–	–	–	–	–
bird	gpt-oss-120b	0.084	0.109	0.051	0.181	74.082	–	–	–	–	–
bird	Grok 4	0.122	0.144	0.073	0.228	74.082	–	–	–	–	–
bird	Llama 3.3 70B	0.122	0.144	0.082	0.218	74.082	–	–	–	–	–
dblp_acm	Claude Opus 5	1.000	1.000	1.000	1.000	72.166	1.000	1.000	1.000	1.000	0.180
dblp_acm	Claude Sonnet 5	1.000	1.000	1.000	1.000	72.166	1.000	1.000	1.000	1.000	0.180
dblp_acm	DeepSeek-chat	1.000	1.000	1.000	1.000	72.166	1.000	1.000	1.000	1.000	0.180
dblp_acm	DeepSeek-R1	0.990	0.996	0.988	1.000	72.166	0.990	0.990	0.965	1.000	0.180
dblp_acm	Gemini 3.1 Pro	1.000	1.000	1.000	1.000	72.166	1.000	1.000	1.000	1.000	0.180
dblp_acm	Gemini 3.8 Flash	1.000	1.000	1.000	1.000	72.166	1.000	1.000	1.000	1.000	0.180
dblp_acm	GPT-5.4-mini	0.990	0.996	0.988	1.000	72.166	0.990	0.990	0.965	1.000	0.180
dblp_acm	GPT-5.5	1.000	1.000	1.000	1.000	72.166	1.000	1.000	1.000	1.000	0.180
dblp_acm	gpt-oss-120b	0.910	0.967	0.945	0.987	72.166	0.899	0.899	0.829	0.960	0.180
dblp_acm	Grok 4	1.000	1.000	1.000	1.000	72.166	1.000	1.000	1.000	1.000	0.180
dblp_acm	Llama 3.3 70B	0.970	0.952	0.893	1.000	72.166	0.970	0.882	0.750	1.000	0.180
dblp_scholar	Claude Opus 5	0.950	0.956	0.901	0.996	70.463	0.950	0.888	0.752	0.981	0.186
dblp_scholar	Claude Sonnet 5	0.970	0.976	0.932	1.000	70.463	0.970	0.937	0.835	1.000	0.186
dblp_scholar	DeepSeek-chat	0.960	0.959	0.907	0.997	70.463	0.961	0.898	0.780	0.991	0.186
dblp_scholar	DeepSeek-R1	0.950	0.969	0.931	0.996	70.463	0.949	0.917	0.818	0.987	0.186
dblp_scholar	Gemini 3.1 Pro	0.940	0.965	0.927	0.993	70.463	0.939	0.906	0.812	0.980	0.186
dblp_scholar	Gemini 3.8 Flash	0.930	0.961	0.914	0.990	70.463	0.928	0.895	0.791	0.971	0.186
dblp_scholar	GPT-5.4-mini	0.940	0.965	0.925	0.992	70.463	0.939	0.906	0.799	0.980	0.186
dblp_scholar	GPT-5.5	0.960	0.959	0.905	0.996	70.463	0.961	0.898	0.784	0.991	0.186
dblp_scholar	gpt-oss-120b	0.910	0.967	0.943	0.986	70.463	0.903	0.903	0.829	0.961	0.186
dblp_scholar	Grok 4	0.950	0.969	0.928	0.996	70.463	0.949	0.917	0.812	0.988	0.186
dblp_scholar	Llama 3.3 70B	0.970	0.976	0.936	1.000	70.463	0.970	0.937	0.847	1.000	0.186

fetaqa	Claude Opus 5	0.401	0.401	0.373	0.430	100.000	–	–	–	–	–
fetaqa	Claude Sonnet 5	0.373	0.373	0.345	0.400	100.000	–	–	–	–	–
fetaqa	DeepSeek-chat	0.434	0.434	0.411	0.462	100.000	–	–	–	–	–
fetaqa	DeepSeek-R1	0.413	0.413	0.383	0.445	100.000	–	–	–	–	–
fetaqa	Gemini 3.1 Pro	0.433	0.433	0.409	0.456	100.000	–	–	–	–	–
fetaqa	Gemini 3.8 Flash	0.445	0.445	0.422	0.470	100.000	–	–	–	–	–
fetaqa	GPT-5.4-mini	0.312	0.312	0.289	0.336	100.000	–	–	–	–	–
fetaqa	GPT-5.5	0.438	0.438	0.412	0.466	100.000	–	–	–	–	–
fetaqa	gpt-oss-120b	0.322	0.322	0.301	0.341	100.000	–	–	–	–	–
fetaqa	Grok 4	0.428	0.428	0.405	0.451	100.000	–	–	–	–	–
fetaqa	Llama 3.3 70B	0.253	0.253	0.226	0.279	100.000	–	–	–	–	–
fintagging	Claude Opus 5	0.909	0.932	0.876	0.978	51.769	–	–	–	–	–
fintagging	Claude Sonnet 5	0.898	0.897	0.832	0.956	51.769	–	–	–	–	–
fintagging	DeepSeek-chat	0.858	0.847	0.774	0.912	51.769	–	–	–	–	–
fintagging	DeepSeek-R1	0.878	0.894	0.828	0.949	51.769	–	–	–	–	–
fintagging	Gemini 3.1 Pro	0.887	0.907	0.841	0.960	51.769	–	–	–	–	–
fintagging	Gemini 3.8 Flash	0.897	0.908	0.845	0.957	51.769	–	–	–	–	–
fintagging	GPT-5.4-mini	0.892	0.863	0.794	0.922	51.769	–	–	–	–	–
fintagging	GPT-5.5	0.893	0.899	0.834	0.957	51.769	–	–	–	–	–
fintagging	gpt-oss-120b	0.863	0.847	0.770	0.916	51.769	–	–	–	–	–
fintagging	Grok 4	0.887	0.876	0.811	0.943	51.769	–	–	–	–	–
fintagging	Llama 3.3 70B	0.858	0.836	0.762	0.910	51.769	–	–	–	–	–
hitab	Claude Opus 5	0.781	0.790	0.708	0.859	94.537	–	–	–	–	–
hitab	Claude Sonnet 5	0.763	0.758	0.671	0.835	94.537	–	–	–	–	–
hitab	DeepSeek-chat	0.814	0.822	0.749	0.889	94.537	–	–	–	–	–
hitab	DeepSeek-R1	0.792	0.787	0.715	0.860	94.537	–	–	–	–	–
hitab	Gemini 3.1 Pro	0.778	0.786	0.704	0.858	94.537	–	–	–	–	–
hitab	Gemini 3.8 Flash	0.748	0.757	0.677	0.830	94.537	–	–	–	–	–
hitab	GPT-5.4-mini	0.784	0.783	0.708	0.855	94.537	–	–	–	–	–
hitab	GPT-5.5	0.757	0.764	0.682	0.839	94.537	–	–	–	–	–
hitab	gpt-oss-120b	0.750	0.752	0.670	0.827	94.537	–	–	–	–	–
hitab	Grok 4	0.746	0.749	0.671	0.825	94.537	–	–	–	–	–
hitab	Llama 3.3 70B	0.784	0.783	0.700	0.853	94.537	–	–	–	–	–
machamp	Claude Opus 5	0.900	0.912	0.826	0.970	32.491	0.904	0.774	0.617	0.908	0.175
machamp	Claude Sonnet 5	0.900	0.923	0.852	0.975	32.491	0.902	0.793	0.642	0.924	0.175
machamp	DeepSeek-chat	0.890	0.892	0.801	0.962	32.491	0.897	0.742	0.604	0.895	0.175
machamp	DeepSeek-R1	0.900	0.897	0.808	0.963	32.491	0.906	0.751	0.594	0.895	0.175
machamp	Gemini 3.1 Pro	0.890	0.908	0.832	0.965	32.491	0.893	0.761	0.618	0.898	0.175
machamp	Gemini 3.8 Flash	0.900	0.912	0.836	0.973	32.491	0.904	0.774	0.608	0.920	0.175
machamp	GPT-5.4-mini	0.890	0.893	0.804	0.962	32.491	0.895	0.739	0.568	0.898	0.175

machamp	GPT-5.5	0.900	0.897	0.812	0.964	32.491	0.906	0.751	0.591	0.895	0.175
machamp	gpt-oss-120b	0.830	0.794	0.664	0.913	32.491	0.835	0.558	0.364	0.761	0.175
machamp	Grok 4	0.890	0.896	0.806	0.962	32.491	0.895	0.747	0.594	0.906	0.175
machamp	Llama 3.3 70B	0.870	0.880	0.791	0.947	32.491	0.874	0.698	0.529	0.846	0.175
magneto__gdc	Claude Opus 5	0.971	0.974	0.946	0.995	93.027	–	–	–	–	–
magneto__gdc	Claude Sonnet 5	0.955	0.955	0.916	0.987	93.027	–	–	–	–	–
magneto__gdc	DeepSeek-chat	0.953	0.956	0.919	0.986	93.027	–	–	–	–	–
magneto__gdc	DeepSeek-R1	0.978	0.981	0.959	1.000	93.027	–	–	–	–	–
magneto__gdc	Gemini 3.1 Pro	0.956	0.957	0.923	0.983	93.027	–	–	–	–	–
magneto__gdc	Gemini 3.8 Flash	0.963	0.967	0.936	0.993	93.027	–	–	–	–	–
magneto__gdc	GPT-5.4-mini	0.912	0.914	0.862	0.962	93.027	–	–	–	–	–
magneto__gdc	GPT-5.5	0.967	0.972	0.940	0.994	93.027	–	–	–	–	–
magneto__gdc	gpt-oss-120b	0.961	0.965	0.933	0.989	93.027	–	–	–	–	–
magneto__gdc	Grok 4	0.939	0.944	0.901	0.979	93.027	–	–	–	–	–
magneto__gdc	Llama 3.3 70B	0.753	0.759	0.694	0.820	93.027	–	–	–	–	–
mmtu	Claude Opus 5	0.882	0.846	0.760	0.930	86.538	–	–	–	–	–
mmtu	Claude Sonnet 5	0.872	0.838	0.750	0.921	86.538	–	–	–	–	–
mmtu	DeepSeek-chat	0.865	0.831	0.743	0.911	86.538	–	–	–	–	–
mmtu	DeepSeek-R1	0.850	0.802	0.710	0.888	86.538	–	–	–	–	–
mmtu	Gemini 3.1 Pro	0.914	0.881	0.794	0.952	86.538	–	–	–	–	–
mmtu	Gemini 3.8 Flash	0.894	0.856	0.773	0.932	86.538	–	–	–	–	–
mmtu	GPT-5.4-mini	0.882	0.851	0.769	0.926	86.538	–	–	–	–	–
mmtu	GPT-5.5	0.917	0.882	0.804	0.950	86.538	–	–	–	–	–
mmtu	gpt-oss-120b	0.830	0.774	0.676	0.868	86.538	–	–	–	–	–
mmtu	Grok 4	0.901	0.869	0.793	0.945	86.538	–	–	–	–	–
mmtu	Llama 3.3 70B	0.807	0.773	0.683	0.852	86.538	–	–	–	–	–
officeqa	Claude Opus 5	0.199	0.199	0.158	0.247	100.000	–	–	–	–	–
officeqa	Claude Sonnet 5	0.171	0.171	0.129	0.211	100.000	–	–	–	–	–
officeqa	DeepSeek-chat	0.075	0.075	0.041	0.116	100.000	–	–	–	–	–
officeqa	DeepSeek-R1	0.147	0.147	0.113	0.189	100.000	–	–	–	–	–
officeqa	Gemini 3.1 Pro	0.119	0.119	0.088	0.157	100.000	–	–	–	–	–
officeqa	Gemini 3.8 Flash	0.194	0.194	0.158	0.235	100.000	–	–	–	–	–
officeqa	GPT-5.4-mini	0.107	0.107	0.082	0.137	100.000	–	–	–	–	–
officeqa	GPT-5.5	0.168	0.168	0.133	0.210	100.000	–	–	–	–	–
officeqa	gpt-oss-120b	0.079	0.079	0.051	0.113	100.000	–	–	–	–	–
officeqa	Grok 4	0.129	0.129	0.095	0.168	100.000	–	–	–	–	–
officeqa	Llama 3.3 70B	0.097	0.097	0.072	0.124	100.000	–	–	–	–	–
officeqa__pro__v2	Claude Opus 5	0.270	0.270	0.210	0.337	90.000	–	–	–	–	–
officeqa__pro__v2	Claude Sonnet 5	0.157	0.157	0.112	0.208	90.000	–	–	–	–	–
officeqa__pro__v2	DeepSeek-chat	0.121	0.121	0.085	0.157	90.000	–	–	–	–	–
officeqa__pro__v2	DeepSeek-R1	0.148	0.148	0.097	0.202	90.000	–	–	–	–	–

officeqa_pro_v2	Gemini 3.1 Pro	0.120	0.120	0.082	0.162	90.000	–	–	–	–	–
officeqa_pro_v2	Gemini 3.8 Flash	0.198	0.198	0.153	0.252	90.000	–	–	–	–	–
officeqa_pro_v2	GPT-5.4-mini	0.123	0.123	0.096	0.155	90.000	–	–	–	–	–
officeqa_pro_v2	GPT-5.5	0.188	0.188	0.128	0.248	90.000	–	–	–	–	–
officeqa_pro_v2	gpt-oss-120b	0.104	0.104	0.065	0.147	90.000	–	–	–	–	–
officeqa_pro_v2	Grok 4	0.119	0.119	0.082	0.164	90.000	–	–	–	–	–
officeqa_pro_v2	Llama 3.3 70B	0.115	0.115	0.082	0.152	90.000	–	–	–	–	–
opensanctions_pairs	Claude Opus 5	0.940	0.906	0.831	0.969	76.249	0.935	0.935	0.878	0.979	0.769
opensanctions_pairs	Claude Sonnet 5	0.730	0.587	0.479	0.701	76.249	0.630	0.636	0.500	0.755	0.769
opensanctions_pairs	DeepSeek-chat	0.760	0.635	0.529	0.741	76.249	0.684	0.691	0.543	0.800	0.769
opensanctions_pairs	DeepSeek-R1	0.850	0.765	0.670	0.863	76.249	0.819	0.819	0.714	0.900	0.769
opensanctions_pairs	Gemini 3.1 Pro	0.970	0.964	0.917	1.000	76.249	0.969	0.976	0.943	1.000	0.769
opensanctions_pairs	Gemini 3.8 Flash	0.860	0.780	0.681	0.882	76.249	0.833	0.833	0.732	0.911	0.769
opensanctions_pairs	GPT-5.4-mini	0.900	0.865	0.781	0.940	76.249	0.891	0.905	0.833	0.960	0.769
opensanctions_pairs	GPT-5.5	0.750	0.608	0.503	0.724	76.249	0.658	0.658	0.525	0.778	0.769
opensanctions_pairs	gpt-oss-120b	0.780	0.666	0.561	0.780	76.249	0.718	0.725	0.588	0.815	0.769
opensanctions_pairs	Grok 4	0.850	0.776	0.676	0.868	76.249	0.824	0.830	0.734	0.913	0.769
opensanctions_pairs	Llama 3.3 70B	0.790	0.682	0.567	0.796	76.249	0.734	0.741	0.619	0.845	0.769
papadakis_dn	Claude Opus 5	0.920	0.955	0.894	0.995	32.601	0.789	0.322	0.131	0.814	0.018
papadakis_dn	Claude Sonnet 5	0.950	0.994	0.987	0.999	32.601	0.857	0.786	0.559	0.939	0.018
papadakis_dn	DeepSeek-chat	0.940	0.993	0.986	0.998	32.601	0.833	0.757	0.534	0.923	0.018
papadakis_dn	DeepSeek-R1	0.940	0.993	0.986	0.998	32.601	0.833	0.757	0.527	0.918	0.018
papadakis_dn	Gemini 3.1 Pro	0.930	0.974	0.927	0.997	32.601	0.811	0.452	0.196	0.886	0.018
papadakis_dn	Gemini 3.8 Flash	0.940	0.992	0.984	0.998	32.601	0.824	0.702	0.442	0.893	0.018
papadakis_dn	GPT-5.4-mini	0.950	0.994	0.988	0.999	32.601	0.857	0.786	0.575	0.949	0.018
papadakis_dn	GPT-5.5	0.950	0.994	0.987	0.999	32.601	0.857	0.786	0.564	0.944	0.018
papadakis_dn	gpt-oss-120b	0.930	0.992	0.984	0.998	32.601	0.788	0.682	0.431	0.890	0.018
papadakis_dn	Grok 4	0.950	0.994	0.988	0.999	32.601	0.857	0.786	0.562	0.951	0.018
papadakis_dn	Llama 3.3 70B	0.940	0.975	0.926	0.998	32.601	0.833	0.463	0.185	0.905	0.018
realhitbench	Claude Opus 5	0.743	0.745	0.662	0.813	98.080	–	–	–	–	–
realhitbench	Claude Sonnet 5	0.681	0.684	0.606	0.769	98.080	–	–	–	–	–
realhitbench	DeepSeek-chat	0.662	0.656	0.577	0.745	98.080	–	–	–	–	–
realhitbench	DeepSeek-R1	0.672	0.673	0.590	0.755	98.080	–	–	–	–	–
realhitbench	Gemini 3.1 Pro	0.699	0.698	0.614	0.780	98.080	–	–	–	–	–
realhitbench	Gemini 3.8 Flash	0.746	0.747	0.668	0.823	98.080	–	–	–	–	–
realhitbench	GPT-5.4-mini	0.661	0.658	0.574	0.742	98.080	–	–	–	–	–
realhitbench	GPT-5.5	0.724	0.721	0.644	0.799	98.080	–	–	–	–	–
realhitbench	gpt-oss-120b	0.603	0.602	0.514	0.691	98.080	–	–	–	–	–
realhitbench	Grok 4	0.640	0.638	0.549	0.725	98.080	–	–	–	–	–
realhitbench	Llama 3.3 70B	0.611	0.616	0.538	0.694	98.080	–	–	–	–	–
smat	Claude Opus 5	0.933	0.931	0.886	0.968	98.100	–	–	–	–	–

smat	Claude Sonnet 5	0.938	0.936	0.894	0.973	98.100	–	–	–	–	–
smat	DeepSeek-chat	0.918	0.916	0.863	0.958	98.100	–	–	–	–	–
smat	DeepSeek-R1	0.930	0.928	0.882	0.968	98.100	–	–	–	–	–
smat	Gemini 3.1 Pro	0.883	0.881	0.822	0.930	98.100	–	–	–	–	–
smat	Gemini 3.8 Flash	0.921	0.919	0.871	0.960	98.100	–	–	–	–	–
smat	GPT-5.4-mini	0.956	0.958	0.922	0.988	98.100	–	–	–	–	–
smat	GPT-5.5	0.940	0.939	0.899	0.975	98.100	–	–	–	–	–
smat	gpt-oss-120b	0.916	0.913	0.862	0.955	98.100	–	–	–	–	–
smat	Grok 4	0.936	0.933	0.887	0.973	98.100	–	–	–	–	–
smat	Llama 3.3 70B	0.806	0.803	0.737	0.856	98.100	–	–	–	–	–
suc	Claude Opus 5	0.986	0.988	0.965	1.000	98.920	–	–	–	–	–
suc	Claude Sonnet 5	0.911	0.908	0.857	0.954	98.920	–	–	–	–	–
suc	DeepSeek-chat	0.858	0.858	0.795	0.910	98.920	–	–	–	–	–
suc	DeepSeek-R1	0.942	0.939	0.893	0.976	98.920	–	–	–	–	–
suc	Gemini 3.1 Pro	0.945	0.939	0.898	0.976	98.920	–	–	–	–	–
suc	Gemini 3.8 Flash	0.976	0.978	0.949	1.000	98.920	–	–	–	–	–
suc	GPT-5.4-mini	0.914	0.910	0.855	0.957	98.920	–	–	–	–	–
suc	GPT-5.5	0.943	0.938	0.895	0.977	98.920	–	–	–	–	–
suc	gpt-oss-120b	0.932	0.930	0.882	0.972	98.920	–	–	–	–	–
suc	Grok 4	0.943	0.938	0.897	0.976	98.920	–	–	–	–	–
suc	Llama 3.3 70B	0.740	0.738	0.656	0.812	98.920	–	–	–	–	–
tab_fact	Claude Opus 5	0.660	0.667	0.571	0.762	99.360	0.673	0.696	0.586	0.781	0.500
tab_fact	Claude Sonnet 5	0.740	0.733	0.641	0.820	99.360	0.698	0.710	0.590	0.806	0.500
tab_fact	DeepSeek-chat	0.840	0.826	0.744	0.901	99.360	0.789	0.789	0.677	0.884	0.500
tab_fact	DeepSeek-R1	0.770	0.763	0.672	0.846	99.360	0.729	0.740	0.617	0.836	0.500
tab_fact	Gemini 3.1 Pro	0.830	0.818	0.746	0.901	99.360	0.785	0.788	0.672	0.876	0.500
tab_fact	Gemini 3.8 Flash	0.860	0.848	0.773	0.913	99.360	0.821	0.821	0.720	0.900	0.500
tab_fact	GPT-5.4-mini	0.820	0.808	0.734	0.891	99.360	0.769	0.772	0.654	0.859	0.500
tab_fact	GPT-5.5	0.840	0.829	0.748	0.903	99.360	0.800	0.803	0.700	0.892	0.500
tab_fact	gpt-oss-120b	0.830	0.815	0.739	0.890	99.360	0.773	0.773	0.647	0.868	0.500
tab_fact	Grok 4	0.850	0.839	0.762	0.912	99.360	0.810	0.812	0.707	0.897	0.500
tab_fact	Llama 3.3 70B	0.780	0.769	0.681	0.853	99.360	0.725	0.732	0.615	0.832	0.500
tabis	Claude Opus 5	0.960	0.964	0.915	0.994	77.069	–	–	–	–	–
tabis	Claude Sonnet 5	0.920	0.917	0.846	0.971	77.069	–	–	–	–	–
tabis	DeepSeek-chat	0.770	0.710	0.602	0.807	77.069	–	–	–	–	–
tabis	DeepSeek-R1	0.970	0.982	0.957	1.000	77.069	–	–	–	–	–
tabis	Gemini 3.1 Pro	0.950	0.970	0.940	0.994	77.069	–	–	–	–	–
tabis	Gemini 3.8 Flash	0.950	0.970	0.940	0.994	77.069	–	–	–	–	–
tabis	GPT-5.4-mini	0.900	0.905	0.835	0.962	77.069	–	–	–	–	–
tabis	GPT-5.5	0.960	0.976	0.946	0.994	77.069	–	–	–	–	–

tabis	gpt-oss-120b	0.920	0.951	0.915	0.983	77.069	–	–	–	–	–
tabis	Grok 4	0.980	0.988	0.966	1.000	77.069	–	–	–	–	–
tabis	Llama 3.3 70B	0.820	0.821	0.734	0.901	77.069	–	–	–	–	–
tablebench	Claude Opus 5	0.846	0.846	0.776	0.912	89.187	–	–	–	–	–
tablebench	Claude Sonnet 5	0.877	0.875	0.813	0.931	89.187	–	–	–	–	–
tablebench	DeepSeek-chat	0.833	0.844	0.782	0.904	89.187	–	–	–	–	–
tablebench	DeepSeek-R1	0.803	0.809	0.733	0.876	89.187	–	–	–	–	–
tablebench	Gemini 3.1 Pro	0.858	0.853	0.781	0.917	89.187	–	–	–	–	–
tablebench	Gemini 3.8 Flash	0.845	0.843	0.767	0.906	89.187	–	–	–	–	–
tablebench	GPT-5.4-mini	0.841	0.835	0.764	0.898	89.187	–	–	–	–	–
tablebench	GPT-5.5	0.852	0.850	0.781	0.916	89.187	–	–	–	–	–
tablebench	gpt-oss-120b	0.813	0.808	0.738	0.875	89.187	–	–	–	–	–
tablebench	Grok 4	0.839	0.837	0.760	0.903	89.187	–	–	–	–	–
tablebench	Llama 3.3 70B	0.660	0.641	0.554	0.725	89.187	–	–	–	–	–
tableeval	Claude Opus 5	0.765	0.784	0.689	0.874	69.689	–	–	–	–	–
tableeval	Claude Sonnet 5	0.733	0.764	0.665	0.852	69.689	–	–	–	–	–
tableeval	DeepSeek-chat	0.711	0.708	0.609	0.802	69.689	–	–	–	–	–
tableeval	DeepSeek-R1	0.773	0.760	0.662	0.856	69.689	–	–	–	–	–
tableeval	Gemini 3.1 Pro	0.790	0.794	0.702	0.884	69.689	–	–	–	–	–
tableeval	Gemini 3.8 Flash	0.791	0.805	0.709	0.894	69.689	–	–	–	–	–
tableeval	GPT-5.4-mini	0.742	0.751	0.654	0.842	69.689	–	–	–	–	–
tableeval	GPT-5.5	0.784	0.794	0.693	0.882	69.689	–	–	–	–	–
tableeval	gpt-oss-120b	0.773	0.774	0.674	0.861	69.689	–	–	–	–	–
tableeval	Grok 4	0.802	0.785	0.685	0.877	69.689	–	–	–	–	–
tableeval	Llama 3.3 70B	0.636	0.618	0.506	0.722	69.689	–	–	–	–	–
tat_qa	Claude Opus 5	0.861	0.846	0.769	0.913	60.725	–	–	–	–	–
tat_qa	Claude Sonnet 5	0.845	0.845	0.777	0.906	60.725	–	–	–	–	–
tat_qa	DeepSeek-chat	0.806	0.774	0.684	0.864	60.725	–	–	–	–	–
tat_qa	DeepSeek-R1	0.815	0.780	0.705	0.862	60.725	–	–	–	–	–
tat_qa	Gemini 3.1 Pro	0.802	0.745	0.652	0.833	60.725	–	–	–	–	–
tat_qa	Gemini 3.8 Flash	0.829	0.815	0.729	0.888	60.725	–	–	–	–	–
tat_qa	GPT-5.4-mini	0.789	0.719	0.630	0.805	60.725	–	–	–	–	–
tat_qa	GPT-5.5	0.789	0.748	0.669	0.827	60.725	–	–	–	–	–
tat_qa	gpt-oss-120b	0.765	0.684	0.586	0.778	60.725	–	–	–	–	–
tat_qa	Grok 4	0.791	0.710	0.619	0.798	60.725	–	–	–	–	–
tat_qa	Llama 3.3 70B	0.752	0.679	0.573	0.778	60.725	–	–	–	–	–
tpcdi_cells	Claude Opus 5	0.462	0.465	0.378	0.549	95.091	–	–	–	–	–
tpcdi_cells	Claude Sonnet 5	0.462	0.465	0.379	0.560	95.091	–	–	–	–	–
tpcdi_cells	DeepSeek-chat	0.462	0.465	0.381	0.561	95.091	–	–	–	–	–
tpcdi_cells	DeepSeek-R1	0.462	0.466	0.370	0.552	95.091	–	–	–	–	–
tpcdi_cells	Gemini 3.1 Pro	0.462	0.465	0.372	0.548	95.091	–	–	–	–	–

tpcdi_cells	Gemini 3.8 Flash	0.462	0.465	0.373	0.555	95.091	–	–	–	–	–
tpcdi_cells	GPT-5.4-mini	0.455	0.458	0.373	0.542	95.091	–	–	–	–	–
tpcdi_cells	GPT-5.5	0.462	0.466	0.376	0.559	95.091	–	–	–	–	–
tpcdi_cells	gpt-oss-120b	0.442	0.453	0.362	0.534	95.091	–	–	–	–	–
tpcdi_cells	Grok 4	0.447	0.456	0.368	0.543	95.091	–	–	–	–	–
tpcdi_cells	Llama 3.3 70B	0.442	0.453	0.366	0.544	95.091	–	–	–	–	–
valentine	Claude Opus 5	1.000	1.000	1.000	1.000	81.736	–	–	–	–	–
valentine	Claude Sonnet 5	0.994	0.989	0.966	1.000	81.736	–	–	–	–	–
valentine	DeepSeek-chat	0.979	0.967	0.925	1.000	81.736	–	–	–	–	–
valentine	DeepSeek-R1	0.990	0.986	0.957	1.000	81.736	–	–	–	–	–
valentine	Gemini 3.1 Pro	1.000	1.000	1.000	1.000	81.736	–	–	–	–	–
valentine	Gemini 3.8 Flash	0.970	0.988	0.973	1.000	81.736	–	–	–	–	–
valentine	GPT-5.4-mini	0.985	0.978	0.943	1.000	81.736	–	–	–	–	–
valentine	GPT-5.5	1.000	1.000	1.000	1.000	81.736	–	–	–	–	–
valentine	gpt-oss-120b	0.979	0.971	0.934	1.000	81.736	–	–	–	–	–
valentine	Grok 4	0.973	0.957	0.910	1.000	81.736	–	–	–	–	–
valentine	Llama 3.3 70B	0.970	0.958	0.915	0.992	81.736	–	–	–	–	–
walmart_amazon	Claude Opus 5	0.980	0.982	0.947	1.000	70.807	0.976	0.912	0.770	1.000	0.094
walmart_amazon	Claude Sonnet 5	0.950	0.962	0.914	0.996	70.807	0.938	0.823	0.644	0.977	0.094
walmart_amazon	DeepSeek-chat	0.950	0.962	0.914	0.996	70.807	0.938	0.823	0.661	0.977	0.094
walmart_amazon	DeepSeek-R1	0.960	0.978	0.940	0.998	70.807	0.950	0.887	0.733	0.989	0.094
walmart_amazon	Gemini 3.1 Pro	0.970	0.967	0.920	1.000	70.807	0.964	0.848	0.669	1.000	0.094
walmart_amazon	Gemini 3.8 Flash	0.990	0.998	0.993	1.000	70.807	0.988	0.988	0.955	1.000	0.094
walmart_amazon	GPT-5.4-mini	0.950	0.975	0.941	0.998	70.807	0.937	0.874	0.708	0.978	0.094
walmart_amazon	GPT-5.5	0.970	0.967	0.920	1.000	70.807	0.964	0.848	0.676	1.000	0.094
walmart_amazon	gpt-oss-120b	0.940	0.973	0.936	0.996	70.807	0.923	0.860	0.690	0.974	0.094
walmart_amazon	Grok 4	0.960	0.978	0.943	0.998	70.807	0.950	0.887	0.733	0.989	0.094
walmart_amazon	Llama 3.3 70B	0.930	0.971	0.935	0.994	70.807	0.909	0.847	0.703	0.966	0.094
wdc_lspc	Claude Opus 5	0.990	0.996	0.986	1.000	75.082	0.991	0.992	0.974	1.000	0.264
wdc_lspc	Claude Sonnet 5	0.980	0.992	0.978	1.000	75.082	0.981	0.984	0.957	1.000	0.264
wdc_lspc	DeepSeek-chat	0.960	0.940	0.876	0.995	75.082	0.964	0.895	0.785	0.989	0.264
wdc_lspc	DeepSeek-R1	1.000	1.000	1.000	1.000	75.082	1.000	1.000	1.000	1.000	0.264
wdc_lspc	Gemini 3.1 Pro	0.980	0.992	0.978	1.000	75.082	0.981	0.984	0.959	1.000	0.264
wdc_lspc	Gemini 3.8 Flash	0.980	0.992	0.978	1.000	75.082	0.981	0.984	0.959	1.000	0.264
wdc_lspc	GPT-5.4-mini	0.970	0.986	0.967	1.000	75.082	0.971	0.973	0.937	1.000	0.264
wdc_lspc	GPT-5.5	1.000	1.000	1.000	1.000	75.082	1.000	1.000	1.000	1.000	0.264
wdc_lspc	gpt-oss-120b	0.980	0.990	0.976	1.000	75.082	0.981	0.981	0.951	1.000	0.264
wdc_lspc	Grok 4	1.000	1.000	1.000	1.000	75.082	1.000	1.000	1.000	1.000	0.264
wdc_lspc	Llama 3.3 70B	0.960	0.940	0.875	0.994	75.082	0.964	0.895	0.784	0.990	0.264
wdc_products	Claude Opus 5	0.980	0.990	0.975	1.000	77.046	0.981	0.982	0.953	1.000	0.280
wdc_products	Claude Sonnet 5	0.970	0.986	0.967	1.000	77.046	0.971	0.974	0.938	1.000	0.280

wdc_products	DeepSeek-chat	0.970	0.977	0.941	1.000	77.046	0.972	0.958	0.905	1.000	0.280
wdc_products	DeepSeek-R1	0.980	0.990	0.975	1.000	77.046	0.981	0.982	0.953	1.000	0.280
wdc_products	Gemini 3.1 Pro	0.980	0.990	0.974	1.000	77.046	0.981	0.982	0.953	1.000	0.280
wdc_products	Gemini 3.8 Flash	0.990	0.995	0.984	1.000	77.046	0.991	0.991	0.967	1.000	0.280
wdc_products	GPT-5.4-mini	0.970	0.986	0.965	1.000	77.046	0.971	0.974	0.935	1.000	0.280
wdc_products	GPT-5.5	0.980	0.979	0.939	1.000	77.046	0.981	0.964	0.895	1.000	0.280
wdc_products	gpt-oss-120b	0.980	0.979	0.939	1.000	77.046	0.981	0.964	0.899	1.000	0.280
wdc_products	Grok 4	0.990	0.995	0.983	1.000	77.046	0.991	0.991	0.969	1.000	0.280
wdc_products	Llama 3.3 70B	0.970	0.977	0.942	1.000	77.046	0.972	0.958	0.899	1.000	0.280
wikitablequestions	Claude Opus 5	0.879	0.878	0.812	0.934	97.686	–	–	–	–	–
wikitablequestions	Claude Sonnet 5	0.863	0.863	0.800	0.926	97.686	–	–	–	–	–
wikitablequestions	DeepSeek-chat	0.843	0.842	0.772	0.901	97.686	–	–	–	–	–
wikitablequestions	DeepSeek-R1	0.857	0.856	0.787	0.917	97.686	–	–	–	–	–
wikitablequestions	Gemini 3.1 Pro	0.855	0.855	0.788	0.914	97.686	–	–	–	–	–
wikitablequestions	Gemini 3.8 Flash	0.881	0.881	0.822	0.936	97.686	–	–	–	–	–
wikitablequestions	GPT-5.4-mini	0.842	0.839	0.767	0.901	97.686	–	–	–	–	–
wikitablequestions	GPT-5.5	0.865	0.862	0.798	0.918	97.686	–	–	–	–	–
wikitablequestions	gpt-oss-120b	0.824	0.821	0.750	0.887	97.686	–	–	–	–	–
wikitablequestions	Grok 4	0.867	0.864	0.801	0.923	97.686	–	–	–	–	–
wikitablequestions	Llama 3.3 70B	0.749	0.747	0.673	0.821	97.686	–	–	–	–	–

Table 29: Official-split estimates under the headline rule averaged over the ten models.

dataset	suite_mean	official_ estimate	suite_f1_ pos	official_precision_ pos	official_recall_ pos	official_f1_ pos	official_positive_ share
abt_buy	0.971	0.987	0.969	0.933	0.950	0.940	0.108
alaska_camera	0.945	0.945	0.939	1.000	0.895	0.939	0.500
alaska_schema	0.929	0.929	–	–	–	–	–
amazon_google	0.935	0.932	0.928	0.609	0.941	0.738	0.102
bird	0.191	0.218	–	–	–	–	–
dblp_acm	0.987	0.992	0.986	0.981	0.980	0.978	0.180
dblp_scholar	0.948	0.966	0.947	0.901	0.922	0.909	0.186
fetaqa	0.387	0.387	–	–	–	–	–
fintagging	0.884	0.882	–	–	–	–	–
hitab	0.772	0.776	–	–	–	–	–
machamp	0.887	0.891	0.892	0.649	0.852	0.735	0.175
magneto_gdc	0.937	0.940	–	–	–	–	–
mmtu	0.874	0.837	–	–	–	–	–
officeqa	0.135	0.135	–	–	–	–	–
officeqa_pro_v2	0.151	0.151	–	–	–	–	–
opensanctions_ pairs	0.835	0.748	0.790	0.993	0.677	0.795	0.769
papadakis_dn	0.940	0.987	0.831	0.527	0.967	0.662	0.018
realhitbench	0.677	0.676	–	–	–	–	–
smat	0.916	0.914	–	–	–	–	–
suc	0.917	0.915	–	–	–	–	–
tab_fact	0.802	0.792	0.761	0.903	0.674	0.767	0.500
tabis	0.918	0.923	–	–	–	–	–
tablebench	0.824	0.822	–	–	–	–	–
tableeval	0.755	0.758	–	–	–	–	–
tat_qa	0.804	0.759	–	–	–	–	–
tpcdi_cells	0.456	0.462	–	–	–	–	–
valentine	0.985	0.981	–	–	–	–	–
walmart_amazon	0.959	0.974	0.949	0.824	0.931	0.872	0.094
wdc_lspc	0.982	0.984	0.983	0.968	0.978	0.972	0.264
wdc_products	0.978	0.986	0.979	0.982	0.968	0.975	0.280
wikitablequestions	0.848	0.846	–	–	–	–	–

Table 30: Cross-dataset pairwise tests under rule best-rho: Friedman $\chi^2 = 88.6$ ($p = 1.0\text{e-}14$), Iman–Davenport $F = 12.0$, Nemenyi CD = 2.71, Kendall's $W = 0.277$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.3710	22	7	2	0.0236	0.7079	no	no
Claude Opus 5	DeepSeek-chat	-3.0000	22	8	1	0.0143	0.4715	yes	no
Claude Opus 5	DeepSeek-R1	-1.1129	16	12	3	0.1238	1.0000	no	no
Claude Opus 5	Gemini 3.1 Pro	-1.6290	20	8	3	0.0487	1.0000	no	no
Claude Opus 5	Gemini 3.8 Flash	-0.5645	19	10	2	0.1958	1.0000	no	no
Claude Opus 5	GPT-5.4-mini	-2.8548	25	6	0	0.0051	0.1823	yes	no
Claude Opus 5	GPT-5.5	0.0484	13	12	6	0.5238	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.6774	26	4	1	0.0004	0.0155	yes	yes
Claude Opus 5	Grok 4	-1.8548	18	10	3	0.0745	1.0000	no	no
Claude Opus 5	Llama 3.3 70B	-5.1613	26	5	0	0.0002	0.0097	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.6290	20	7	4	0.0368	0.9928	no	no
Claude Sonnet 5	DeepSeek-R1	0.2581	15	14	2	0.4385	1.0000	no	no
Claude Sonnet 5	Gemini 3.1 Pro	-0.2581	15	14	2	0.9142	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.8065	11	17	3	0.1059	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-1.4839	20	8	3	0.0499	1.0000	no	no
Claude Sonnet 5	GPT-5.5	1.4194	9	19	3	0.0368	0.9928	no	no
Claude Sonnet 5	gpt-oss-120b	-3.3065	22	8	1	0.0038	0.1423	yes	no
Claude Sonnet 5	Grok 4	-0.4839	19	10	2	0.4623	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-3.7903	25	4	2	0.0001	0.0042	yes	yes
DeepSeek-chat	DeepSeek-R1	1.8871	9	21	1	0.0315	0.8832	no	no
DeepSeek-chat	Gemini 3.1 Pro	1.3710	11	18	2	0.1239	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	2.4355	6	22	3	0.0057	0.1998	no	no
DeepSeek-chat	GPT-5.4-mini	0.1452	11	16	4	0.3777	1.0000	no	no
DeepSeek-chat	GPT-5.5	3.0484	5	22	4	0.0024	0.0902	yes	no
DeepSeek-chat	gpt-oss-120b	-1.6774	21	9	1	0.0387	0.9928	no	no
DeepSeek-chat	Grok 4	1.1452	12	16	3	0.2898	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.1613	21	6	4	0.0006	0.0256	no	yes
DeepSeek-R1	Gemini 3.1 Pro	-0.5161	16	14	1	0.5113	1.0000	no	no
DeepSeek-R1	Gemini 3.8 Flash	0.5484	11	18	2	0.1287	1.0000	no	no
DeepSeek-R1	GPT-5.4-mini	-1.7419	22	8	1	0.0624	1.0000	no	no
DeepSeek-R1	GPT-5.5	1.1613	9	18	4	0.1386	1.0000	no	no
DeepSeek-R1	gpt-oss-120b	-3.5645	26	4	1	0.0000	0.0019	yes	yes
DeepSeek-R1	Grok 4	-0.7419	15	12	4	0.3414	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-4.0484	26	2	3	0.0000	0.0004	yes	yes
Gemini 3.1 Pro	Gemini 3.8 Flash	1.0645	9	17	5	0.0488	1.0000	no	no
Gemini 3.1 Pro	GPT-5.4-mini	-1.2258	20	8	3	0.0267	0.7756	no	no
Gemini 3.1 Pro	GPT-5.5	1.6774	9	18	4	0.0159	0.5087	no	no
Gemini 3.1 Pro	gpt-oss-120b	-3.0484	23	4	4	0.0004	0.0169	yes	yes
Gemini 3.1 Pro	Grok 4	-0.2258	14	15	2	0.7316	1.0000	no	no
Gemini 3.1 Pro	Llama 3.3 70B	-3.5323	25	6	0	0.0001	0.0052	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-2.2903	22	8	1	0.0159	0.5087	no	no
Gemini 3.8 Flash	GPT-5.5	0.6129	13	16	2	0.8677	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-4.1129	26	4	1	0.0001	0.0033	yes	yes
Gemini 3.8 Flash	Grok 4	-1.2903	18	11	2	0.0698	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-4.5968	26	4	1	0.0001	0.0038	yes	yes
GPT-5.4-mini	GPT-5.5	2.9032	5	25	1	0.0022	0.0842	yes	no
GPT-5.4-mini	gpt-oss-120b	-1.8226	23	8	0	0.0018	0.0708	no	no
GPT-5.4-mini	Grok 4	1.0000	12	16	3	0.4927	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-2.3065	26	3	2	0.0000	0.0025	no	yes
GPT-5.5	gpt-oss-120b	-4.7258	27	2	2	0.0000	0.0004	yes	yes
GPT-5.5	Grok 4	-1.9032	20	7	4	0.0108	0.3669	no	no
GPT-5.5	Llama 3.3 70B	-5.2097	27	4	0	0.0000	0.0008	yes	yes
gpt-oss-120b	Grok 4	2.8226	5	26	0	0.0001	0.0043	yes	yes
gpt-oss-120b	Llama 3.3 70B	-0.4839	19	12	0	0.2475	1.0000	no	no

Grok 4	Llama 3.3 70B	-3.3065	25	5	1	0.0001	0.0055	yes	yes

Table 31: Cross-dataset pairwise tests under rule exact match: Friedman $\chi^2 = 91.5$ ($p = 2.7\text{e-}15$), Iman–Davenport $F = 12.6$, Nemenyi CD = 2.71, Kendall's $W = 0.277$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.5000	21	7	3	0.0266	0.7453	no	no
Claude Opus 5	DeepSeek-chat	-3.0161	22	6	3	0.0049	0.1955	yes	no
Claude Opus 5	DeepSeek-R1	-1.2258	15	9	7	0.0873	1.0000	no	no
Claude Opus 5	Gemini 3.1 Pro	-2.0161	21	6	4	0.0133	0.4573	no	no
Claude Opus 5	Gemini 3.8 Flash	-1.0484	19	8	4	0.0550	1.0000	no	no
Claude Opus 5	GPT-5.4-mini	-3.2742	25	5	1	0.0057	0.2158	yes	no
Claude Opus 5	GPT-5.5	-0.3387	13	9	9	0.2972	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.7419	26	3	2	0.0004	0.0157	yes	yes
Claude Opus 5	Grok 4	-1.9677	18	9	4	0.0559	1.0000	no	no
Claude Opus 5	Llama 3.3 70B	-5.5323	26	5	0	0.0001	0.0060	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.5161	19	5	7	0.0153	0.4892	no	no
Claude Sonnet 5	DeepSeek-R1	0.2742	12	14	5	0.3506	1.0000	no	no
Claude Sonnet 5	Gemini 3.1 Pro	-0.5161	14	13	4	0.8598	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.4516	11	13	7	0.3876	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-1.7742	19	7	5	0.1052	1.0000	no	no
Claude Sonnet 5	GPT-5.5	1.1613	7	15	9	0.0397	1.0000	no	no
Claude Sonnet 5	gpt-oss-120b	-3.2419	22	7	2	0.0089	0.3190	yes	no
Claude Sonnet 5	Grok 4	-0.4677	18	9	4	0.4680	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-4.0323	25	4	2	0.0001	0.0045	yes	yes
DeepSeek-chat	DeepSeek-R1	1.7903	8	19	4	0.0202	0.6265	no	no
DeepSeek-chat	Gemini 3.1 Pro	1.0000	10	15	6	0.1854	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	1.9677	6	19	6	0.0131	0.4573	no	no
DeepSeek-chat	GPT-5.4-mini	-0.2581	12	14	5	0.4859	1.0000	no	no
DeepSeek-chat	GPT-5.5	2.6774	6	19	6	0.0052	0.2018	no	no
DeepSeek-chat	gpt-oss-120b	-1.7258	19	9	3	0.0827	1.0000	no	no
DeepSeek-chat	Grok 4	1.0484	12	15	4	0.2936	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.5161	22	3	6	0.0001	0.0065	no	yes
DeepSeek-R1	Gemini 3.1 Pro	-0.7903	16	12	3	0.6024	1.0000	no	no
DeepSeek-R1	Gemini 3.8 Flash	0.1774	12	15	4	0.4211	1.0000	no	no
DeepSeek-R1	GPT-5.4-mini	-2.0484	20	8	3	0.1157	1.0000	no	no
DeepSeek-R1	GPT-5.5	0.8871	8	16	7	0.1471	1.0000	no	no
DeepSeek-R1	gpt-oss-120b	-3.5161	26	3	2	0.0001	0.0041	yes	yes
DeepSeek-R1	Grok 4	-0.7419	14	10	7	0.3978	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-4.3065	26	2	3	0.0000	0.0004	yes	yes
Gemini 3.1 Pro	Gemini 3.8 Flash	0.9677	7	17	7	0.0608	1.0000	no	no
Gemini 3.1 Pro	GPT-5.4-mini	-1.2581	20	8	3	0.0516	1.0000	no	no
Gemini 3.1 Pro	GPT-5.5	1.6774	5	16	10	0.0253	0.7351	no	no
Gemini 3.1 Pro	gpt-oss-120b	-2.7258	21	3	7	0.0008	0.0336	yes	yes
Gemini 3.1 Pro	Grok 4	0.0484	11	14	6	0.9922	1.0000	no	no
Gemini 3.1 Pro	Llama 3.3 70B	-3.5161	27	4	0	0.0001	0.0044	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-2.2258	21	8	2	0.0385	1.0000	no	no
Gemini 3.8 Flash	GPT-5.5	0.7097	10	15	6	0.6092	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-3.6935	24	2	5	0.0001	0.0040	yes	yes
Gemini 3.8 Flash	Grok 4	-0.9194	15	10	6	0.2232	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-4.4839	26	4	1	0.0001	0.0045	yes	yes
GPT-5.4-mini	GPT-5.5	2.9355	5	24	2	0.0064	0.2363	yes	no
GPT-5.4-mini	gpt-oss-120b	-1.4677	20	10	1	0.0134	0.4573	no	no
GPT-5.4-mini	Grok 4	1.3065	9	16	6	0.3613	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-2.2581	23	4	4	0.0001	0.0053	no	yes
GPT-5.5	gpt-oss-120b	-4.4032	26	1	4	0.0000	0.0005	yes	yes
GPT-5.5	Grok 4	-1.6290	16	6	9	0.0480	1.0000	no	no
GPT-5.5	Llama 3.3 70B	-5.1935	27	4	0	0.0000	0.0010	yes	yes
gpt-oss-120b	Grok 4	2.7742	3	24	4	0.0002	0.0066	yes	yes
gpt-oss-120b	Llama 3.3 70B	-0.7903	20	8	3	0.0217	0.6516	no	no

Grok 4	Llama 3.3 70B	-3.5645	24	4	3	0.0001	0.0045	yes	yes

Table 32: Cross-dataset pairwise tests under rule knee: Friedman $\chi^2 = 92.7$ ($p = 1.6\text{e-}15$), Iman–Davenport $F = 12.8$, Nemenyi CD = 2.71, Kendall's $W = 0.290$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.4677	22	7	2	0.0213	0.6213	no	no
Claude Opus 5	DeepSeek-chat	-3.1935	22	8	1	0.0086	0.2935	yes	no
Claude Opus 5	DeepSeek-R1	-1.2742	17	11	3	0.1017	1.0000	no	no
Claude Opus 5	Gemini 3.1 Pro	-1.6613	20	8	3	0.0385	0.9627	no	no
Claude Opus 5	Gemini 3.8 Flash	-0.6613	20	9	2	0.1123	1.0000	no	no
Claude Opus 5	GPT-5.4-mini	-2.9839	25	6	0	0.0042	0.1559	yes	no
Claude Opus 5	GPT-5.5	0.0484	14	11	6	0.3935	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.7742	26	4	1	0.0004	0.0168	yes	yes
Claude Opus 5	Grok 4	-1.9516	18	10	3	0.0523	1.0000	no	no
Claude Opus 5	Llama 3.3 70B	-5.3226	26	5	0	0.0002	0.0077	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.7258	20	7	4	0.0261	0.7036	no	no
Claude Sonnet 5	DeepSeek-R1	0.1935	15	14	2	0.5238	1.0000	no	no
Claude Sonnet 5	Gemini 3.1 Pro	-0.1935	15	14	2	0.8677	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.8065	11	17	3	0.1553	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-1.5161	20	8	3	0.0455	1.0000	no	no
Claude Sonnet 5	GPT-5.5	1.5161	9	19	3	0.0318	0.8264	no	no
Claude Sonnet 5	gpt-oss-120b	-3.3065	22	8	1	0.0056	0.1999	yes	no
Claude Sonnet 5	Grok 4	-0.4839	19	10	2	0.4988	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-3.8548	25	4	2	0.0001	0.0039	yes	yes
DeepSeek-chat	DeepSeek-R1	1.9194	9	21	1	0.0170	0.5283	no	no
DeepSeek-chat	Gemini 3.1 Pro	1.5323	9	20	2	0.0976	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	2.5323	6	22	3	0.0064	0.2251	no	no
DeepSeek-chat	GPT-5.4-mini	0.2097	10	17	4	0.3270	1.0000	no	no
DeepSeek-chat	GPT-5.5	3.2419	5	22	4	0.0018	0.0730	yes	no
DeepSeek-chat	gpt-oss-120b	-1.5806	21	9	1	0.0425	1.0000	no	no
DeepSeek-chat	Grok 4	1.2419	12	16	3	0.2809	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.1290	21	6	4	0.0004	0.0169	no	yes
DeepSeek-R1	Gemini 3.1 Pro	-0.3871	16	14	1	0.4988	1.0000	no	no
DeepSeek-R1	Gemini 3.8 Flash	0.6129	11	18	2	0.1553	1.0000	no	no
DeepSeek-R1	GPT-5.4-mini	-1.7097	22	8	1	0.0682	1.0000	no	no
DeepSeek-R1	GPT-5.5	1.3226	8	19	4	0.1143	1.0000	no	no
DeepSeek-R1	gpt-oss-120b	-3.5000	26	4	1	0.0000	0.0023	yes	yes
DeepSeek-R1	Grok 4	-0.6774	15	12	4	0.4044	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-4.0484	26	2	3	0.0000	0.0003	yes	yes
Gemini 3.1 Pro	Gemini 3.8 Flash	1.0000	9	17	5	0.0488	1.0000	no	no
Gemini 3.1 Pro	GPT-5.4-mini	-1.3226	20	8	3	0.0218	0.6213	no	no
Gemini 3.1 Pro	GPT-5.5	1.7097	8	19	4	0.0143	0.4566	no	no
Gemini 3.1 Pro	gpt-oss-120b	-3.1129	23	4	4	0.0005	0.0192	yes	yes
Gemini 3.1 Pro	Grok 4	-0.2903	14	15	2	0.8064	1.0000	no	no
Gemini 3.1 Pro	Llama 3.3 70B	-3.6613	26	5	0	0.0001	0.0043	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-2.3226	22	8	1	0.0207	0.6213	no	no
Gemini 3.8 Flash	GPT-5.5	0.7097	12	17	2	0.8677	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-4.1129	27	3	1	0.0001	0.0029	yes	yes
Gemini 3.8 Flash	Grok 4	-1.2903	19	10	2	0.0761	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-4.6613	26	4	1	0.0001	0.0035	yes	yes
GPT-5.4-mini	GPT-5.5	3.0323	5	25	1	0.0025	0.0959	yes	no
GPT-5.4-mini	gpt-oss-120b	-1.7903	23	8	0	0.0032	0.1208	no	no
GPT-5.4-mini	Grok 4	1.0323	11	17	3	0.3884	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-2.3387	25	4	2	0.0001	0.0032	no	yes
GPT-5.5	gpt-oss-120b	-4.8226	27	2	2	0.0000	0.0004	yes	yes
GPT-5.5	Grok 4	-2.0000	21	6	4	0.0128	0.4209	no	no
GPT-5.5	Llama 3.3 70B	-5.3710	27	4	0	0.0000	0.0007	yes	yes
gpt-oss-120b	Grok 4	2.8226	5	26	0	0.0001	0.0042	yes	yes
gpt-oss-120b	Llama 3.3 70B	-0.5484	20	11	0	0.0957	1.0000	no	no

Grok 4	Llama 3.3 70B	-3.3710	26	4	1	0.0001	0.0039	yes	yes

Table 33: Cross-dataset pairwise tests under rule top1: Friedman $\chi^2 = 91.5$ ($p = 2.7\text{e-}15$), Iman–Davenport $F = 12.6$, Nemenyi CD = 2.71, Kendall's $W = 0.277$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.5000	21	7	3	0.0266	0.7453	no	no
Claude Opus 5	DeepSeek-chat	-3.0161	22	6	3	0.0049	0.1955	yes	no
Claude Opus 5	DeepSeek-R1	-1.2258	15	9	7	0.0873	1.0000	no	no
Claude Opus 5	Gemini 3.1 Pro	-2.0161	21	6	4	0.0133	0.4573	no	no
Claude Opus 5	Gemini 3.8 Flash	-1.0484	19	8	4	0.0550	1.0000	no	no
Claude Opus 5	GPT-5.4-mini	-3.2742	25	5	1	0.0057	0.2158	yes	no
Claude Opus 5	GPT-5.5	-0.3387	13	9	9	0.2972	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.7419	26	3	2	0.0004	0.0157	yes	yes
Claude Opus 5	Grok 4	-1.9677	18	9	4	0.0559	1.0000	no	no
Claude Opus 5	Llama 3.3 70B	-5.5323	26	5	0	0.0001	0.0060	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.5161	19	5	7	0.0153	0.4892	no	no
Claude Sonnet 5	DeepSeek-R1	0.2742	12	14	5	0.3506	1.0000	no	no
Claude Sonnet 5	Gemini 3.1 Pro	-0.5161	14	13	4	0.8598	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.4516	11	13	7	0.3876	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-1.7742	19	7	5	0.1052	1.0000	no	no
Claude Sonnet 5	GPT-5.5	1.1613	7	15	9	0.0397	1.0000	no	no
Claude Sonnet 5	gpt-oss-120b	-3.2419	22	7	2	0.0089	0.3190	yes	no
Claude Sonnet 5	Grok 4	-0.4677	18	9	4	0.4680	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-4.0323	25	4	2	0.0001	0.0045	yes	yes
DeepSeek-chat	DeepSeek-R1	1.7903	8	19	4	0.0202	0.6265	no	no
DeepSeek-chat	Gemini 3.1 Pro	1.0000	10	15	6	0.1854	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	1.9677	6	19	6	0.0131	0.4573	no	no
DeepSeek-chat	GPT-5.4-mini	-0.2581	12	14	5	0.4859	1.0000	no	no
DeepSeek-chat	GPT-5.5	2.6774	6	19	6	0.0052	0.2018	no	no
DeepSeek-chat	gpt-oss-120b	-1.7258	19	9	3	0.0827	1.0000	no	no
DeepSeek-chat	Grok 4	1.0484	12	15	4	0.2936	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.5161	22	3	6	0.0001	0.0065	no	yes
DeepSeek-R1	Gemini 3.1 Pro	-0.7903	16	12	3	0.6024	1.0000	no	no
DeepSeek-R1	Gemini 3.8 Flash	0.1774	12	15	4	0.4211	1.0000	no	no
DeepSeek-R1	GPT-5.4-mini	-2.0484	20	8	3	0.1157	1.0000	no	no
DeepSeek-R1	GPT-5.5	0.8871	8	16	7	0.1471	1.0000	no	no
DeepSeek-R1	gpt-oss-120b	-3.5161	26	3	2	0.0001	0.0041	yes	yes
DeepSeek-R1	Grok 4	-0.7419	14	10	7	0.3978	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-4.3065	26	2	3	0.0000	0.0004	yes	yes
Gemini 3.1 Pro	Gemini 3.8 Flash	0.9677	7	17	7	0.0608	1.0000	no	no
Gemini 3.1 Pro	GPT-5.4-mini	-1.2581	20	8	3	0.0516	1.0000	no	no
Gemini 3.1 Pro	GPT-5.5	1.6774	5	16	10	0.0253	0.7351	no	no
Gemini 3.1 Pro	gpt-oss-120b	-2.7258	21	3	7	0.0008	0.0336	yes	yes
Gemini 3.1 Pro	Grok 4	0.0484	11	14	6	0.9922	1.0000	no	no
Gemini 3.1 Pro	Llama 3.3 70B	-3.5161	27	4	0	0.0001	0.0044	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-2.2258	21	8	2	0.0385	1.0000	no	no
Gemini 3.8 Flash	GPT-5.5	0.7097	10	15	6	0.6092	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-3.6935	24	2	5	0.0001	0.0040	yes	yes
Gemini 3.8 Flash	Grok 4	-0.9194	15	10	6	0.2232	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-4.4839	26	4	1	0.0001	0.0045	yes	yes
GPT-5.4-mini	GPT-5.5	2.9355	5	24	2	0.0064	0.2363	yes	no
GPT-5.4-mini	gpt-oss-120b	-1.4677	20	10	1	0.0134	0.4573	no	no
GPT-5.4-mini	Grok 4	1.3065	9	16	6	0.3613	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-2.2581	23	4	4	0.0001	0.0053	no	yes
GPT-5.5	gpt-oss-120b	-4.4032	26	1	4	0.0000	0.0005	yes	yes
GPT-5.5	Grok 4	-1.6290	16	6	9	0.0480	1.0000	no	no
GPT-5.5	Llama 3.3 70B	-5.1935	27	4	0	0.0000	0.0010	yes	yes
gpt-oss-120b	Grok 4	2.7742	3	24	4	0.0002	0.0066	yes	yes
gpt-oss-120b	Llama 3.3 70B	-0.7903	20	8	3	0.0217	0.6516	no	no

Grok 4	Llama 3.3 70B	-3.5645	24	4	3	0.0001	0.0045	yes	yes

Table 34: Cross-dataset pairwise tests under rule top2: Friedman $\chi^2 = 90.5$ ($p = 4.2\text{e-}15$), Iman–Davenport $F = 12.4$, Nemenyi CD = 2.71, Kendall's $W = 0.275$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.4516	20	7	4	0.0280	0.7559	no	no
Claude Opus 5	DeepSeek-chat	-3.0484	22	6	3	0.0049	0.1908	yes	no
Claude Opus 5	DeepSeek-R1	-1.1129	15	10	6	0.1007	1.0000	no	no
Claude Opus 5	Gemini 3.1 Pro	-1.8226	21	6	4	0.0141	0.4223	no	no
Claude Opus 5	Gemini 3.8 Flash	-0.9032	16	9	6	0.1750	1.0000	no	no
Claude Opus 5	GPT-5.4-mini	-3.0806	24	5	2	0.0066	0.2372	yes	no
Claude Opus 5	GPT-5.5	-0.0645	13	9	9	0.3603	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.6129	26	4	1	0.0006	0.0235	yes	yes
Claude Opus 5	Grok 4	-1.8065	19	9	3	0.0394	0.9841	no	no
Claude Opus 5	Llama 3.3 70B	-5.3387	25	5	1	0.0002	0.0075	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.5968	20	5	6	0.0098	0.3389	no	no
Claude Sonnet 5	DeepSeek-R1	0.3387	11	15	5	0.3709	1.0000	no	no
Claude Sonnet 5	Gemini 3.1 Pro	-0.3710	14	12	5	0.9608	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.5484	11	13	7	0.4095	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-1.6290	20	7	4	0.0856	1.0000	no	no
Claude Sonnet 5	GPT-5.5	1.3871	7	17	7	0.0097	0.3389	no	no
Claude Sonnet 5	gpt-oss-120b	-3.1613	22	7	2	0.0059	0.2173	yes	no
Claude Sonnet 5	Grok 4	-0.3548	15	11	5	0.5693	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-3.8871	25	4	2	0.0001	0.0055	yes	yes
DeepSeek-chat	DeepSeek-R1	1.9355	8	20	3	0.0126	0.3895	no	no
DeepSeek-chat	Gemini 3.1 Pro	1.2258	9	17	5	0.0791	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	2.1452	5	19	7	0.0101	0.3389	no	no
DeepSeek-chat	GPT-5.4-mini	-0.0323	11	13	7	0.3767	1.0000	no	no
DeepSeek-chat	GPT-5.5	2.9839	4	20	7	0.0008	0.0304	yes	yes
DeepSeek-chat	gpt-oss-120b	-1.5645	18	8	5	0.1014	1.0000	no	no
DeepSeek-chat	Grok 4	1.2419	11	16	4	0.1724	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.2903	23	3	5	0.0001	0.0064	no	yes
DeepSeek-R1	Gemini 3.1 Pro	-0.7097	15	13	3	0.7832	1.0000	no	no
DeepSeek-R1	Gemini 3.8 Flash	0.2097	13	15	3	0.4859	1.0000	no	no
DeepSeek-R1	GPT-5.4-mini	-1.9677	21	8	2	0.1157	1.0000	no	no
DeepSeek-R1	GPT-5.5	1.0484	8	17	6	0.0805	1.0000	no	no
DeepSeek-R1	gpt-oss-120b	-3.5000	25	4	2	0.0001	0.0041	yes	yes
DeepSeek-R1	Grok 4	-0.6935	13	11	7	0.5356	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-4.2258	26	2	3	0.0000	0.0004	yes	yes
Gemini 3.1 Pro	Gemini 3.8 Flash	0.9194	8	15	8	0.1051	1.0000	no	no
Gemini 3.1 Pro	GPT-5.4-mini	-1.2581	20	8	3	0.0516	1.0000	no	no
Gemini 3.1 Pro	GPT-5.5	1.7581	5	18	8	0.0164	0.4753	no	no
Gemini 3.1 Pro	gpt-oss-120b	-2.7903	22	3	6	0.0006	0.0245	yes	yes
Gemini 3.1 Pro	Grok 4	0.0161	11	13	7	0.9922	1.0000	no	no
Gemini 3.1 Pro	Llama 3.3 70B	-3.5161	25	5	1	0.0001	0.0055	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-2.1774	21	9	1	0.0594	1.0000	no	no
Gemini 3.8 Flash	GPT-5.5	0.8387	10	15	6	0.5100	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-3.7097	24	2	5	0.0001	0.0040	yes	yes
Gemini 3.8 Flash	Grok 4	-0.9032	17	11	3	0.1913	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-4.4355	26	4	1	0.0001	0.0055	yes	yes
GPT-5.4-mini	GPT-5.5	3.0161	5	25	1	0.0053	0.2013	yes	no
GPT-5.4-mini	gpt-oss-120b	-1.5323	21	9	1	0.0102	0.3389	no	no
GPT-5.4-mini	Grok 4	1.2742	10	15	6	0.4437	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-2.2581	23	3	5	0.0001	0.0042	no	yes
GPT-5.5	gpt-oss-120b	-4.5484	25	1	5	0.0000	0.0006	yes	yes
GPT-5.5	Grok 4	-1.7419	17	6	8	0.0270	0.7559	no	no
GPT-5.5	Llama 3.3 70B	-5.2742	27	4	0	0.0000	0.0013	yes	yes
gpt-oss-120b	Grok 4	2.8065	3	24	4	0.0001	0.0065	yes	yes
gpt-oss-120b	Llama 3.3 70B	-0.7258	19	8	4	0.0302	0.7855	no	no

Grok 4	Llama 3.3 70B	-3.5323	23	4	4	0.0001	0.0058	yes	yes

Table 35: Cross-dataset pairwise tests under rule top3: Friedman $\chi^2 = 92.2$ ($p = 2.0\text{e-}15$), Iman–Davenport $F = 12.7$, Nemenyi CD = 2.71, Kendall's $W = 0.288$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.5645	22	7	2	0.0202	0.5659	no	no
Claude Opus 5	DeepSeek-chat	-3.2903	23	7	1	0.0054	0.1992	yes	no
Claude Opus 5	DeepSeek-R1	-1.2419	16	12	3	0.1146	1.0000	no	no
Claude Opus 5	Gemini 3.1 Pro	-1.8871	21	7	3	0.0138	0.4425	no	no
Claude Opus 5	Gemini 3.8 Flash	-0.9839	21	8	2	0.0626	1.0000	no	no
Claude Opus 5	GPT-5.4-mini	-3.0806	25	6	0	0.0045	0.1747	yes	no
Claude Opus 5	GPT-5.5	-0.0484	14	11	6	0.3316	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.8387	26	4	1	0.0004	0.0180	yes	yes
Claude Opus 5	Grok 4	-1.9516	19	9	3	0.0377	0.9436	no	no
Claude Opus 5	Llama 3.3 70B	-5.4194	26	5	0	0.0001	0.0061	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.7258	20	7	4	0.0163	0.4758	no	no
Claude Sonnet 5	DeepSeek-R1	0.3226	15	14	2	0.4502	1.0000	no	no
Claude Sonnet 5	Gemini 3.1 Pro	-0.3226	15	14	2	0.7613	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.5806	11	17	3	0.2516	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-1.5161	20	8	3	0.0744	1.0000	no	no
Claude Sonnet 5	GPT-5.5	1.5161	9	19	3	0.0334	0.8678	no	no
Claude Sonnet 5	gpt-oss-120b	-3.2742	22	8	1	0.0059	0.2123	yes	no
Claude Sonnet 5	Grok 4	-0.3871	19	10	2	0.3830	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-3.8548	25	4	2	0.0001	0.0039	yes	yes
DeepSeek-chat	DeepSeek-R1	2.0484	8	22	1	0.0153	0.4745	no	no
DeepSeek-chat	Gemini 3.1 Pro	1.4032	9	20	2	0.0900	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	2.3065	7	21	3	0.0102	0.3375	no	no
DeepSeek-chat	GPT-5.4-mini	0.2097	10	17	4	0.2555	1.0000	no	no
DeepSeek-chat	GPT-5.5	3.2419	5	22	4	0.0015	0.0597	yes	no
DeepSeek-chat	gpt-oss-120b	-1.5484	22	8	1	0.0467	1.0000	no	no
DeepSeek-chat	Grok 4	1.3387	12	16	3	0.2168	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.1290	21	6	4	0.0004	0.0182	no	yes
DeepSeek-R1	Gemini 3.1 Pro	-0.6452	17	13	1	0.5498	1.0000	no	no
DeepSeek-R1	Gemini 3.8 Flash	0.2581	12	17	2	0.2356	1.0000	no	no
DeepSeek-R1	GPT-5.4-mini	-1.8387	22	8	1	0.0775	1.0000	no	no
DeepSeek-R1	GPT-5.5	1.1935	9	18	4	0.1549	1.0000	no	no
DeepSeek-R1	gpt-oss-120b	-3.5968	26	4	1	0.0001	0.0035	yes	yes
DeepSeek-R1	Grok 4	-0.7097	14	13	4	0.4740	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-4.1774	26	2	3	0.0000	0.0003	yes	yes
Gemini 3.1 Pro	Gemini 3.8 Flash	0.9032	9	17	5	0.0559	1.0000	no	no
Gemini 3.1 Pro	GPT-5.4-mini	-1.1935	20	8	3	0.0455	1.0000	no	no
Gemini 3.1 Pro	GPT-5.5	1.8387	7	20	4	0.0097	0.3285	no	no
Gemini 3.1 Pro	gpt-oss-120b	-2.9516	23	4	4	0.0005	0.0222	yes	yes
Gemini 3.1 Pro	Grok 4	-0.0645	13	16	2	0.9922	1.0000	no	no
Gemini 3.1 Pro	Llama 3.3 70B	-3.5323	26	5	0	0.0001	0.0041	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-2.0968	22	8	1	0.0296	0.7980	no	no
Gemini 3.8 Flash	GPT-5.5	0.9355	11	18	2	0.6734	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-3.8548	26	4	1	0.0001	0.0035	yes	yes
Gemini 3.8 Flash	Grok 4	-0.9677	17	12	2	0.1497	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-4.4355	26	4	1	0.0001	0.0038	yes	yes
GPT-5.4-mini	GPT-5.5	3.0323	6	24	1	0.0052	0.1984	yes	no
GPT-5.4-mini	gpt-oss-120b	-1.7581	22	9	0	0.0066	0.2321	no	no
GPT-5.4-mini	Grok 4	1.1290	11	17	3	0.3777	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-2.3387	25	4	2	0.0001	0.0038	no	yes
GPT-5.5	gpt-oss-120b	-4.7903	27	2	2	0.0000	0.0005	yes	yes
GPT-5.5	Grok 4	-1.9032	21	6	4	0.0159	0.4758	no	no
GPT-5.5	Llama 3.3 70B	-5.3710	27	4	0	0.0000	0.0009	yes	yes
gpt-oss-120b	Grok 4	2.8871	5	26	0	0.0001	0.0049	yes	yes
gpt-oss-120b	Llama 3.3 70B	-0.5806	20	11	0	0.0500	1.0000	no	no

Grok 4	Llama 3.3 70B	-3.4677	26	4	1	0.0001	0.0039	yes	yes

Table 36: Cross-dataset pairwise tests under rule top4: Friedman $\chi^2 = 93.7$ ($p = 9.9\text{e-}16$), Iman–Davenport $F = 13.0$, Nemenyi CD = 2.71, Kendall's $W = 0.293$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.4032	22	7	2	0.0213	0.6174	no	no
Claude Opus 5	DeepSeek-chat	-3.0000	22	8	1	0.0077	0.2690	yes	no
Claude Opus 5	DeepSeek-R1	-1.3226	17	11	3	0.0880	1.0000	no	no
Claude Opus 5	Gemini 3.1 Pro	-1.5484	20	8	3	0.0333	0.8329	no	no
Claude Opus 5	Gemini 3.8 Flash	-0.5484	19	10	2	0.1168	1.0000	no	no
Claude Opus 5	GPT-5.4-mini	-3.2258	25	6	0	0.0042	0.1601	yes	no
Claude Opus 5	GPT-5.5	-0.0645	14	10	7	0.3265	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.6290	26	4	1	0.0004	0.0164	yes	yes
Claude Opus 5	Grok 4	-1.8548	18	10	3	0.0535	1.0000	no	no
Claude Opus 5	Llama 3.3 70B	-5.4677	26	5	0	0.0002	0.0069	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.5968	20	7	4	0.0235	0.6574	no	no
Claude Sonnet 5	DeepSeek-R1	0.0806	15	14	2	0.4740	1.0000	no	no
Claude Sonnet 5	Gemini 3.1 Pro	-0.1452	15	14	2	0.9844	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.8548	11	17	3	0.1891	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-1.8226	20	8	3	0.0477	1.0000	no	no
Claude Sonnet 5	GPT-5.5	1.3387	10	18	3	0.0303	0.7867	no	no
Claude Sonnet 5	gpt-oss-120b	-3.2258	22	7	2	0.0059	0.2181	yes	no
Claude Sonnet 5	Grok 4	-0.4516	19	10	2	0.5240	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-4.0645	25	4	2	0.0001	0.0034	yes	yes
DeepSeek-chat	DeepSeek-R1	1.6774	9	21	1	0.0153	0.4898	no	no
DeepSeek-chat	Gemini 3.1 Pro	1.4516	9	20	2	0.0976	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	2.4516	6	22	3	0.0064	0.2316	no	no
DeepSeek-chat	GPT-5.4-mini	-0.2258	13	14	4	0.6950	1.0000	no	no
DeepSeek-chat	GPT-5.5	2.9355	5	22	4	0.0016	0.0638	yes	no
DeepSeek-chat	gpt-oss-120b	-1.6290	21	8	2	0.0445	0.9796	no	no
DeepSeek-chat	Grok 4	1.1452	12	16	3	0.2898	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.4677	23	4	4	0.0001	0.0056	no	yes
DeepSeek-R1	Gemini 3.1 Pro	-0.2258	16	14	1	0.7021	1.0000	no	no
DeepSeek-R1	Gemini 3.8 Flash	0.7742	11	18	2	0.1991	1.0000	no	no
DeepSeek-R1	GPT-5.4-mini	-1.9032	22	8	1	0.0522	1.0000	no	no
DeepSeek-R1	GPT-5.5	1.2581	8	19	4	0.0739	1.0000	no	no
DeepSeek-R1	gpt-oss-120b	-3.3065	26	4	1	0.0000	0.0023	yes	yes
DeepSeek-R1	Grok 4	-0.5323	13	14	4	0.5429	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-4.1452	26	2	3	0.0000	0.0003	yes	yes
Gemini 3.1 Pro	Gemini 3.8 Flash	1.0000	8	18	5	0.0405	0.9307	no	no
Gemini 3.1 Pro	GPT-5.4-mini	-1.6774	22	6	3	0.0114	0.3776	no	no
Gemini 3.1 Pro	GPT-5.5	1.4839	8	19	4	0.0196	0.6091	no	no
Gemini 3.1 Pro	gpt-oss-120b	-3.0806	23	4	4	0.0005	0.0191	yes	yes
Gemini 3.1 Pro	Grok 4	-0.3065	15	14	2	0.7762	1.0000	no	no
Gemini 3.1 Pro	Llama 3.3 70B	-3.9194	27	4	0	0.0001	0.0035	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-2.6774	22	8	1	0.0197	0.6091	no	no
Gemini 3.8 Flash	GPT-5.5	0.4839	13	16	2	0.8140	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-4.0806	27	3	1	0.0001	0.0029	yes	yes
Gemini 3.8 Flash	Grok 4	-1.3065	19	10	2	0.0899	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-4.9194	26	4	1	0.0001	0.0035	yes	yes
GPT-5.4-mini	GPT-5.5	3.1613	5	25	1	0.0020	0.0788	yes	no
GPT-5.4-mini	gpt-oss-120b	-1.4032	20	11	0	0.0097	0.3288	no	no
GPT-5.4-mini	Grok 4	1.3710	11	17	3	0.2898	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-2.2419	25	4	2	0.0001	0.0032	no	yes
GPT-5.5	gpt-oss-120b	-4.5645	27	2	2	0.0000	0.0004	yes	yes
GPT-5.5	Grok 4	-1.7903	19	7	5	0.0235	0.6574	no	no
GPT-5.5	Llama 3.3 70B	-5.4032	27	4	0	0.0000	0.0007	yes	yes
gpt-oss-120b	Grok 4	2.7742	5	26	0	0.0001	0.0049	yes	yes
gpt-oss-120b	Llama 3.3 70B	-0.8387	22	9	0	0.0377	0.9059	no	no

Grok 4	Llama 3.3 70B	-3.6129	26	4	1	0.0001	0.0035	yes	yes

Table 37: Cross-dataset pairwise tests under rule top5: Friedman $\chi^2 = 92.7$ ($p = 1.6\text{e-}15$), Iman–Davenport $F = 12.8$, Nemenyi CD = 2.71, Kendall's $W = 0.290$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.4677	22	7	2	0.0213	0.6213	no	no
Claude Opus 5	DeepSeek-chat	-3.1935	22	8	1	0.0086	0.2935	yes	no
Claude Opus 5	DeepSeek-R1	-1.2742	17	11	3	0.1017	1.0000	no	no
Claude Opus 5	Gemini 3.1 Pro	-1.6613	20	8	3	0.0385	0.9627	no	no
Claude Opus 5	Gemini 3.8 Flash	-0.6613	20	9	2	0.1123	1.0000	no	no
Claude Opus 5	GPT-5.4-mini	-2.9839	25	6	0	0.0042	0.1559	yes	no
Claude Opus 5	GPT-5.5	0.0484	14	11	6	0.3935	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.7742	26	4	1	0.0004	0.0168	yes	yes
Claude Opus 5	Grok 4	-1.9516	18	10	3	0.0523	1.0000	no	no
Claude Opus 5	Llama 3.3 70B	-5.3226	26	5	0	0.0002	0.0077	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.7258	20	7	4	0.0261	0.7036	no	no
Claude Sonnet 5	DeepSeek-R1	0.1935	15	14	2	0.5238	1.0000	no	no
Claude Sonnet 5	Gemini 3.1 Pro	-0.1935	15	14	2	0.8677	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.8065	11	17	3	0.1553	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-1.5161	20	8	3	0.0455	1.0000	no	no
Claude Sonnet 5	GPT-5.5	1.5161	9	19	3	0.0318	0.8264	no	no
Claude Sonnet 5	gpt-oss-120b	-3.3065	22	8	1	0.0056	0.1999	yes	no
Claude Sonnet 5	Grok 4	-0.4839	19	10	2	0.4988	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-3.8548	25	4	2	0.0001	0.0039	yes	yes
DeepSeek-chat	DeepSeek-R1	1.9194	9	21	1	0.0170	0.5283	no	no
DeepSeek-chat	Gemini 3.1 Pro	1.5323	9	20	2	0.0976	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	2.5323	6	22	3	0.0064	0.2251	no	no
DeepSeek-chat	GPT-5.4-mini	0.2097	10	17	4	0.3270	1.0000	no	no
DeepSeek-chat	GPT-5.5	3.2419	5	22	4	0.0018	0.0730	yes	no
DeepSeek-chat	gpt-oss-120b	-1.5806	21	9	1	0.0425	1.0000	no	no
DeepSeek-chat	Grok 4	1.2419	12	16	3	0.2809	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.1290	21	6	4	0.0004	0.0169	no	yes
DeepSeek-R1	Gemini 3.1 Pro	-0.3871	16	14	1	0.4988	1.0000	no	no
DeepSeek-R1	Gemini 3.8 Flash	0.6129	11	18	2	0.1553	1.0000	no	no
DeepSeek-R1	GPT-5.4-mini	-1.7097	22	8	1	0.0682	1.0000	no	no
DeepSeek-R1	GPT-5.5	1.3226	8	19	4	0.1143	1.0000	no	no
DeepSeek-R1	gpt-oss-120b	-3.5000	26	4	1	0.0000	0.0023	yes	yes
DeepSeek-R1	Grok 4	-0.6774	15	12	4	0.4044	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-4.0484	26	2	3	0.0000	0.0003	yes	yes
Gemini 3.1 Pro	Gemini 3.8 Flash	1.0000	9	17	5	0.0488	1.0000	no	no
Gemini 3.1 Pro	GPT-5.4-mini	-1.3226	20	8	3	0.0218	0.6213	no	no
Gemini 3.1 Pro	GPT-5.5	1.7097	8	19	4	0.0143	0.4566	no	no
Gemini 3.1 Pro	gpt-oss-120b	-3.1129	23	4	4	0.0005	0.0192	yes	yes
Gemini 3.1 Pro	Grok 4	-0.2903	14	15	2	0.8064	1.0000	no	no
Gemini 3.1 Pro	Llama 3.3 70B	-3.6613	26	5	0	0.0001	0.0043	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-2.3226	22	8	1	0.0207	0.6213	no	no
Gemini 3.8 Flash	GPT-5.5	0.7097	12	17	2	0.8677	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-4.1129	27	3	1	0.0001	0.0029	yes	yes
Gemini 3.8 Flash	Grok 4	-1.2903	19	10	2	0.0761	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-4.6613	26	4	1	0.0001	0.0035	yes	yes
GPT-5.4-mini	GPT-5.5	3.0323	5	25	1	0.0025	0.0959	yes	no
GPT-5.4-mini	gpt-oss-120b	-1.7903	23	8	0	0.0032	0.1208	no	no
GPT-5.4-mini	Grok 4	1.0323	11	17	3	0.3884	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-2.3387	25	4	2	0.0001	0.0032	no	yes
GPT-5.5	gpt-oss-120b	-4.8226	27	2	2	0.0000	0.0004	yes	yes
GPT-5.5	Grok 4	-2.0000	21	6	4	0.0128	0.4209	no	no
GPT-5.5	Llama 3.3 70B	-5.3710	27	4	0	0.0000	0.0007	yes	yes
gpt-oss-120b	Grok 4	2.8226	5	26	0	0.0001	0.0042	yes	yes
gpt-oss-120b	Llama 3.3 70B	-0.5484	20	11	0	0.0957	1.0000	no	no

Grok 4	Llama 3.3 70B	-3.3710	26	4	1	0.0001	0.0039	yes	yes

Table 38: Cross-dataset pairwise tests under rule v1 rubric: Friedman $\chi^2 = 88.1$ ($p = 1.3\text{e-}14$), Iman–Davenport $F = 11.9$, Nemenyi CD = 2.71, Kendall's $W = 0.275$.

a	b	mean rank diff	wins a	wins b	ties	Wilcoxon p	Holm p	Nemenyi	sig .05
Claude Opus 5	Claude Sonnet 5	-1.4194	22	7	2	0.0275	0.7965	no	no
Claude Opus 5	DeepSeek-chat	-2.7581	21	8	2	0.0151	0.4676	yes	no
Claude Opus 5	DeepSeek-R1	-1.1129	16	12	3	0.1167	1.0000	no	no
Claude Opus 5	Gemini 3.1 Pro	-1.7097	20	7	4	0.0465	1.0000	no	no
Claude Opus 5	Gemini 3.8 Flash	-0.4355	18	10	3	0.2279	1.0000	no	no
Claude Opus 5	GPT-5.4-mini	-2.9839	25	6	0	0.0048	0.1715	yes	no
Claude Opus 5	GPT-5.5	0.0806	13	11	7	0.5172	1.0000	no	no
Claude Opus 5	gpt-oss-120b	-4.5806	26	4	1	0.0003	0.0141	yes	yes
Claude Opus 5	Grok 4	-1.7419	18	10	3	0.0900	1.0000	no	no
Claude Opus 5	Llama 3.3 70B	-5.1613	25	6	0	0.0002	0.0097	yes	yes
Claude Sonnet 5	DeepSeek-chat	-1.3387	20	7	4	0.0488	1.0000	no	no
Claude Sonnet 5	DeepSeek-R1	0.3065	15	14	2	0.4620	1.0000	no	no
Claude Sonnet 5	Gemini 3.1 Pro	-0.2903	14	15	2	0.9844	1.0000	no	no
Claude Sonnet 5	Gemini 3.8 Flash	0.9839	11	17	3	0.0957	1.0000	no	no
Claude Sonnet 5	GPT-5.4-mini	-1.5645	20	8	3	0.0395	0.9491	no	no
Claude Sonnet 5	GPT-5.5	1.5000	9	19	3	0.0368	0.9491	no	no
Claude Sonnet 5	gpt-oss-120b	-3.1613	21	8	2	0.0038	0.1461	yes	no
Claude Sonnet 5	Grok 4	-0.3226	18	11	2	0.5898	1.0000	no	no
Claude Sonnet 5	Llama 3.3 70B	-3.7419	25	4	2	0.0001	0.0039	yes	yes
DeepSeek-chat	DeepSeek-R1	1.6452	9	21	1	0.0365	0.9491	no	no
DeepSeek-chat	Gemini 3.1 Pro	1.0484	12	16	3	0.1524	1.0000	no	no
DeepSeek-chat	Gemini 3.8 Flash	2.3226	6	22	3	0.0064	0.2251	no	no
DeepSeek-chat	GPT-5.4-mini	-0.2258	13	14	4	0.6241	1.0000	no	no
DeepSeek-chat	GPT-5.5	2.8387	6	21	4	0.0025	0.0988	yes	no
DeepSeek-chat	gpt-oss-120b	-1.8226	20	9	2	0.0351	0.9480	no	no
DeepSeek-chat	Grok 4	1.0161	13	15	3	0.3270	1.0000	no	no
DeepSeek-chat	Llama 3.3 70B	-2.4032	22	5	4	0.0004	0.0165	no	yes
DeepSeek-R1	Gemini 3.1 Pro	-0.5968	17	13	1	0.4986	1.0000	no	no
DeepSeek-R1	Gemini 3.8 Flash	0.6774	11	18	2	0.1287	1.0000	no	no
DeepSeek-R1	GPT-5.4-mini	-1.8710	22	8	1	0.0454	1.0000	no	no
DeepSeek-R1	GPT-5.5	1.1935	8	19	4	0.1073	1.0000	no	no
DeepSeek-R1	gpt-oss-120b	-3.4677	26	4	1	0.0000	0.0018	yes	yes
DeepSeek-R1	Grok 4	-0.6290	14	13	4	0.3987	1.0000	no	no
DeepSeek-R1	Llama 3.3 70B	-4.0484	26	2	3	0.0000	0.0004	yes	yes
Gemini 3.1 Pro	Gemini 3.8 Flash	1.2742	7	18	6	0.0317	0.8885	no	no
Gemini 3.1 Pro	GPT-5.4-mini	-1.2742	21	7	3	0.0114	0.3776	no	no
Gemini 3.1 Pro	GPT-5.5	1.7903	6	21	4	0.0086	0.2928	no	no
Gemini 3.1 Pro	gpt-oss-120b	-2.8710	23	4	4	0.0003	0.0138	yes	yes
Gemini 3.1 Pro	Grok 4	-0.0323	15	14	2	0.6592	1.0000	no	no
Gemini 3.1 Pro	Llama 3.3 70B	-3.4516	26	5	0	0.0001	0.0052	yes	yes
Gemini 3.8 Flash	GPT-5.4-mini	-2.5484	22	8	1	0.0114	0.3776	no	no
Gemini 3.8 Flash	GPT-5.5	0.5161	14	15	2	0.8140	1.0000	no	no
Gemini 3.8 Flash	gpt-oss-120b	-4.1452	26	4	1	0.0001	0.0029	yes	yes
Gemini 3.8 Flash	Grok 4	-1.3065	18	11	2	0.0937	1.0000	no	no
Gemini 3.8 Flash	Llama 3.3 70B	-4.7258	26	4	1	0.0001	0.0040	yes	yes
GPT-5.4-mini	GPT-5.5	3.0645	5	25	1	0.0013	0.0505	yes	no
GPT-5.4-mini	gpt-oss-120b	-1.5968	22	9	0	0.0042	0.1558	no	no
GPT-5.4-mini	Grok 4	1.2419	12	16	3	0.3468	1.0000	no	no
GPT-5.4-mini	Llama 3.3 70B	-2.1774	25	4	2	0.0001	0.0029	no	yes
GPT-5.5	gpt-oss-120b	-4.6613	27	2	2	0.0000	0.0005	yes	yes
GPT-5.5	Grok 4	-1.8226	18	8	5	0.0212	0.6353	no	no
GPT-5.5	Llama 3.3 70B	-5.2419	27	4	0	0.0000	0.0009	yes	yes
gpt-oss-120b	Grok 4	2.8387	5	26	0	0.0001	0.0038	yes	yes
gpt-oss-120b	Llama 3.3 70B	-0.5806	19	12	0	0.1169	1.0000	no	no

Grok 4	Llama 3.3 70B	-3.4194	26	4	1	0.0001	0.0052	yes	yes

Table 39: The (J, ρ) Pareto front.

w_em	w_num_tol	w_num_decay	w_tok_prec	w_tok_rec	w_tok_f1	w_edit_sim	w_rouge_l	w_jaccard	J	rho	auc
1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.437	0.828	0.989
0.650	0.150	0.100	0.000	0.000	0.000	0.100	0.000	0.000	0.432	0.864	0.994
0.400	0.200	0.100	0.000	0.000	0.050	0.000	0.200	0.050	0.431	0.878	0.994
0.400	0.100	0.150	0.100	0.000	0.000	0.100	0.150	0.000	0.431	0.884	0.994
0.050	0.050	0.350	0.250	0.050	0.100	0.100	0.050	0.000	0.417	0.884	0.994
0.200	0.000	0.250	0.250	0.050	0.050	0.150	0.050	0.000	0.416	0.884	0.994

Table 40: Mean score per dataset and model under the headline rule (v1 rubric).

dataset	Gemini 3.8 Flash	Claude Opus 5	GPT-5.5	DeepSeek-R1	Claude Sonnet 5	Gemini 3.1 Pro	Grok 4	GPT-5.4-mini	DeepSeek-chat	gpt-oss-120b	Llama 3.3 70B
abt_buy	0.980	1.000	1.000	0.970	0.970	0.980	0.960	0.960	0.960	0.940	0.960
alaska_camera	0.890	0.820	1.000	0.990	0.980	0.810	0.940	0.970	1.000	1.000	0.990
alaska_schema	0.918	0.945	0.928	0.957	0.929	0.928	0.929	0.958	0.951	0.934	0.840
amazon_google	0.930	0.940	0.940	0.950	0.960	0.930	0.940	0.930	0.960	0.860	0.950
bird	0.293	0.293	0.185	0.216	0.233	0.183	0.122	0.178	0.187	0.084	0.122
dblp_acm	1.000	1.000	1.000	0.990	1.000	1.000	1.000	0.990	1.000	0.910	0.970
dblp_scholar	0.930	0.950	0.960	0.950	0.970	0.940	0.950	0.940	0.960	0.910	0.970
fetaqa	0.445	0.401	0.438	0.413	0.373	0.433	0.428	0.312	0.434	0.322	0.253
fintagging	0.897	0.909	0.893	0.878	0.898	0.887	0.887	0.892	0.858	0.863	0.858
hitab	0.748	0.781	0.757	0.792	0.763	0.778	0.746	0.784	0.814	0.750	0.784
machamp	0.900	0.900	0.900	0.900	0.900	0.890	0.890	0.890	0.890	0.830	0.870
magneto_gdc	0.963	0.971	0.967	0.978	0.955	0.956	0.939	0.912	0.953	0.961	0.753
mmtu	0.894	0.882	0.917	0.850	0.872	0.914	0.901	0.882	0.865	0.830	0.807
officeqa	0.194	0.199	0.168	0.147	0.171	0.119	0.129	0.107	0.075	0.079	0.097
officeqa_pro_v2	0.198	0.270	0.188	0.148	0.157	0.120	0.119	0.123	0.121	0.104	0.115
opensanctions_pairs	0.860	0.940	0.750	0.850	0.730	0.970	0.850	0.900	0.760	0.780	0.790
papadakis_dn	0.940	0.920	0.950	0.940	0.950	0.930	0.950	0.950	0.940	0.930	0.940
realhitbench	0.746	0.743	0.724	0.672	0.681	0.699	0.640	0.661	0.662	0.603	0.611
smat	0.921	0.933	0.940	0.930	0.938	0.883	0.936	0.956	0.918	0.916	0.806
suc	0.976	0.986	0.943	0.942	0.911	0.945	0.943	0.914	0.858	0.932	0.740
tab_fact	0.860	0.660	0.840	0.770	0.740	0.830	0.850	0.820	0.840	0.830	0.780
tabis	0.950	0.960	0.960	0.970	0.920	0.950	0.980	0.900	0.770	0.920	0.820
tablebench	0.845	0.846	0.852	0.803	0.877	0.858	0.839	0.841	0.833	0.813	0.660
tableeval	0.791	0.765	0.784	0.773	0.733	0.790	0.802	0.742	0.711	0.773	0.636
tat_qa	0.829	0.861	0.789	0.815	0.845	0.802	0.791	0.789	0.806	0.765	0.752
tpcdi_cells	0.462	0.462	0.462	0.462	0.462	0.462	0.447	0.455	0.462	0.442	0.442
valentine	0.970	1.000	1.000	0.990	0.994	1.000	0.973	0.985	0.979	0.979	0.970
walmart_amazon	0.990	0.980	0.970	0.960	0.950	0.970	0.960	0.950	0.950	0.940	0.930
wdc_lspc	0.980	0.990	1.000	1.000	0.980	0.980	1.000	0.970	0.960	0.980	0.960
wdc_products	0.990	0.980	0.980	0.980	0.970	0.980	0.990	0.970	0.970	0.980	0.970
wikitablequestions	0.881	0.879	0.865	0.857	0.863	0.855	0.867	0.842	0.843	0.824	0.749

Table 41: Per-family optimum.

family	best weights	J own best	J global best
EM	exact match 1.00	0.358	0.358
SM	numeric tolerance 0.15, numeric decay 0.60, edit similarity 0.25	0.544	0.541
TQA	exact match 1.00	0.523	0.523
FIN	numeric tolerance 0.15, numeric decay 0.05, token precision 0.25, token recall 0.05, ROUGE-L 0.15, Jaccard 0.35	0.881	0.738

Table 42: The 30 best candidates with bootstrap statistics.

rank_full	w_em	w_ num_ tol	w_ num_ decay	w_ tok_ prec	w_ tok_rec	w_ tok_f1	w_ edit_ sim	w_ rouge_l	w_ jaccard	J_ boot_ mean	J_ boot_ sd	J_ boot_lo	J_ boot_ hi	p_vs_ best	in_ plateau	in_ practical_ plateau	times_ selected	J_ oob_ mean
41	0.100	0.450	0.100	0.050	0.050	0.000	0.000	0.200	0.050	0.370	0.025	0.319	0.418	0.636	yes	yes	27	0.341
4	0.400	0.200	0.100	0.000	0.000	0.050	0.000	0.200	0.050	0.370	0.026	0.318	0.418	0.656	yes	yes	22	0.340
19	0.200	0.250	0.100	0.050	0.150	0.100	0.000	0.150	0.000	0.369	0.025	0.321	0.417	0.634	yes	yes	8	0.341
88	0.000	0.400	0.000	0.050	0.150	0.100	0.100	0.200	0.000	0.369	0.025	0.319	0.416	0.631	yes	yes	12	0.341
26	0.100	0.300	0.100	0.200	0.050	0.050	0.000	0.100	0.100	0.369	0.025	0.320	0.416	0.640	yes	yes	4	0.340
46	0.150	0.350	0.100	0.050	0.100	0.050	0.050	0.150	0.000	0.369	0.025	0.321	0.416	0.644	yes	yes	8	0.340
76	0.000	0.350	0.000	0.250	0.200	0.050	0.050	0.050	0.050	0.369	0.025	0.319	0.416	0.627	yes	yes	16	0.340
49	0.200	0.200	0.050	0.050	0.050	0.150	0.050	0.150	0.100	0.369	0.025	0.318	0.415	0.639	yes	yes	4	0.339
73	0.050	0.300	0.100	0.050	0.050	0.200	0.000	0.050	0.200	0.369	0.025	0.320	0.415	0.618	yes	yes	3	0.340
142	0.100	0.250	0.000	0.000	0.100	0.150	0.150	0.250	0.000	0.369	0.025	0.317	0.416	0.609	yes	yes	5	0.340
23	0.050	0.300	0.100	0.100	0.100	0.050	0.050	0.150	0.100	0.369	0.025	0.319	0.416	0.623	yes	yes	6	0.340
51	0.050	0.250	0.100	0.050	0.050	0.150	0.050	0.150	0.150	0.369	0.025	0.318	0.415	0.632	yes	yes	3	0.340
7	0.200	0.050	0.200	0.000	0.050	0.100	0.000	0.250	0.150	0.369	0.026	0.320	0.419	0.632	yes	yes	9	0.341
215	0.050	0.300	0.000	0.100	0.200	0.050	0.150	0.150	0.000	0.369	0.025	0.318	0.416	0.619	yes	yes	3	0.340
31	0.300	0.200	0.000	0.050	0.050	0.250	0.100	0.050	0.000	0.369	0.025	0.318	0.416	0.631	yes	yes	7	0.339
114	0.100	0.200	0.100	0.000	0.050	0.200	0.000	0.050	0.300	0.369	0.025	0.320	0.416	0.618	yes	yes	6	0.340
122	0.100	0.250	0.000	0.200	0.150	0.100	0.150	0.000	0.050	0.368	0.025	0.318	0.415	0.605	yes	yes	1	0.340
16	0.200	0.250	0.200	0.150	0.050	0.150	0.000	0.000	0.000	0.368	0.026	0.318	0.418	0.611	yes	yes	14	0.340
64	0.150	0.300	0.050	0.150	0.000	0.050	0.100	0.100	0.100	0.368	0.025	0.319	0.414	0.613	yes	yes	2	0.338
112	0.050	0.250	0.000	0.200	0.100	0.000	0.150	0.100	0.150	0.368	0.025	0.318	0.415	0.606	yes	yes	5	0.340
90	0.050	0.300	0.050	0.000	0.100	0.100	0.050	0.050	0.300	0.368	0.025	0.319	0.413	0.598	yes	yes	3	0.338
50	0.100	0.250	0.100	0.050	0.050	0.250	0.050	0.050	0.100	0.368	0.025	0.319	0.415	0.601	yes	yes	2	0.339
183	0.050	0.300	0.050	0.150	0.050	0.050	0.150	0.200	0.000	0.368	0.025	0.317	0.416	0.606	yes	yes	5	0.339
45	0.200	0.250	0.150	0.050	0.100	0.200	0.050	0.000	0.000	0.368	0.025	0.320	0.417	0.616	yes	yes	3	0.339
35	0.250	0.100	0.100	0.050	0.100	0.300	0.000	0.050	0.050	0.368	0.025	0.319	0.417	0.613	yes	yes	4	0.339
39	0.200	0.250	0.000	0.100	0.100	0.000	0.050	0.100	0.200	0.368	0.025	0.318	0.414	0.595	yes	yes	4	0.338
125	0.150	0.200	0.050	0.050	0.000	0.150	0.100	0.100	0.200	0.368	0.025	0.318	0.413	0.614	yes	yes	2	0.338
160	0.100	0.100	0.100	0.150	0.100	0.350	0.000	0.100	0.000	0.368	0.025	0.319	0.417	0.599	yes	yes	4	0.339
103	0.150	0.100	0.150	0.050	0.050	0.300	0.050	0.150	0.000	0.368	0.025	0.319	0.417	0.611	yes	yes	1	0.339
80	0.000	0.250	0.150	0.000	0.050	0.100	0.050	0.200	0.200	0.368	0.025	0.320	0.416	0.598	yes	yes	2	0.340

Table 43: Bradley–Terry rank (Borda rank in brackets) under each reported rule, Friedman p and Kendall's W among datasets.

rule	Gemini 3.8 Flash	Claude Opus 5	GPT-5.5	DeepSeek-R1	Claude Sonnet 5	Gemini 3.1 Pro	Grok 4	GPT-5.4-mini	DeepSeek-chat	gpt-oss-120b	Llama 3.3 70B	Friedman p	W
exact match	3 (3)	1 (1)	2 (2)	4 (4)	7 (5)	5 (7)	6 (6)	8 (9)	9 (8)	10 (10)	11 (11)	2.7e-15	0.28
v1 rubric	1 (3)	2 (2)	3 (1)	4 (4)	5 (5)	6 (6)	7 (7)	8 (9)	9 (8)	10 (10)	11 (11)	1.3e-14	0.27
top1	3 (3)	1 (1)	2 (2)	4 (4)	7 (5)	5 (7)	6 (6)	8 (9)	9 (8)	10 (10)	11 (11)	2.7e-15	0.28
top2	3 (3)	1 (1)	2 (2)	4 (4)	7 (5)	5 (7)	6 (6)	8 (9)	9 (8)	10 (10)	11 (11)	4.2e-15	0.27
top3	2 (3)	1 (1)	3 (2)	4 (4)	5 (5)	6 (6)	7 (7)	8 (8)	9 (9)	10 (10)	11 (11)	2.0e-15	0.29
top4	1 (3)	2 (1)	3 (2)	4 (4)	6 (5)	5 (6)	7 (7)	8 (9)	9 (8)	10 (10)	11 (11)	9.9e-16	0.29
top5	1 (3)	2 (2)	3 (1)	4 (4)	5 (5)	6 (6)	7 (7)	8 (8)	9 (9)	10 (10)	11 (11)	1.6e-15	0.29
best-rho	1 (3)	2 (2)	3 (1)	4 (4)	5 (5)	6 (6)	7 (7)	8 (8)	9 (9)	10 (10)	11 (11)	1.0e-14	0.28
knee	1 (3)	2 (2)	3 (1)	4 (4)	5 (5)	6 (6)	7 (7)	8 (8)	9 (9)	10 (10)	11 (11)	1.6e-15	0.29

Table 44: Anchor correctness per dataset.

dataset	v1	bestJ	bestrho
abt_buy	0.860	0.860	0.827
alaska_camera	0.857	0.857	0.827
alaska_schema	0.877	0.836	0.913
amazon_google	0.863	0.863	0.827
bird	0.889	0.820	0.910
dblp_acm	0.860	0.860	0.825
dblp_scholar	0.858	0.858	0.828
fetaqa	0.913	0.811	0.931
fintagging	0.874	0.848	0.879
hitab	0.906	0.826	0.921
machamp	0.857	0.857	0.825
magneto_gdc	0.872	0.838	0.910
mmtu	0.868	0.836	0.891
officeqa	0.894	0.769	0.900
officeqa_pro_v2	0.885	0.757	0.898
opensanctions_pairs	0.860	0.860	0.828
papadakis_dn	0.891	0.891	0.838
realhitbench	0.899	0.824	0.918
smat	0.914	0.839	0.926
suc	0.857	0.826	0.881
tab_fact	0.862	0.862	0.826
tabis	0.639	0.649	0.595
tablebench	0.898	0.823	0.917
tableeval	0.866	0.814	0.887
tat_qa	0.885	0.796	0.887
tpcdi_cells	0.905	0.812	0.908
valentine	0.846	0.844	0.891
walmart_amazon	0.867	0.867	0.828
wdc_lspc	0.856	0.856	0.825
wdc_products	0.856	0.856	0.825
wikitablequestions	0.877	0.821	0.911

Table 45: Kendall τ between the Bradley–Terry rankings of the rules.

rule	exact match	v1 rubric	top1	top2	top3	top4	top5	best-rho	knee
exact match	1.000	0.855	1.000	1.000	0.891	0.891	0.855	0.855	0.855
v1 rubric	0.855	1.000	0.855	0.855	0.964	0.964	1.000	1.000	1.000
top1	1.000	0.855	1.000	1.000	0.891	0.891	0.855	0.855	0.855
top2	1.000	0.855	1.000	1.000	0.891	0.891	0.855	0.855	0.855
top3	0.891	0.964	0.891	0.891	1.000	0.927	0.964	0.964	0.964
top4	0.891	0.964	0.891	0.891	0.927	1.000	0.964	0.964	0.964
top5	0.855	1.000	0.855	0.855	0.964	0.964	1.000	1.000	1.000
best-rho	0.855	1.000	0.855	0.855	0.964	0.964	1.000	1.000	1.000
knee	0.855	1.000	0.855	0.855	0.964	0.964	1.000	1.000	1.000

Table 46: One-step sensitivity.

point	n_neighbours	J_centre	J_min	J_max	J_mean	flatness_range	share_neighbours_better
v1_rubric	32.0000	0.4110	0.4052	0.4187	0.4120	0.0135	0.6250
best	8.0000	0.4372	0.4178	0.4382	0.4250	0.0205	0.1250